



TECHNISCHE UNIVERSITÄT
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Faculty of Electrical Engineering and Information Technology
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Electronic Media Technology

MASTER THESIS

Impact of Acoustic Simulation Fidelity on Deep Learning-Based Sound Localization

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Major: Research in Media Engineering

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Ilmenau, August 31, 2026

Abstract

Deep learning-based sound source localization and detection (SSLD) typically requires large amounts of labeled training data covering diverse acoustic conditions. Since acquiring such data through real-world measurements is costly and time-consuming, simulated room impulse responses provide an efficient alternative. However, increasing the physical fidelity of acoustic simulations also increases computational cost, and it remains unclear which aspects of simulation accuracy are relevant for generalization to real environments. This thesis investigates the influence of acoustic simulation fidelity on the real-world performance of a deep learning-based SSLD model. Multiple training datasets were generated using different room-simulation configurations. A convolutional recurrent neural network using third-order Ambisonics input features was trained to detect and localize up to three simultaneously active, stationary speech sources. Generalization was evaluated on independent datasets derived from measured higher-order Ambisonic room impulse responses. It was found that the presence of early reflections in the simulation data is considerably more important than accurately modeling their spatial properties or the characteristics of late reverberation. A more accurate microphone simulation can improve performance, although the gain is comparatively small relative to the substantial increase in computational cost. Overall, the results indicate that highly detailed room simulations are not required to achieve good generalization to real acoustic environments. Computationally efficient simulation methods can provide an effective basis for generating SSLD training data.

Keywords: sound source localization and detection, deep learning, room acoustics simulation, room impulse response, image-source method, higher-order Ambisonics

Acknowledgments

I would like to thank Florian Klein for suggesting the collaboration with the Aalto Acoustics Lab and for putting me in contact with Johannes Arend in the first place.

I am especially grateful to my two supervisors, Stephan Werner and Johannes Arend. Stephan, thank you for making the collaboration with the Aalto Acoustics Lab so easy and uncomplicated, and for trusting Johannes and the other colleagues at the lab. Johannes, thank you for giving me a lot of freedom during the thesis while still guiding me whenever I needed help. I really appreciated you introducing me to the various room simulation and Ambisonics libraries and tools, as well as sharing your expertise in virtual acoustics with me.

I would also like to thank all the other colleagues at the Aalto Acoustics Lab. I am especially grateful to Nils Meyer-Kahlen for guiding me during Johannes' absence, giving me many suggestions and possible directions for the thesis, and pointing me toward a lot of papers and research. I really appreciated your awareness of recent developments, and your help with Ambisonics. Thank you to Sergio de las Heras for always being there to listen whenever I felt stuck, and to Thomas Deppisch for the valuable feedback while proofreading my thesis. I would also like to thank everyone else at the lab for the fresh ideas, discussions, and input during lunch and coffee breaks.

Finally, I would like to thank my family. I am especially grateful to my parents for supporting me throughout my studies and for making my exchange in Finland possible. And thank you to my grandma for making sure I always had something to eat during the final week before the submission deadline.

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1 Introduction

The localization of sound sources is an important task in many applications, such as robotics, human–computer interaction, hearing aids, surveillance systems, and immersive audio technologies. In recent years, methods from the field of deep learning (DL) have shown strong performance in sound source localization and detection (SSLD) tasks. However, training such models usually requires large amounts of labeled audio data that cover a wide range of acoustic conditions, like different room geometries, source–receiver positions, noise levels, and reverberation characteristics. Collecting such datasets through real-world recordings is costly, time-consuming, and often impractical.

To address this challenge, many approaches rely on simulated acoustic data. Simulation enables automatic and efficient generation of large-scale datasets with controllable parameters. Several simulation techniques exist that differ in their level of physical realism and computational cost. In principle, simulations with higher physical fidelity may produce acoustic signals that more closely resemble real-world recordings and therefore reduce the simulation-to-real domain gap. At the same time, greater physical accuracy generally comes at the cost of increased computational complexity and longer data-generation times. It is therefore unclear whether highly detailed acoustic simulations are necessary for training DL models that generalize effectively to real acoustic environments.

The main objective of this thesis is to investigate the influence of acoustic simulation fidelity on the real-world performance of DL models for SSLD. In particular, the study aims to determine whether increasing the physical realism of simulated training data leads to measurable improvements in performance on real recordings, and whether such improvements justify the additional computational cost. By comparing models trained on datasets generated using simulations with different levels of fidelity, this work seeks to identify an appropriate trade-off between simulation accuracy, computational efficiency, and model generalization.

First, this thesis introduces the fundamental principles of sound propagation, room acoustics, and spatial audio in the Ambisonics domain. It then examines existing approaches to acoustic simulation, with their underlying assumptions and limitations. Then, the current state of SSLD is reviewed, focusing especially on DL-based methods. Based on this literature review, suitable simulation methods and a model architecture are selected for the experimental

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study. Subsequently, multiple training datasets are generated using different acoustic simulation parameters, representing varying levels of simulation complexity and physical realism. The selected DL model is then trained separately on these datasets. To evaluate generalization beyond simulated conditions, the trained models are tested on various datasets derived from measured room impulse responses captured in diverse acoustic environments. The performance is evaluated using appropriate quantitative metrics.

2 Fundamentals of Acoustics

2.1 Propagation of Sound

Sound is a mechanical wave that can only propagate in a material medium. A free field describes the ideal case in which sound propagates without reflections from boundaries. In such a field, the propagation is mainly determined by geometric spreading and by losses in the medium. Although a perfect free field is an idealization, it is a useful model for estimating sound propagation.

2.1.1 Sound Pressure and Sound Intensity

Sound appears as small fluctuations of pressure around the ambient atmospheric pressure, as illustrated in Fig. 2.1. These fluctuations are produced by local compressions and rarefactions of the air. The acoustic pressure $p(\mathbf{x}, t)$ is therefore defined as the difference between the instantaneous pressure and the static pressure at a position $\mathbf{x}$ and time t .

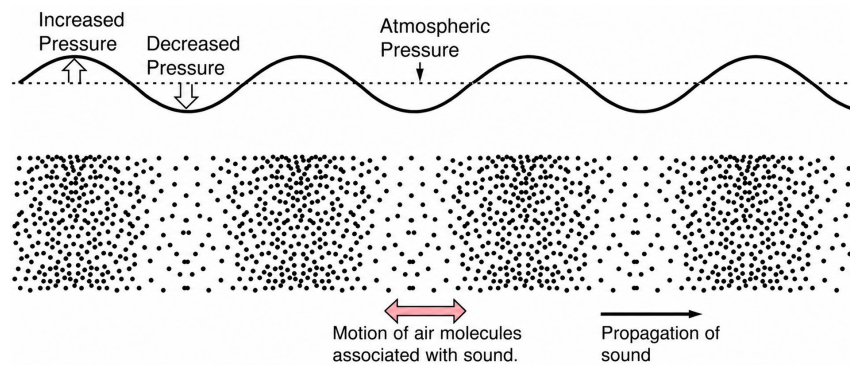


Figure 2.1: Compressions and rarefactions of air molecules during sound propagation [68].

Sound pressure and sound intensity describe different properties of a sound field. Sound pressure is a scalar quantity that characterizes the local pressure fluctuation in the medium, whereas sound intensity is a vector quantity that describes the local rate and direction of acoustic energy transport per unit area and therefore has both magnitude and direction. The

relationship between the magnitude of the sound intensity and the sound pressure for a plane progressive wave is given by

$$I = \frac{p_{\text{rms}}^2}{\rho_0 c}, \quad (2.1)$$

where p_{rms} is the root-mean-square sound pressure, ρ_0 is the density of air, and c is the speed of sound. Sound intensity is quantified in watts per square meter (W m^{-2}) [89].

The intensity vector points in the direction of acoustic energy flow, while its magnitude represents the acoustic power transmitted per unit area. Its calculation requires the sound pressure $p(\mathbf{x}, t)$ and the particle-velocity vector $\mathbf{u}(\mathbf{x}, t)$, which describes the motion of particles in the medium. Since both pressure and particle velocity oscillate, the instantaneous intensity can vary rapidly. The net transport of acoustic energy is therefore described by the time-averaged intensity, also known as the active intensity [22].

$$\mathbf{I}_a := \langle \mathbf{I} \rangle_t = \frac{1}{T} \int_0^T p(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) dt \quad (2.2)$$

For signals represented in the complex time–frequency domain, typically obtained using the short-time Fourier transform (STFT), the active acoustic intensity is defined as the real part of the complex pressure–velocity product [83]:

$$\mathbf{I}_a(\nu, \ell) = \text{Re} \{ p(\nu, \ell) \mathbf{u}^*(\nu, \ell) \}, \quad (2.3)$$

where ν and ℓ denote the frequency-bin and time-frame indices, respectively, and $(\cdot)^*$ denotes complex conjugation. The so-called reactive acoustic intensity is obtained from the imaginary part of the same product:

$$\mathbf{I}_r(\nu, \ell) = \text{Im} \{ p(\nu, \ell) \mathbf{u}^*(\nu, \ell) \}. \quad (2.4)$$

While active intensity describes the net transport of acoustic energy, reactive intensity represents the non-propagating, oscillatory sound energy flux [22, 83].

2.1.2 Plane and Spherical Waves

The shape of the wavefront depends on the type of source and on the distance from it. Two basic cases are especially important: plane waves and spherical waves.

A plane wave is characterized by parallel wavefronts, with all points on each wavefront having the same phase. If the wave travels in the positive x -direction, the pressure depends only on x and time t . The harmonic pressure field can then be written as

$$p(x, t) = A e^{i(\omega t - kx)}, \quad (2.5)$$

2 Fundamentals of Acoustics

where A is the complex pressure amplitude, $\omega = 2\pi f$ is the angular frequency, f is the frequency in hertz, $k = \omega/c$ is the acoustic wavenumber, c is the speed of sound, and i denotes the imaginary unit [45].

An ideal point source in a free field radiating equally in all directions produces spherical wavefronts. In this case, the pressure depends on the radial distance r from the source and on time t . For an outgoing harmonic spherical wave, the pressure can be written as

$$p(r, t) = \frac{A}{r} e^{i(\omega t - kr)}. \quad (2.6)$$

The factor $1/r$ describes spherical spreading. It follows from conservation of acoustic power: as the wave propagates outward, the same total power is spread over the surface of an expanding sphere, whose area increases as $4\pi r^2$. Therefore, the intensity decreases proportionally to $1/r^2$ (see Fig. 2.2). Since the intensity is proportional to the square of the sound pressure according to Eq. (2.1), the sound pressure decreases proportionally to $1/r$. In level terms, this corresponds to a decrease of approximately 6 dB for every doubling of distance if additional losses such as atmospheric absorption are neglected [45, 75].

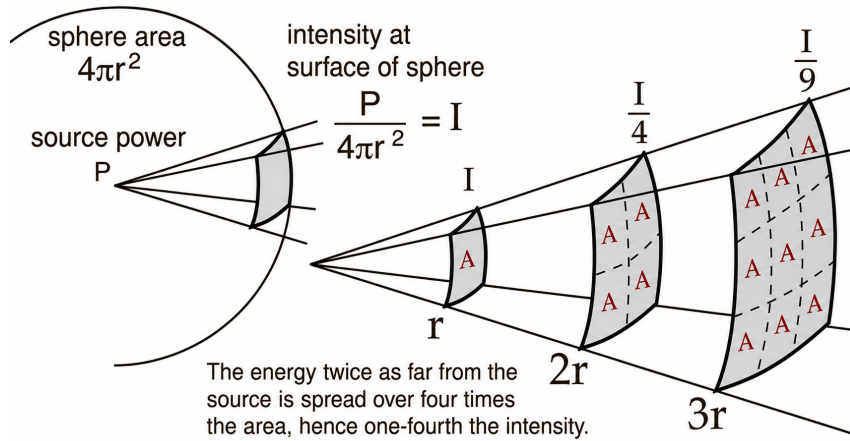


Figure 2.2: Spherical spreading of sound from an ideal point source in a free field [67].

Unlike a plane wave, the particle velocity in a spherical wave is not exactly in phase with the acoustic pressure. Consequently, relations derived under the plane-wave assumption are not strictly valid for spherical waves. However, at a distance sufficiently far from the acoustic source, the curvature of the wavefront becomes small over the region of interest, and the spherical wave can be locally approximated as a plane wave [45, 75].

2.1.3 Sound in Air

The speed of sound in air is not a fixed constant, since it depends on the thermodynamic properties of the gas. For an ideal gas, using Laplace's adiabatic assumption, the speed of sound is given by

$$c = \sqrt{\gamma R_s T}, \quad (2.7)$$

where c is the speed of sound, γ is the ratio of specific heats, R_s is the specific gas constant, and T is the absolute temperature. This expression shows that, for ordinary conditions in air, temperature is the main factor controlling the speed of sound. The value of γ is usually taken to be 1.4 for air [75]. For dry air at normal atmospheric pressure, the following linear approximation is often used:

$$c \approx 331.3 + 0.6 \vartheta \quad \text{m s}^{-1}, \quad (2.8)$$

where ϑ is the temperature in degrees Celsius. At a typical room temperature of 20°C, this results in approximately

$$c \approx 343 \text{ m s}^{-1}. \quad (2.9)$$

Humidity also influences the speed of sound, with higher humidity causing a slight increase. In most practical applications, however, this effect is small and typically less than about 1.5 % [75].

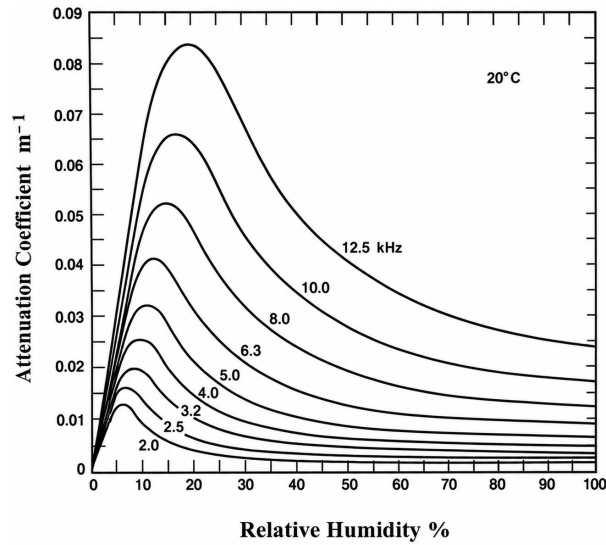


Figure 2.3: Air attenuation coefficient as a function of relative humidity for different frequencies at 20°C [6].

Sound propagation in air is affected by atmospheric absorption. Part of the acoustic energy is converted into heat due to viscous and thermal losses. In addition, molecular relaxation processes in the air contribute to the attenuation. These effects depend strongly on temperature, relative humidity, atmospheric pressure and frequency. Higher frequencies are usually absorbed more strongly than lower frequencies (Fig. 2.3), which means that the spectral con-

tent of a sound signal changes with distance [42]. Relative humidity differs between indoor and outdoor environments and varies with geographic location and season. A study of office buildings reported typical indoor relative humidity levels ranging from approximately 30 % to 60 % [44].

2.1.4 Interaction of Sound with Materials

When an acoustic wave encounters a material boundary, several phenomena may occur, including absorption, reflection, diffusion, transmission, and diffraction (Fig. 2.4). The relative contribution of each mechanism depends on the material properties, surface geometry, frequency, and angle of incidence. Consequently, acoustic materials cannot be described by a single universal behavior. Instead, their responses are typically characterized using specific equations and approximations associated with each phenomenon.

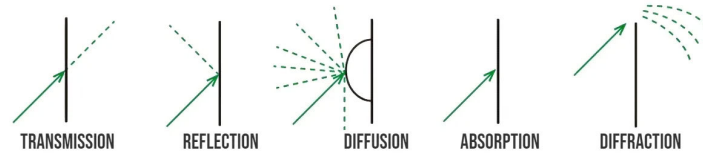


Figure 2.4: Behavior of sound at material boundaries [71].

Reflection and Absorption

When a plane sound wave strikes a flat, uniform wall, part of the acoustic energy is generally reflected back into the room. The reflected wave may differ from the incident wave in both amplitude and phase. This change is described by the complex reflection factor

$$R = |R|e^{i\chi}, \quad (2.10)$$

where $|R|$ represents the ratio of reflected to incident pressure amplitude and χ is the phase shift upon reflection. In general, the reflection factor depends on frequency and the angle of incidence.

The sound absorption coefficient is defined as the fraction of the incident sound intensity that is not reflected. According to Eq. (2.1), the sound intensity is proportional to the square of the sound pressure. Consequently, the absorption coefficient can be expressed in terms of the pressure reflection coefficient R as

$$\alpha = 1 - |R|^2. \quad (2.11)$$

Thus, $\alpha = 0$ corresponds to perfect reflection, whereas $\alpha = 1$ corresponds to complete absorption. For $R = 1$, the reflected pressure wave is in phase with the incident wave, corresponding to an acoustically hard boundary. For $R = -1$, the reflection occurs with a phase inversion, and the wall is referred to as acoustically soft. In both cases $|R| = 1$, so no acoustic energy is absorbed [51].

Sound absorption depends strongly on material properties such as thickness and porosity, as well as on sound frequency and angle of incidence. In practice, it is often characterized by the frequency-dependent normal-incidence absorption coefficient which can be obtained by an impedance tube measurement according to ISO 10534-2 [38].

Scattering and Diffusion

Scattering occurs when sound is redirected in multiple directions after interacting with a surface. Unlike specular reflections, where the angle of reflection equals the angle of incidence, scattering distributes sound energy over a range of directions. Surface roughness, irregular geometry, edges, and objects with dimensions comparable to the wavelength can all contribute to scattering [15]. ISO 17497-1 defines the measurement of a scattering coefficient that quantifies how much of the reflected sound energy is scattered away from the specular direction. If the scattered sound energy is distributed uniformly in all directions, the reflection is considered perfectly diffuse. The directional uniformity of the scattered sound can be quantified by a diffusion coefficient. Since both scattering and diffusion depend on the relationship between surface geometry and acoustic wavelength, they are frequency dependent [39].

Diffraction

Diffraction describes the bending or spreading of sound waves around obstacles, through openings, or at the edges of finite surfaces. The significance of diffraction depends on the relationship between wavelength and object size. When the size of an obstacle is much larger than the wavelength, the sound field can often be approximated by sound rays, and a clear acoustic shadow may occur behind the obstacle. However, when the obstacle size is comparable to or smaller than the wavelength, this ray-based description becomes less accurate. In this case, edges and openings act as secondary sources that radiate sound into the shadow region. As a result, sound can still reach a receiver even when the direct path is blocked. Diffraction therefore becomes stronger at larger wavelengths. Consequently, low-frequency sound is diffracted more strongly around objects, while high-frequency sound is shadowed. Scattering and diffraction are closely related phenomena and share the same physical basis [49].

Transmission and Refraction

Transmission occurs when part of the incident sound energy passes through a material or partition into another medium or space. The amount of transmitted sound is described by transmission loss, which compares the sound power incident on a partition with the sound power transmitted through it. Transmission loss depends mainly on the material's mass, stiffness, and damping, and it also varies with frequency [31]. In building acoustics, airborne sound transmission is evaluated according to ISO 717-1 [41].

When sound passes from one medium to another in which the speed of sound is different, its propagation direction may change. This change in direction is known as refraction. This bending of sound is similar to the refraction of light and depends on the sound velocities in the two media. Refraction can also occur in air due to temperature or wind gradients, causing sound rays to curve [49].

2.2 Room Acoustics

While sound propagation in a free field can often be described in a relatively simple and mathematically well-defined way, the situation becomes much more complex in enclosed spaces. In rooms, sound waves interact repeatedly with walls, objects, and other boundary surfaces. These interactions include reflection, scattering, absorption, and diffraction, as described in Sec. 2.1.4. Each interaction modifies the sound field, and the modified waves then propagate further and interact again with other surfaces. As a result, many sound components overlap in space and time, leading to a complex sound field that depends strongly on the room geometry, the material properties of the boundaries, and the positions of the source and receiver.

2.2.1 Room Impulse Response

The room impulse response (RIR) characterizes the acoustic behavior of an environment by capturing how an acoustic signal propagates from a source to a receiver. The RIR reflects the geometry, surface materials, and volume of the room. An anechoic signal can be enriched with the acoustic properties of a room by convolving it with the RIR [50]. In physical spaces, the RIR includes contributions from direct sound, early reflections, and dense late reverberation (see Fig. 2.5). Early reflections are temporally sparse, whereas late reverberation consists of dense and overlapping reflections and is therefore often modeled as a diffuse, stochastic sound field with a decaying energy envelope.

The transition from early reflections to late reverberation is referred to as the mixing time. The physical mixing time describes the point at which individual reflections begin to overlap and form a dense reverberant sound field. This transition can be estimated from the statistical distribution of the RIR sample values, which increasingly resembles a Gaussian distribution as the reverberant sound field becomes more diffuse [105]. The perceptual mixing time, on the other hand, refers to the point at which individual reflections are no longer perceived as distinct auditory events. It may occur earlier than the physical mixing time due to perceptual effects such as masking and the limited temporal resolution of the auditory system. The mixing time depends on various room properties, particularly on the room volume [55].

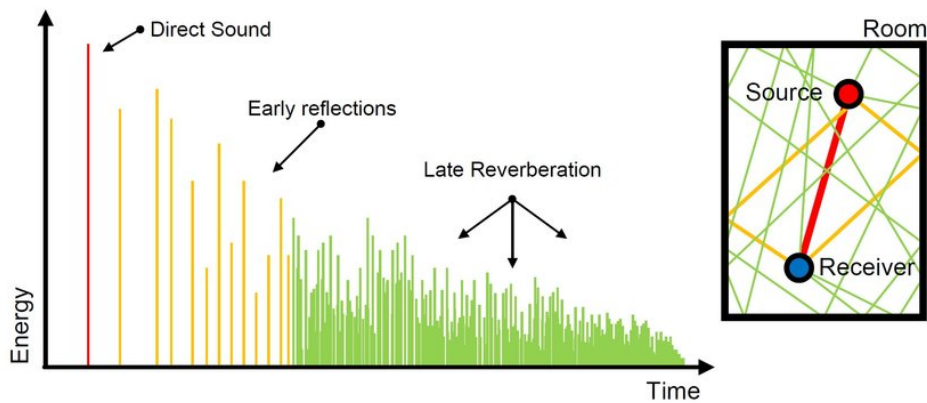


Figure 2.5: Components of a room impulse response [73].

There are several approaches for measuring the RIR. Early methods relied on impulsive sound sources such as starter pistols or balloon pops to approximate a Dirac delta impulse and thereby excite a broad range of frequencies in the room. While straightforward, these approaches generally suffer from low signal-to-noise ratio (SNR), limited repeatability, and weak low-frequency excitation, especially in large or highly damped spaces. More recently, the exponential sine sweep method has become the preferred approach in many applications [23]. In this method, a sine sweep with exponentially increasing frequency is reproduced by a loudspeaker and recorded in the room. The recording is then convolved with the time-inverse sweep signal to obtain the RIR. By sweeping energy over time across the frequency spectrum, it achieves a higher effective SNR and more reliable excitation of low frequencies [50]. RIRs can also be approximated using room-acoustic simulations, as described in Chap. 4. However, this requires knowledge or assumptions about the room geometry and the acoustic properties of the surface materials.

While omnidirectional RIRs are sufficient for deriving scalar reverberation metrics such as reverberation time, modern 3D-audio applications require directional RIRs that capture the spatial distribution of incident sound. These measurements are commonly referred to as spatial room impulse responses (SRIRs). In general, an SRIR can be understood as an ensemble of RIRs measured at multiple known receiver positions, typically using a microphone array. The

relative time and level differences between the individual RIRs contain information about the directions and propagation paths of sound within the room. Extracting this spatial information requires an appropriate decoding method that depends on the geometry and characteristics of the microphone array [111]. One example is an Ambisonic microphone array, as covered in Sec. 3.2. SRIRs have a wide range of applications, including the analysis of sound propagation in enclosed spaces and the auralization of acoustically realistic sound scenes.

2.2.2 Reverberation Time

Reverberation time (RT) is one of the most commonly used parameters for describing the acoustic properties of a room. It characterizes how long sound energy remains in the room after the excitation source has stopped. In an ideal diffuse sound field, the reverberation time T_{60} is defined as the time required for the reverberant sound energy to decay by 60 dB. Since a full 60 dB decay is often not available in measured RIRs due to background noise and limited dynamic range, T_{60} is usually estimated from a smaller decay range and then extrapolated to 60 dB. Two commonly used estimates are T_{20} and T_{30} , which are obtained from decay ranges of 20 dB and 30 dB, respectively [40].

The RT can be determined using different methods. One popular method is to calculate it from the RIR, denoted by $h(t)$, using the backward integration method proposed by Schroeder [97]. In this method, the energy decay curve (EDC) is calculated by integrating the squared impulse response backward in time t :

$$E(t) = \int_t^\infty h^2(\tau) d\tau. \quad (2.12)$$

The EDC is then converted to a decibel scale and normalized to its initial value:

$$L(t) = 10 \log_{10} \left(\frac{E(t)}{E(0)} \right). \quad (2.13)$$

A straight line is fitted to $L(t)$, starting at -5 dB to exclude the direct sound contribution from the estimate. For T_{20} , the fit is performed over the range from -5 to -25 dB, while for T_{30} the range extends from -5 to -35 dB. The slope of this fitted line represents the decay rate of the sound energy in the room. If the slope is denoted by s (in dB per second), the RT is calculated as

$$T_{20} = -\frac{60}{s_{20}}, \quad T_{30} = -\frac{60}{s_{30}}. \quad (2.14)$$

Since sound absorption at room boundaries and objects is frequency-dependent, the RT is usually not evaluated as a single broadband value. Instead, the RIR is filtered into octave or third-octave frequency bands before the decay curve is calculated. The RT is then determined separately for each band [40].

In addition to estimating the RT from a measured RIR, it can be approximated analytically from the geometry and absorption properties of a room. Two of the most widely used analytical expressions are the Sabine (2.15) and Eyring (2.16) equations. Both equations assume that the sound field is approximately diffuse and that the acoustic absorption is distributed sufficiently uniformly over the room boundaries [91, 21, 10].

$$RT_{\text{Sabine}} = \frac{0.161V}{S\bar{\alpha} + 4mV} \quad (2.15)$$

$$RT_{\text{Eyring}} = \frac{0.161V}{-S \ln(1 - \bar{\alpha}) + 4mV} \quad (2.16)$$

Here, V is the room volume, S is the total surface area, m is the air-attenuation coefficient, and $\bar{\alpha}$ is the area-weighted mean absorption coefficient, given by

$$\bar{\alpha} = \frac{\sum_i \alpha_i S_i}{S}. \quad (2.17)$$

Sabine's equation was obtained empirically from reverberation measurements in rooms while varying the amount of acoustic absorption [91]. One limitation of the equation is that it predicts a finite RT even when all room surfaces are perfectly absorbing ($\bar{\alpha} = 1$). Eyring addressed this limitation by deriving an alternative expression based on the fraction of sound energy remaining after successive reflections. For a mean absorption coefficient $\bar{\alpha}$, the fraction of energy remaining after each reflection is $1 - \bar{\alpha}$. For rooms with low average absorption, the Sabine and Eyring equations yield nearly identical results because $-\ln(1 - \bar{\alpha}) \approx \bar{\alpha}$. As the average absorption increases, the difference between the two predictions becomes increasingly significant. As $\bar{\alpha} \rightarrow 1$, almost no sound energy remains after a reflection and the Eyring equation predicts $RT \rightarrow 0$, making it more suitable for highly absorptive rooms with short RTs [21]. Nevertheless, both equations rely on simplifying assumptions and may not accurately predict the RT in all rooms.

2.2.3 Direct-to-Reverberant Ratio

The direct-to-reverberant ratio (DRR) describes the relationship between the sound energy arriving directly from the source and the energy arriving through reflections from the room. Together with the RT, it provides information about the acoustic characteristics and the source–receiver position. The DRR is obtained from the ratio of the energy of the direct sound component to the energy of the reverberant component and is defined as

$$DRR = 10 \log_{10} \left(\frac{\sum_{l \in \mathcal{D}} h^2(l)}{\sum_{l \in \mathcal{R}} h^2(l)} \right) \quad [\text{dB}], \quad (2.18)$$

where $h(l)$ denotes the room impulse response, $\mathcal{D}$ denotes the set of samples l corresponding to the direct sound, and $\mathcal{R}$ denotes the set of samples corresponding to the reverberant sound [11].

A high DRR indicates that the direct sound dominates over the reverberant sound, whereas a low or negative DRR indicates a stronger contribution from room reflections. The DRR generally decreases as the distance between the sound source and receiver increases, as the direct sound level decreases with distance while the reverberant sound energy is more constant throughout the space.

3 Ambisonics

Ambisonics is a spatial audio format based on the spherical harmonic decomposition of a sound field. Unlike channel-based audio formats, Ambisonics describes the directional structure of a sound field independently of a specific loudspeaker or playback configuration. This makes it particularly suitable for virtual reality and other spatial audio applications, because the encoded sound field can be rotated, processed, and decoded flexibly for different audio systems [123, 33]. Originally introduced in the 1970s by Michael Gerzon, Ambisonics was conceived as a compact and theoretically grounded method for reproducing sound fields with high spatial fidelity. Although early implementations were limited by computational and playback constraints, the method has received renewed attention due to advances in digital signal processing, the development of higher-order microphone arrays, and its adoption in immersive audio standards such as MPEG-H and MPEG-I for immersive media delivery [123]. Ambisonics represents a three-dimensional sound field at a single listening position by expanding its directional dependence in spherical harmonics. The spherical harmonics provide a convenient coordinate system for describing how sound energy is distributed across directions.

3.1 Spherical Harmonics

3.1.1 Mathematical Definition

The spherical harmonics (SHs) form an orthogonal set of basis functions on the unit sphere. They describe angular variation as a function of colatitude θ and azimuth ϕ , both measured in radians. The complex SHs are commonly written as

$$Y_n^m(\theta, \phi) = \sqrt{\frac{2n+1}{4\pi} \frac{(n-m)!}{(n+m)!}} P_n^m(\cos \theta) e^{im\phi}, \quad (3.1)$$

where the square-root term is a normalization factor, $P_n^m(\cdot)$ denotes the associated Legendre function including the Condon-Shortley phase, $n \in \mathbb{N}_0$ is the SH order, and $m \in \mathbb{Z}$ is the degree or index within that order, with $-n \leq m \leq n$ [86, 5]. Tab. A.1 presents expressions for the SHs up to fourth order.

3 Ambisonics

Due to their complex exponential dependence on the azimuthal angle ϕ , the magnitude of the SHs does not vary with ϕ . Therefore, plotting only the magnitude would not reveal their azimuthal structure. Instead, the SHs can be represented by corresponding real-valued functions constructed from their real and imaginary parts. The balloon plot in Fig. 3.1 shows these real-valued SH patterns up to fourth order $n = 4$, with the value of each function represented as the distance from the origin. Cyan indicates positive values and magenta negative values. The zeroth-order harmonic Y_0^0 is constant over the sphere and therefore corresponds to a monopole-like pattern. The first-order harmonics have dipole-like shapes, while higher orders lead to more complex patterns with additional lobes and nulls. Increasing n therefore increases the spatial detail [86]. For each order n , there are $2n + 1$ degrees m , with $m = -n, \dots, n$. Including all orders from 0 to N , the total number of SH components C is

$$C = \sum_{n=0}^N (2n + 1) = (N + 1)^2. \quad (3.2)$$

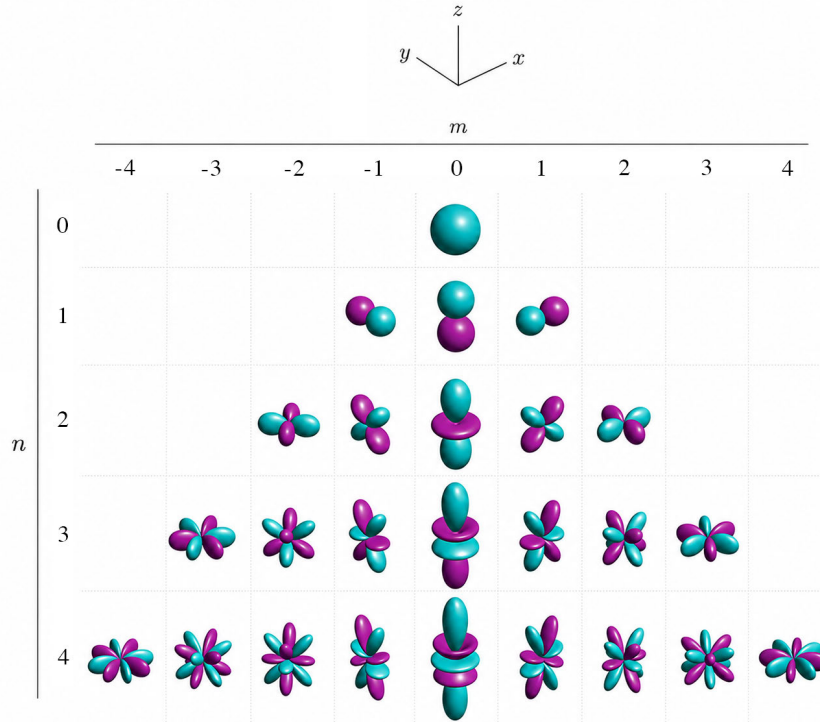


Figure 3.1: Balloon plot of the spherical harmonics for order $n = 0$ in the top row through $n = 4$ in the bottom row. The real-valued functions $Y_n^0(\theta, \phi)$ are shown in the central column. The imaginary parts, $\text{Im}\{Y_n^m(\theta, \phi)\}$, for $-n \leq m \leq -1$, are shown in the left-hand columns, while the real parts, $\text{Re}\{Y_n^m(\theta, \phi)\}$, for $1 \leq m \leq n$, are shown in the right-hand columns. The viewing direction is indicated by the orientation of the axes at the top of the figure. Colors denote the sign of the spherical harmonic functions: cyan shades represent positive values, and magenta shades represent negative values [86].

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SHs are orthogonal on the unit sphere, meaning that their inner product is zero for different modes:

$$\int_0^{2\pi} \int_0^\pi [Y_n^m(\theta, \phi)]^* Y_{n'}^{m'}(\theta, \phi) \sin \theta d\theta d\phi = \delta_{nn'} \delta_{mm'}. \quad (3.3)$$

Because of this orthogonality, a spatial function $g(\theta, \phi)$ defined on the unit sphere can be decomposed into SHs:

$$g(\theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n g_n^m Y_n^m(\theta, \phi). \quad (3.4)$$

In practice, the expansion cannot be continued to infinite order. Instead, it is truncated at a maximum order N :

$$g(\theta, \phi) \approx \sum_{n=0}^N \sum_{m=-n}^n g_n^m Y_n^m(\theta, \phi). \quad (3.5)$$

This truncation limits the spatial resolution. A low-order representation captures only broad directional information, whereas a higher-order representation can describe more detailed angular variations. The coefficients g_n^m describe how strongly each SH contributes to the spatial function [86]. They can be obtained by projecting $g(\theta, \phi)$ onto the corresponding SH:

$$g_n^m = \int_0^{2\pi} \int_0^\pi g(\theta, \phi) [Y_n^m(\theta, \phi)]^* \sin \theta d\theta d\phi. \quad (3.6)$$

In Ambisonics, each SH coefficient is represented by an individual audio channel. To ensure compatibility and consistent signal exchange between different Ambisonic systems, both the channel ordering and the normalization convention of the SHs must be clearly defined. A common channel ordering convention is Ambisonic Channel Number (ACN). ACN assigns a unique channel index to each SH component according to its order n and degree m (see Tab. A.1) [3]:

$$\text{ACN}_{\text{idx}} = n^2 + n + m. \quad (3.7)$$

Since Ambisonic audio channels are real-valued time-domain signals, practical Ambisonic systems use a real-valued SH basis rather than the complex-valued basis introduced in (3.1). This real basis is obtained from the complex SHs using Euler's formula, $e^{im\phi} = \cos(m\phi) + i\sin(m\phi)$. The cosine and sine terms are then used as real basis functions corresponding to the real and imaginary parts of the complex harmonics. In addition to the channel ordering, the amplitude scaling of the basis functions must be fixed by a normalization convention. The resulting real-valued SHs are defined as

$$\mathcal{Y}_n^m(\theta, \phi) = (-1)^m \sqrt{\frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!}} P_n^{|m|}(\cos \theta) \begin{cases} \sqrt{2} \sin(|m|\phi), & m < 0, \\ 1, & m = 0, \\ \sqrt{2} \cos(m\phi), & m > 0, \end{cases} \quad (3.8)$$

where the factor $(-1)^m$ cancels out the Condon-Shortley phase included in the associated Legendre polynomials. The square-root factor corresponds to the N3D normalization convention, which normalizes all SHs to have the same total energy over the unit sphere. The sign of m determines whether the azimuthal dependence is represented by the sine or cosine term. The additional factor $\sqrt{2}$ for nonzero degrees ensures that the resulting sine and cosine components retain the same energy as the corresponding complex SHs [18].

3.1.2 Pseudointensity Vectors

Using the Ambisonic Channel Number (ACN) channel notation, the first-order Ambisonics (FOA) signal in the complex time–frequency domain is written as

$$\mathbf{A}_1(\nu, \ell) = \begin{bmatrix} W(\nu, \ell) & Y(\nu, \ell) & Z(\nu, \ell) & X(\nu, \ell) \end{bmatrix}^T, \quad (3.9)$$

where ν and ℓ denote the frequency-bin and time-frame indices, respectively. The channel W is the omnidirectional component, while Y , Z , and X are dipole components oriented along the left, upward, and front directions, respectively.

As described in Sec. 2.1.1, acoustic intensity can be estimated from the sound pressure and particle velocity. FOA signals provide approximations of both quantities at the recording position: the omnidirectional component is proportional to the sound pressure, whereas the three first-order components are negatively proportional to the Cartesian components of the particle velocity. Consequently, the resulting quantity retains the directional properties of acoustic intensity. However, because the Ambisonic signals are generally not calibrated in physical units and constant scaling factors are commonly omitted, it does not represent the absolute acoustic intensity. It is therefore often referred to as “pseudointensity” [74, 83]. Derived from Eqs. (2.3) and (2.4), the active and reactive first-order pseudointensity vectors (FO-PIVs) can be calculated as:

$$\hat{\mathbf{I}}_a(\nu, \ell) = -\operatorname{Re} \left\{ W(\nu, \ell) \begin{bmatrix} X^*(\nu, \ell) \\ Y^*(\nu, \ell) \\ Z^*(\nu, \ell) \end{bmatrix} \right\}, \quad (3.10)$$

$$\hat{\mathbf{I}}_r(\nu, \ell) = -\operatorname{Im} \left\{ W(\nu, \ell) \begin{bmatrix} X^*(\nu, \ell) \\ Y^*(\nu, \ell) \\ Z^*(\nu, \ell) \end{bmatrix} \right\}. \quad (3.11)$$

In theory, the sound direction of arrival (DOA) can be estimated as the opposite direction of the active intensity vector [84]. In practice, however, the estimates obtained across all time-frequency bins are inconsistent in reverberant environments [85].

The FO-PIV can be generalized to higher-order Ambisonics (HOA), resulting in the higher-order pseudointensity vector (HO-PIV). In contrast to the FO-PIV, the HO-PIV does not correspond to a three-dimensional Cartesian direction. Instead, it provides a higher-dimensional and more abstract representation of the directional sound-field information. It is obtained by multiplying the omnidirectional Ambisonics coefficient $a_0(\nu, \ell)$ (previously denoted by $W(\nu, \ell)$) with the complex conjugates of the remaining $C - 1$ channel coefficients, where $C = (N + 1)^2$ is the total number of channels for Ambisonics order N [83]:

$$\hat{\mathbf{I}}_N(\nu, \ell) := -a_0(\nu, \ell) \begin{bmatrix} a_1^*(\nu, \ell) \\ a_2^*(\nu, \ell) \\ \vdots \\ a_{C-1}^*(\nu, \ell) \end{bmatrix}. \quad (3.12)$$

3.2 Spherical Microphone Arrays

Spherical microphone arrays (Fig. 3.2) provide a practical way to capture three-dimensional sound fields. In these arrays, the microphones are placed on, or close to, a spherical surface. Instead of treating the microphone signals as separate sensor signals, spherical array processing transforms the measured microphone pressures into SH components. These components can then be used to obtain Ambisonic signals [86, 3].



Figure 3.2: Examples of spherical microphone arrays. From left to right: the Rode NT-SF1, gfai tech Sphere48 AC Pro, ZYLIA ZM-1, and MH Acoustics Eigenmike em32 [90, 28, 124, 63].

Let $p_q(k)$ denote the frequency-domain pressure measured by the q -th microphone, where k denotes the acoustic wavenumber. The microphone is located at the angular position (θ_q, ϕ_q) . The purpose of the array processing stage is to estimate the SH coefficients of the sound field from these finite microphone measurements. For an accurate estimation, the number of microphones must be at least equal to the number of Ambisonic channels C [86].

For a continuous function $g(\theta, \phi)$ on the sphere, the SH coefficients are obtained by integration over the full sphere, as given in Eq. (3.6). However, a real microphone array can only measure the sound field at a finite number of positions. Therefore, the continuous integral has to be approximated by a discrete weighted sum. If the function is sampled at Q directions (θ_q, ϕ_q) ,

the SH coefficients can be approximated as

$$g_n^m \approx \sum_{q=1}^Q w_q g(\theta_q, \phi_q) [\mathcal{Y}_n^m(\theta_q, \phi_q)]^*, \quad (3.13)$$

where w_q are quadrature weights. These weights account for the spatial distribution of the sampling points on the sphere [86]. If the microphones are approximately uniformly distributed, the weights can be approximated by Eq. (3.14), since the surface area of the unit sphere is 4π .

$$w_q \approx \frac{4\pi}{Q} \quad (3.14)$$

The same idea can be applied to the microphone signals. The measured SH pressure coefficients are estimated as

$$p_n^m(k, r) \approx \sum_{q=1}^Q w_q p_q(k, r, \theta_q, \phi_q) [\mathcal{Y}_n^m(\theta_q, \phi_q)]^*. \quad (3.15)$$

Here, k is the acoustic wavenumber and r is the radius of the spherical microphone array. The coefficients $p_n^m(k, r)$ describe the pressure measured on the array surface in the SH domain. These measured coefficients are not yet the final Ambisonics sound-field coefficients. This is because the finite radius of the array introduces an order- and frequency-dependent radial response [86]. This effect must be compensated for before the measured coefficients can be interpreted as coefficients of the incident sound field, as described in the following section.

3.2.1 Open and Rigid Spherical Arrays

Spherical microphone arrays are commonly divided into open and rigid arrays. In an open array, the microphones are positioned on a spherical geometry, while the mounting structure has a negligible influence on the surrounding sound field. In contrast, a rigid spherical array consists of microphones mounted on the surface of a hard sphere. The important practical difference is that a rigid sphere influences the pressure field around the array. The incoming sound is scattered by the sphere before it reaches the microphones (see Fig. 3.3).

The modal strength $b_n(kr)$, also referred to as the radial response, describes how strongly an SH component of order n is represented in the pressure field at acoustic wavenumber k , measured on a sphere of radius r . The modal strength appears because the incident sound-field coefficients are defined with respect to the array center, while the microphones measure pressure on the spherical surface. Assuming a plane-wave sound field, the relationship in the SH domain can be written as

$$p_n^m(k, r) = b_n(kr) a_n^m(k), \quad (3.16)$$

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where $a_n^m(k)$ denotes the incident sound-field coefficient, and $p_n^m(k, r)$ is the corresponding pressure coefficient obtained from the microphone signals. The modal strength depends on the array type. For an open spherical microphone array, it is expressed as

$$b_n^{\text{open}}(kr) = 4\pi i^n j_n(kr), \quad (3.17)$$

with $j_n(\cdot)$ denoting the spherical Bessel function of order n . For an ideal spherical array, the modal strength $b_n(kr)$ is independent of the degree m , since the geometry is rotationally symmetric. Consequently, all $(2n+1)$ SH components of the same order share the same radial response [86].

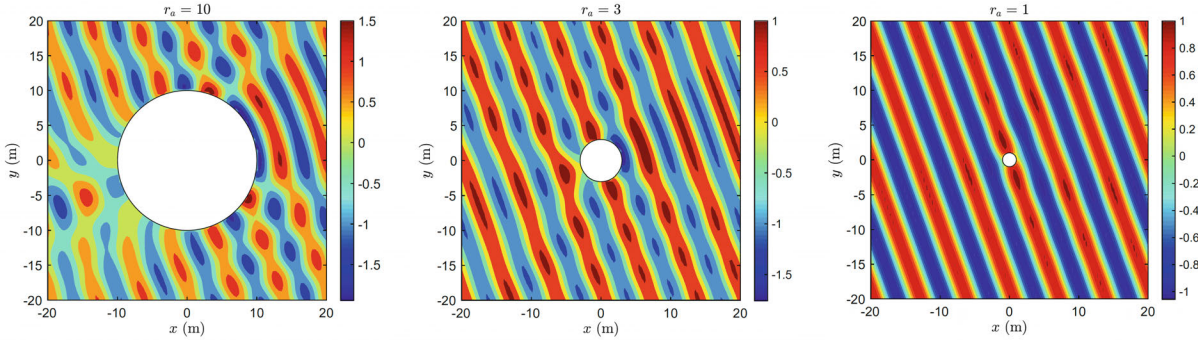


Figure 3.3: Diffraction of an incident sound field by a rigid sphere for different sphere radii. Larger rigid spheres cause stronger scattering and therefore modify the surrounding pressure field more significantly [86].

For a rigid spherical array, the modal strength is modified by scattering from the hard sphere. In this case, the modal strength is given by

$$b_n^{\text{rigid}}(kr) = 4\pi i^n \left[j_n(kr) - \frac{j_n'(kr)}{h_n^{(2)'}(kr)} h_n^{(2)}(kr) \right], \quad (3.18)$$

where $h_n^{(2)}(\cdot)$ denotes the spherical Hankel function of the second kind, and the primes indicate differentiation with respect to the argument kr . The first term inside the brackets corresponds to the incident-field contribution, while the second term accounts for the field scattered by the rigid sphere [86].

Fig. 3.4 shows the modal strengths of open and rigid spherical arrays up to SH order $n = 3$. At higher orders, the attenuation at low frequencies becomes increasingly pronounced. The open-array response follows the spherical Bessel function and therefore contains zeros at specific values of kr . At these frequencies, the corresponding SH components cannot be reliably recovered. For a rigid spherical array, the scattering caused by the rigid sphere modifies the radial response. As a result, the deep nulls that occur in the open-array response are avoided.

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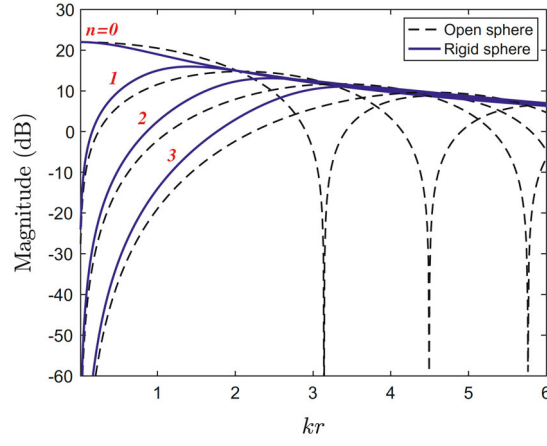


Figure 3.4: Magnitude of the modal strengths for open and rigid spherical microphone arrays up to spherical harmonic order $n = 3$ [86].

Since Ambisonics encoding requires the incident coefficients $a_n^m(k)$, not the measured pressure coefficients $p_n^m(k, r)$, the radial response has to be compensated by applying inverse radial filters:

$$a_n^m(k) = \frac{p_n^m(k, r)}{b_n(kr)}. \quad (3.19)$$

If $b_n(kr)$ is small, this division becomes numerically unstable. The inverse filter then requires a very large gain, which also amplifies microphone noise and other measurement inaccuracies. In practice, this gain is therefore limited by a regularization term. A commonly used regularization method is Tikhonov regularization, where the inverse radial filter is given by

$$b_{n,\text{reg}}^{-1}(kr) = \frac{b_n^*(kr)}{|b_n(kr)|^2 + \beta^2}, \quad \text{with} \quad \beta^2 = \frac{1 - \sqrt{1 - (QG_{\text{max}}^2)^{-1}}}{1 + \sqrt{1 - (QG_{\text{max}}^2)^{-1}}}. \quad (3.20)$$

Here, $(\cdot)^*$ denotes the complex conjugate, β is the regularization parameter, G_{max} specifies the maximum allowed amplification gain, and Q is the number of microphone sensors [79].

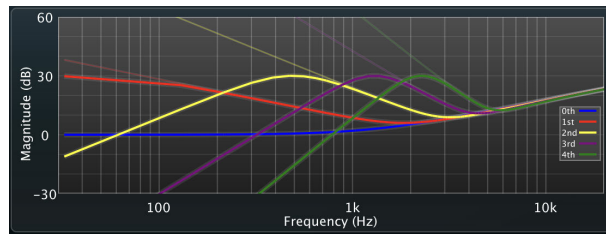


Figure 3.5: Magnitude responses of the inverse radial filters for an Eigenmike em32 up to the fourth spherical harmonic order. Screenshot of the SPARTA Array2SH plugin [58].

For rigid spherical arrays, the modal strength does not exhibit additional zeros at nonzero frequencies and provides slightly higher gain at low frequencies. Consequently, less compensation is required in the inverse radial filters, making rigid spherical arrays more robust and often

preferable for broadband higher-order Ambisonic recordings [86]. Fig. 3.5 shows the frequency responses of the inverse radial filters calculated for an Eigenmike em32, a rigid spherical microphone array with 32 uniformly distributed microphones and therefore supporting Ambisonics up to fourth order [63]. The dark curves represent the unregularized filters, while the bright curves show the filters after applying Tikhonov regularization with G_{max} set to 15 dB.

3.2.2 Beamforming

Beamforming describes a technique to emphasize sound arriving from a chosen direction while attenuating sound from other directions. In the Ambisonic domain, this can be interpreted as forming a directional microphone pattern from a weighted sum of SH components. By choosing suitable weights, the beam can be steered towards a desired direction and shaped into different directivity patterns. The minimum possible beam width is determined by the Ambisonics order, with narrower beams at higher orders [86].

The steering vector defines the directional response of the beamformer. For a look direction (θ_0, ϕ_0) , the steering vector contains the SH basis functions evaluated in that direction:

$$\mathbf{y}(\theta_0, \phi_0) = [\mathcal{Y}_0^0(\theta_0, \phi_0) \quad \mathcal{Y}_1^{-1}(\theta_0, \phi_0) \quad \cdots \quad \mathcal{Y}_N^N(\theta_0, \phi_0)]^T. \quad (3.21)$$

Fig. 3.6 shows an example of a fourth-order cardioid beamformer using the Eigenmike em32. At low frequencies, the directivity is reduced because the higher-order components are strongly attenuated by the array response (see modal strengths in Sec. 3.2.1) and therefore contribute less to the beam pattern. Additionally, at low frequencies, the wavelengths are large compared to the array diameter. As a result, the phase and level differences between the microphone signals become small, which reduces the spatial resolving capability of the array and results in a wider main lobe. At high frequencies above approximately 8 kHz, spatial aliasing causes artifacts in the directivity pattern.

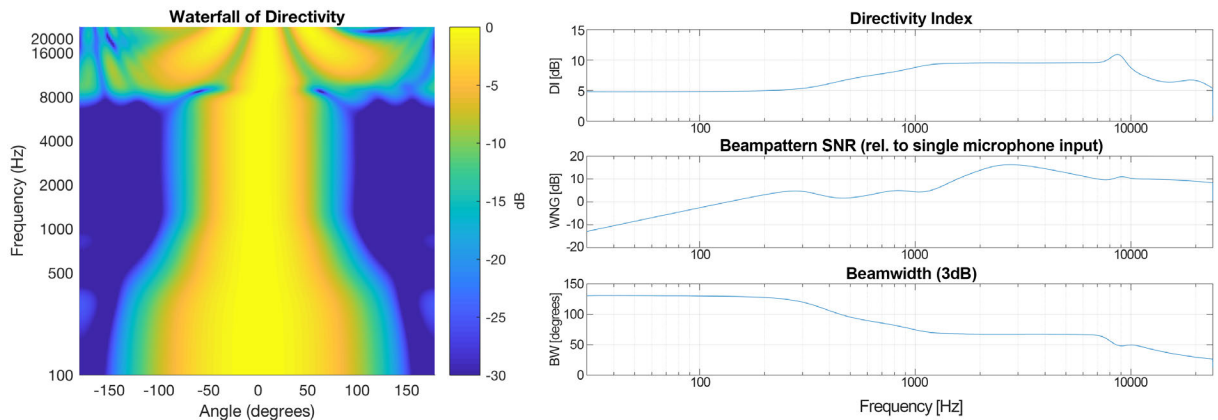


Figure 3.6: Directivity pattern of a fourth-order cardioid beamformer using the Eigenmike em32 [8].

4 Acoustic Room Simulation

Acoustic room simulation aims to reproduce sound propagation in enclosed spaces as realistically as possible. By modeling direct sound, room reflections, absorption, and other acoustic phenomena, simulation methods can approximate the acoustic response of real rooms. These methods are widely used to predict RIRs, evaluate acoustic designs, create realistic virtual auditory environments and, more recently, generate large-scale synthetic datasets for training DL models.

4.1 Image-Source Method

The image-source method (ISM) is the most common technique for simulating reflections in room acoustics and is based on the assumption of perfectly specular reflections. Allen and Berkley introduced the use of the ISM for computerized RIR simulation [4]. Its main advantage is its ability to model specular reflections accurately and computationally efficiently, particularly in rectangular (“shoebox”) rooms where the source mirroring procedure is straightforward.

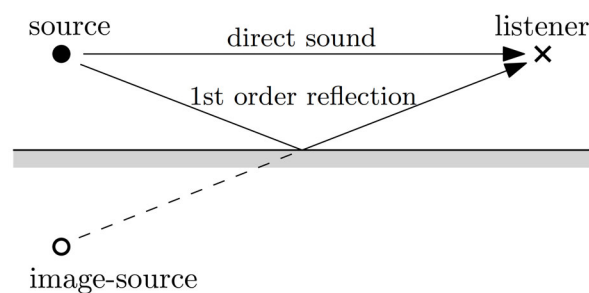


Figure 4.1: First-order image-source reflection [70].

4.1.1 Geometrical Principle

The basic principle of the ISM is based on geometrical acoustics. Instead of explicitly modeling reflections from room boundaries, virtual sound sources (image sources) are created by mirroring the original source across reflective surfaces (see Fig. 4.1). For a simple rectangular room, each rigid wall generates a reflected image of the sound source. Higher-order reflections are

obtained by treating the image source as a “mother” sound source and recursively mirroring it across additional surfaces. The number of possible image sources depends on the number of walls, the geometry of the room, and the maximum order of the computed reflections. For a perfectly rectangular room (with 90° angles), the number of image sources S can be estimated as

$$S = \sum_{k=1}^K N(N-1)^{k-1}, \quad (4.1)$$

where K is the maximum reflection order considered and N is the number of reflecting surfaces. Generally speaking, the number of image sources increases exponentially with the maximum order of the simulation. Consequently, the ISM is computationally efficient for modeling sparse early reflections, but becomes impractical for diffuse late reverberation with many overlapping reflections.

Each image source corresponds to a distinct reflection path between the original source and the receiver. The reflected path can therefore be represented by the direct path from the corresponding image source to the receiver, with its amplitude scaled by the reflection coefficients of the walls encountered along the path. According to Eq. (2.11), the magnitude of a wall reflection coefficient can be estimated from its absorption coefficient α as

$$|R| = \sqrt{1 - \alpha}. \quad (4.2)$$

This relation determines only the magnitude of R . Any phase shift introduced by the boundary must be specified separately if complex, frequency-dependent reflection coefficients are considered.

The contribution of an image source can be described using the spherical-wave expression given by Eq. (2.6). The distance from the image source to the receiver accounts for spherical spreading and propagation delay, while the reflection coefficients describe the loss introduced at each wall. For a propagation path involving j reflections, the acoustic pressure at the receiver at time t is given by

$$p_{\text{IS}}(t) = \frac{A}{r_{\text{IS}}} \left(\prod_{n=1}^j R_n \right) e^{i(\omega t - k r_{\text{IS}})}, \quad (4.3)$$

where A is the complex pressure amplitude of the source, r_{IS} is the distance from the image source to the receiver, and R_n is the reflection coefficient associated with the n -th reflecting wall. As in Eq. (2.6), ω denotes the angular frequency, k is the acoustic wavenumber, and i is the imaginary unit [116].

4.1.2 Audibility Test

For non-rectangular rooms, an audibility test must be performed for all image sources to determine which image sources contribute to the sound field at a given source and receiver position. Each image source is interpreted as the final element of a reflection chain. The indices associated with an image source specify the sequence of walls encountered by the corresponding ray path, while the number of indices defines the reflection order. The audibility test is conducted by backtracking from the receiver to the original source through all chains of image sources. Starting from the receiver position, a straight line is drawn to the image sources of the highest reflection order. The last index of the image source identifies the wall that was struck last. The line between the image source and receiver must intersect the last struck wall surface (not its imaginary extension) for the image source to be considered valid. If this condition is satisfied, the intersection point on the wall is now treated as the receiver position, and the same procedure is repeated for the preceding image source in the reflection chain. This procedure is repeated recursively through every image source of the chain until the original source is reached. An image source is considered audible only if all of its prior image sources were also valid [116]. An example of this recursive audibility test is illustrated in Fig. 4.2.

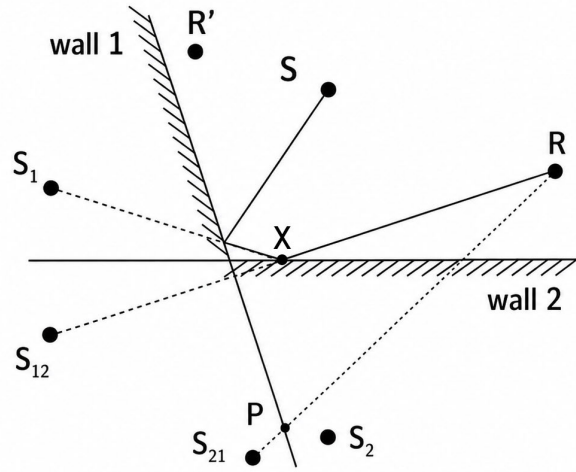


Figure 4.2: Image-source audibility test. The points R denote receiver positions, and $S_{n_1 \dots n_j}$ denotes an image source of order j . The image source S_{21} is not audible at R , because $\overline{S_{21}R}$ does not intersect wall 1. In contrast, S_{12} is audible at R , since $\overline{S_{12}R}$ intersects wall 2 and $\overline{S_1X}$ intersects wall 1 [116].

Consequently, audibility depends on the receiver position and cannot be generalized across different receivers. For each audible image source, a contribution is stored in the sound pressure impulse response.

4.1.3 Room Impulse Response Simulation

One of the main applications of the ISM is room impulse response simulation. The RIR simulation using the ISM proceeds as follows. First, image sources are generated up to a chosen reflection order. For non-rectangular geometries, the corresponding propagation paths are checked using the audibility test described in the previous section, and non-audible image sources are removed. Each audible image source then corresponds to a possible sound propagation path and adds a delayed and attenuated copy of the source impulse to the RIR. The propagation delay is determined by the path length between the image source and the receiver, while the amplitude is determined by geometrical spreading and the reflection coefficients of the surfaces involved as in Eq. (4.3). Finally, the simulated RIR is formed by summing the direct sound and the contributions from all audible image sources. Since material absorption is generally frequency dependent, broadband RIR simulation is commonly performed using frequency-dependent reflection coefficients. The ISM is therefore evaluated separately for several frequency bands, using the corresponding absorption properties of the room boundaries in each band [57]. Optionally, spectral losses due to atmospheric absorption are also taken into account (see Sec. 2.1.3). The resulting band-limited impulse responses are subsequently combined to obtain a broadband RIR. As the number of image sources increases rapidly with reflection order, the ISM is often only used for modeling early reflections. Late reverberation can be modeled using alternative approaches, for example as diffuse noise with an exponentially decaying envelope [53].

However, the ISM is only a simplified representation of real-world acoustics and therefore has several limitations. First, the standard ISM is generally restricted to relatively simple room geometries and does not account for furniture or other objects within the room. Furthermore, real wall materials often exhibit angle-dependent absorption characteristics, whereas the standard ISM typically assumes simplified, angle-independent absorption coefficients. Other acoustic phenomena, such as sound diffusion and diffraction, are also neglected because sound propagation is modeled by idealized geometrical rays [95]. The significance of these limitations depends on the intended application and the required level of simulation accuracy. To address some of these limitations, several extensions of the ISM have been proposed, including methods incorporating diffuse surface scattering and diffraction effects [20, 65].

4.2 Other Methods

Besides the very popular ISM, several other simulation techniques exist. Ray tracing (Fig. 4.3) is another widely used acoustic simulation technique based on geometrical acoustics and is conceptually similar to ray tracing in computer graphics. Sound propagation is modeled by

a large number of rays emitted from a source and tracked as they reflect from surfaces and lose energy through absorption and other propagation effects. The acoustic response is approximated by the rays that reach a receiver. Unlike the ISM, which explicitly constructs virtual source positions corresponding to specular reflections, ray tracing follows the propagation paths directly and can therefore be more flexible when dealing with complex geometries and irregular surfaces. Variants of the method can also incorporate phenomena such as diffuse scattering. Similar to the ISM, ray tracing assumes that sound waves can be locally approximated as plane waves and that their propagation can be modeled by geometrical paths. This approximation is generally most accurate at higher frequencies, where the acoustic wavelength is small compared to the dimensions of the room and its geometry [116].

Wave-based simulation methods (Fig. 4.4) describe sound by solving the wave equation, or a simplified form of it, throughout the room. In contrast to geometrical methods, which treat sound as a collection of rays, wave-based methods model the sound field itself using a discretized representation that includes both pressure and phase. This allows them to reproduce wave phenomena such as diffraction, interference, and room modes. As the required spatial and temporal resolution increases with decreasing wavelength, the computational cost rises rapidly with frequency. Wave-based methods are therefore commonly applied at low frequencies, where wave effects are most significant, and combined with geometrical methods at higher frequencies, where sound propagation can be approximated by rays [76].

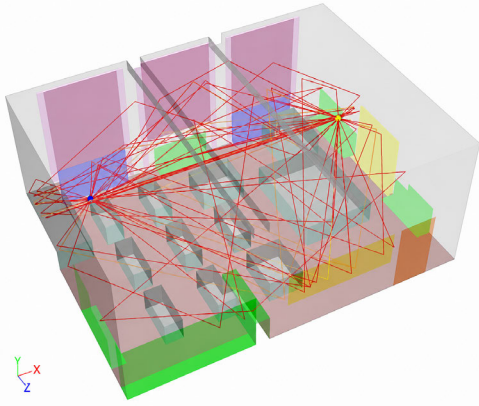


Figure 4.3: Ray tracing [72].

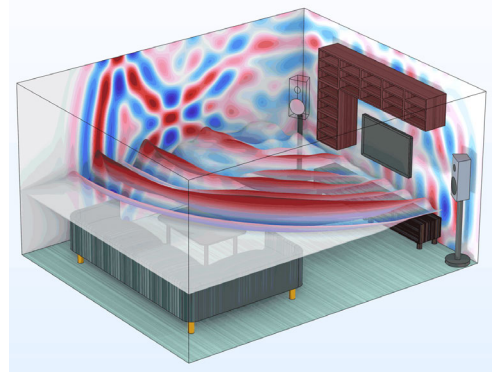


Figure 4.4: Wave-based simulation [13].

The choice of simulation technique always involves a trade-off between accuracy and computational cost. Hybrid approaches combine the strengths of different simulation methods.

5 Sound Source Localization and Detection

Sound source localization and detection (SSLD) consists of two main tasks: sound activity detection (SAD), which estimates the presence and number of sound sources, and sound source localization (SSL), which estimates their DOAs [122]. The combination of these tasks makes SSLD a complex and challenging problem. Various approaches have been proposed to address it, ranging from classical signal-processing-based methods to deep-learning-based approaches, which have gained increased attention in recent years.

5.1 Multiple Signal Classification

Multiple signal classification (MUSIC) is a high-resolution DOA-estimation algorithm capable of simultaneously localizing multiple sound sources using a microphone array. It is based on the principle that the measured sound field can be decomposed into two orthogonal components: a signal subspace, which contains the contributions of the acoustic sources, and a noise subspace, which ideally contains only spatially white noise [25]. The basic MUSIC algorithm is only applicable to narrowband and uncorrelated signals [35].

The MUSIC algorithm first estimates the spatial covariance matrix (SCM) of the Q microphone signals. The resulting matrix has dimensions $Q \times Q$. For a given frequency bin ν , let $\mathbf{x}(\nu, \ell) \in \mathbb{C}^Q$ denote the vector of complex-valued sensor signals at time frame ℓ . Using L time frames as signal snapshots, the SCM is estimated as

$$\hat{\Phi}_{\mathbf{x}\mathbf{x}}(\nu) = \frac{1}{L} \sum_{\ell=1}^L \mathbf{x}(\nu, \ell) \mathbf{x}^H(\nu, \ell), \quad (5.1)$$

where $(\cdot)^H$ denotes the Hermitian transpose. The SCM is then decomposed by eigenvalue decomposition. The resulting Q eigenvalues are sorted in descending order. Ideally, the largest eigenvalues correspond to the signal subspace, while the remaining eigenvalues correspond to the noise subspace. If the number of sources N_{src} is known, the eigenvectors associated with the $Q - N_{src}$ smallest eigenvalues are collected to form the noise-subspace [25], which will be

denoted here as the matrix $\mathbf{E}_{noise}$. When the number of sources is unknown, it must first be estimated. A common approach is to analyze the ordered eigenvalues using model-selection criteria such as the Akaike information criterion (AIC) or the minimum description length (MDL). These methods estimate the number of sources by minimizing a cost function that distinguishes the dominant eigenvalues associated with the signal subspace from the approximately equal eigenvalues associated with the noise subspace. In theory, an array of Q sensors allows MUSIC to resolve up to $(Q - 1)$ sources. In practice, however, the achievable number is usually lower because MUSIC is sensitive to noise and reverberation. Under such conditions, the separation between the signal- and noise-subspace eigenvalues becomes less distinct, which can make source-number estimation using criteria such as AIC or MDL unstable [94].

The main idea of MUSIC is that the steering vector of a true source direction is orthogonal to the noise subspace. For a given frequency bin ν and candidate direction (θ, ϕ) , the steering vector $\mathbf{a}(\nu, \theta, \phi)$ describes the relative phase shifts across the microphone array caused by the propagation delays between the microphones. The projection energy of the steering vector onto the noise subspace is given by

$$\|\mathbf{a}^H(\nu, \theta, \phi)\mathbf{E}_{noise}\|^2 = \mathbf{a}^H(\nu, \theta, \phi)\mathbf{E}_{noise}\mathbf{E}_{noise}^H\mathbf{a}(\nu, \theta, \phi). \quad (5.2)$$

When the candidate direction coincides with the direction of an actual source, the projection energy in Eq. (5.2) approaches zero. MUSIC converts these small values into peaks by taking their inverse. This gives the MUSIC pseudospectrum:

$$P_{\text{MUSIC}}(\nu, \theta, \phi) = \frac{1}{\mathbf{a}^H(\nu, \theta, \phi)\mathbf{E}_{noise}\mathbf{E}_{noise}^H\mathbf{a}(\nu, \theta, \phi)}. \quad (5.3)$$

The pseudospectrum is evaluated over a grid of candidate directions using the corresponding steering vectors. The locations of the N_{src} largest peaks are then selected as the estimated DOAs [25].

MUSIC can also be formulated in the SH domain, often referred to as SHD-MUSIC. In this case the SCM is calculated from the SH coefficients, given by Eq. (3.6), and the steering vector is replaced by the SH steering vector defined in Eq. (3.21) [48]. SHD-MUSIC was found to perform even slightly better than the conventional MUSIC algorithm [54].

Extending the MUSIC algorithm to broadband signals is not straightforward, as the conventional formulation assumes narrowband sources. A common approach is to split the broadband signal into several narrowband frequency bins and calculate the MUSIC pseudospectrum for each bin separately. The resulting pseudospectra are then averaged across the frequency range [87]. However, this approach does not preserve the phase relationships between the different frequency components. One method designed to address this limitation is the poly-

nomial eigenvalue decomposition MUSIC algorithm [35]. A detailed discussion of this method is beyond the scope of this work.

5.2 Deep-Learning-Based Approaches

In recent years, deep learning (DL) has become one of the dominant directions in sound source localization research. It has been strongly influenced by the Detection and Classification of Acoustic Scenes and Events (DCASE) community. Each year, the community organizes a range of challenges in which researchers from around the world compete and contribute to the development of improved DL methods for audio analysis. Since the introduction of the sound event localization and detection (SELD) challenge, this task has grown substantially in both scope and complexity. Current SELD systems are expected to perform sound event detection, classification, localization, and distance estimation for overlapping and moving sound sources from different classes [16]. Driven by this challenge, as well as by broader research in the field, several methods for DL-based SELD have emerged in recent years.

5.2.1 Input Feature Representations

Input features describe the preprocessed data that is provided to a model as the basis for learning. Feature engineering is therefore at least as important as the choice of model architecture. If the selected features do not contain information that is sufficiently correlated with the desired output, the model cannot learn the underlying task effectively. There are numerous ways to derive suitable input features from audio signals. In general, input features can be divided into implicit and explicit features.

Implicit Features

Implicit features refer to raw data that implicitly contains the information a model needs to learn from. Using such inputs is often associated with end-to-end learning, where the model learns the feature extraction process directly from the data rather than relying on hand-crafted features [115, 30]. However, this data also includes a large amount of irrelevant information. The main challenge for the model is to identify which parts of the input are relevant and correlated with the target output. This can slow down training convergence, as the model must first learn to extract the relevant information from the raw input [82]. In addition, variations between samples can be large, meaning that models based on implicit features typically require more training data and larger model architectures. One advantage of implicit features is that they do not require prior knowledge of which parts of the data are most relevant. Instead,

the model learns this automatically and may even discover cues or correlations that were not initially considered important.

In the context of SSLD, examples of implicit features include raw audio channel signals in the time domain [115], or magnitude and phase spectra of the microphone signals in the frequency domain [1].

Explicit Features

Explicit features describe features in which the relevant information has already been extracted from the raw data. The main challenge is to design or select features that are strongly correlated with the target output. Providing the model with features that are highly relevant to the output can lead to more stable, faster, and more efficient training. Since explicit features contain only selected information from the raw data, their dimensionality is usually much lower than that of implicit features. As a result, the model typically requires fewer parameters and less training data. A potential disadvantage of explicit features is that they may discard information that could have been useful for the model to learn from. Therefore, their effectiveness strongly depends on the quality of the feature extraction process and on prior knowledge about the task [9].

In the context of SSLD, explicit features are commonly designed to encode spatial cues known to be informative for DOA estimation. Examples include cross-correlation features, which encode time-difference information between microphone pairs [14], and beamforming-based representations [119, 52]. More recent approaches include spatial cue-augmented log-spectrogram (SALSA) features, which combine multichannel log-spectrograms with spatial cues derived from the principal eigenvector of the SCM [69]. SALSA-based features have been widely used and have shown strong performance in the SELD challenge [81, 118, 117]. The SCM itself can also serve as an input feature [37]. For HOA, several additional features have been investigated using up to fourth-order SH signals, including the HO-PIV, spatially localized pseudointensity vector, and relative harmonic coefficients [83].

5.2.2 Model Architectures

The architecture of a model is highly dependent on the selected input features, the amount of training data, and the underlying task. In the context of SSLD, many different architectures have been developed. Most of them employ methods inspired by computer vision, like convolutional neural networks (CNNs) [1] to extract relevant information and reduce the dimensionality of high-dimensional acoustic features, such as frequency-dependent representations. Recurrent neural networks (RNNs) [2] and Transformer architectures [98] are commonly

used to model temporal context. Most SSLD models combine several of these components, such as SELDnet, the baseline model for the DCASE SELD challenge [81]. It first uses convolutional blocks to extract relevant information from features like SALSA. A gated recurrent unit (GRU) then models the temporal context, before a fully connected (FC) layer maps the learned representation to the output (see Fig. 5.1). Current SSLD models typically use a similar architecture.

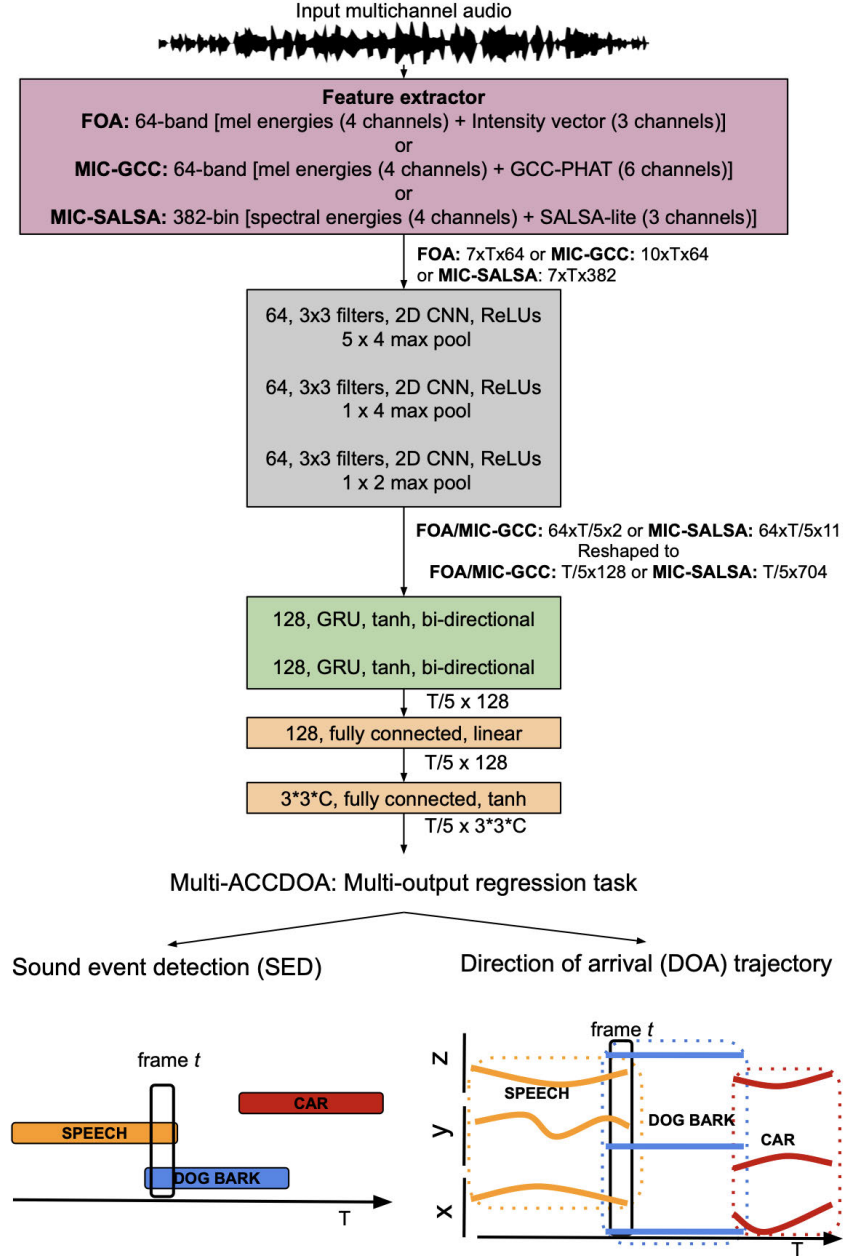


Figure 5.1: SELDnet model architecture [81].

5.2.3 Output Representations

Like the architecture, the model output representation is highly dependent on the requirements of the task. Focusing only on SSLD, the model must be able to detect whether a sound source is active and where it is coming from. This problem can either be treated as a classification or regression task [30].

Classification-Based Representations

When SSLD is treated as a classification task, each possible DOA angle is represented as a separate output class [120]. Therefore, the angular space must be discretized into a finite number of classes. For example, using a resolution of 1° on a single horizontal plane would require 360 classes. When elevation is also considered, separate output branches may be used for azimuth and elevation, or the full spherical field may be discretized into larger angular regions, for example at a resolution of 10° , to limit the number of output classes. Therefore, a suitable segmentation strategy is required. Uniformly dividing a sphere is not straightforward, since equal intervals in azimuth and elevation do not correspond to equal-area regions on the sphere [37].

For single-source DOA classification, the final layer is commonly implemented with a softmax activation, predicting exactly one output class. In contrast, multi-source DOA estimation is usually formulated as a multi-label classification problem, where each DOA class is treated as an independent binary decision and sigmoid activations are used at the output layer [30]. The target labels can be encoded either as one-hot vectors or as soft labels. With one-hot encoding, the spatial relationship between DOA classes is not explicitly represented, so an estimate in a neighboring angular bin can be penalized similarly to an estimate far away from the true direction. To reduce this problem, Gaussian label coding or soft label distributions assign non-zero target values to classes close to the true DOA, thereby incorporating the spatial closeness of adjacent directions into the training target [32, 24, 107].

The classification output is often interpreted as a spatial pseudospectrum. If the number of sources is known, the final DOA estimates can be obtained by selecting the corresponding number of highest peaks. If the number of sources is unknown, peaks above a user-defined threshold are selected. Alternatively, a dedicated network branch can be trained to predict the number of sources [30].

Regression-Based Representations

When SSL is treated as a regression task, the model predicts continuous directional values rather than distinct classes. These values can, for example, represent the azimuth and elevation angles in degrees. However, this representation introduces wrap-around problems at the $360^\circ/0^\circ$ boundary. To address this issue, the azimuth and elevation angles are either transformed into a sine–cosine representation [108] or the DOA is represented as a Cartesian unit vector [102]. The model predictions are then converted back into DOA angles. A remaining problem is the estimation of the number of active sources, since a regression value is predicted for each output track. One possible solution is to use separate models or output branches for SAD and SSL. This requires two loss functions that need to be balanced carefully. It also increases the complexity and size of the model and can make training less stable, for example if one task is learned faster than the other. A more compact solution is the activity-coupled Cartesian direction of arrival (ACCDOA) output representation.

ACCDOA converts the direction of arrival into a unit-vector representation. Active sound events are represented by vectors with a length of one, while inactive sound events are represented by zero-length vectors. Thus, the ACCDOA representation jointly encodes sound-event activity and direction in a single vector. After training, a threshold τ (typically around 0.5) is applied to the predicted vector lengths, and predictions below τ are considered inactive [101]. This ACCDOA approach has outperformed the two-branch architecture in the SELD task of the DCASE challenge in the year 2020, establishing itself as the new baseline method in the following editions [80]. An extension of ACCDOA, known as multi-ACCDOA, enables the prediction of multiple overlapping sound events belonging to the same class. Since the assignment of sound events belonging to the same class to output tracks is arbitrary, multi-ACCDOA uses a permutation-invariant training (PIT) loss, where predicted tracks are matched with ground-truth tracks by selecting the assignment that minimizes the overall loss using the Hungarian algorithm. The PIT loss for the multi-ACCDOA format can be written as

$$\mathcal{L}^{\text{PIT}} = \frac{1}{L} \sum_{\ell} \min_{\alpha \in \text{Perm}(\ell)} l_{\alpha, \ell}^{\text{ACCDOA}}, \quad (5.4)$$

$$l_{\alpha, \ell}^{\text{ACCDOA}} = \frac{1}{MC} \sum_m^M \sum_c^C \text{MSE}(\mathbf{P}_{\alpha, mcl}^*, \hat{\mathbf{P}}_{mcl}), \quad (5.5)$$

where $\mathbf{P}_{\alpha, mcl}^*$ denotes the ACCDOA target corresponding to permutation α , and $\hat{\mathbf{P}}_{mcl}$ denotes the ACCDOA prediction for track m , class c , and frame ℓ . Mean squared error (MSE) is commonly used as the regression loss [101]. The PIT loss handles the arbitrary ordering of tracks by assigning each ground-truth event to the predicted track with the lowest ACCDOA loss [102].

6 Methodology

The goal of this thesis is to investigate how acoustic room simulation affects the performance of DL-based SSLD. To examine this, an SSLD model is trained on simulated acoustic data and evaluated using real-world measurements. The experiments are based on a simplified task derived from the SELD challenge [16]. The task is restricted to the single sound class “speech”, as it is one of the most relevant signal types in many practical applications. Each recording contains between zero and three simultaneously active speech sources, each at a static position. The model has to determine how many sources are active and estimate their DOAs.

6.1 Datasets and Data Generation

6.1.1 Existing RIR Datasets

Finding suitable datasets of room recordings is challenging, as their applicability strongly depends on the microphone configuration used for recording. A practical solution is to operate in the Ambisonics domain, since signals from various microphone arrays can be encoded into a common spatial representation, as discussed in Chap. 3. RIRs are particularly well suited for this purpose, since they capture the acoustic characteristics of an environment and can be convolved with arbitrary audio signals to simulate how those signals would sound in the corresponding room (see Sec. 2.2.1). Several publicly available RIR datasets are provided in the Ambisonics format. To the best of our knowledge, Tab. 6.1 summarizes the datasets that are currently publicly available.

For the present work, an HOA dataset is required to ensure that the spatial resolution imposed by the Ambisonics order does not become a limiting factor. However, most existing datasets are available only at low Ambisonics orders, while HOA datasets are generally small. This is problematic for training deep learning models, which typically require large and diverse datasets to achieve good generalization. Furthermore, publicly available simulated datasets provide no control over the underlying simulation parameters. Since this work aims to compare simulations generated with different parameter configurations, the most suitable approach is to

generate the required data. The HARP dataset [92] provides a suitable basis for this purpose, as its source code is publicly available.

Dataset	Order	# RIRs	Scenes / Variation	Simulation
BUT-ReverbDB [109]	0th	1300	11 real rooms	Measured
MESH-RIR [46]	0th	4400	1 room, 2D grid	Measured
GWA [110]	0th	2M	≈ 6000 realistic interiors	Hybrid Wave + Ray
dEchorate [11]	Linear Array	1800	Variable acoustic conditions	Measured
AIR [43]	Binaural	200+	4 rooms	Measured
Treybig SDM [114]	7-mic SDM array	18,535	1 room, 9 acoustic configurations, 5 sources, dense 2D grids	Measured
OpenAIR [100]	1st	50	50 extreme real rooms	Measured
C4DM [104]	1st	700	3 rooms, multiple positions	Measured
TAU-SRIR [78]	1st	114	9 rooms, multiple positions	Measured
SoundSpaces [12]	1st	17.6M	≈ 100 realistic interiors	Geometric (ray tracing)
STARSS22 [81]	1st	121 clips	11 rooms, real sound scenes, 13 classes, frame-wise labels	Measured scenes
STARSS23 [103]	1st	168 clips	16 rooms, real sound scenes, 13 classes, audio-video, frame-wise labels	Measured scenes
PAN-AR [17]	2nd	21	4 real rooms, including ambient noise and images	Measured HOA
MOTUS [29]	4th	3320	1 room, 830 furniture arrangements	Measured
ARNI-SRIR [59]	4th	–	5 variable acoustic conditions, 6-DoF	Measured
Arni em32 [26]	4th	585	3 variable-acoustics conditions, 3 sources, 65 receiver positions	Measured
HOMULA-RIR [64]	HO mics + ULA	25	1 seminar room, multiple positions	Measured
HARP [92]	7th	100K+	Wide variety of synthetic shoebox rooms	Image-source method (ISM)
HiFi-HARP [93]	7th	100K+	≈ 2000 realistic interiors	Hybrid Wave + Ray
Treble10 [66]	8th	3.1K	10 furnished rooms, 5 sources, dense receiver grids at 3 heights	Hybrid wave + Ray

Table 6.1: Overview of existing room impulse response datasets. Adapted from [93].

6.1.2 Simulated Training RIRs

Ambisonic room impulse responses (ARIRs) were generated with the HARP Python implementation [92] and used for training an SSLD model. HARP is based on the open-source Python library pyroomacoustics [96].

Microphone Simulation

An Eigenmike em32 [8] is simulated using 32 virtual receivers positioned at the center of the virtual microphone array. Each receiver is assigned a frequency-dependent directivity response based on measured array transfer functions (ATFs) of an Eigenmike em32, obtained from 1,200 approximately uniformly distributed source directions (see Fig. 6.1) [77]. Because the measured ATFs inherently captured the relative time delays between microphone channels, the virtual receivers are placed at the array center rather than positioned at the physical microphone locations. The resulting array signals are encoded into the Ambisonics domain using the discrete SH transform and inverse radial filtering, as described in Sec. 3.2. Tikhonov regularization is applied according to Eq. (3.20), with the maximum gain set to 15 dB.

6 Methodology

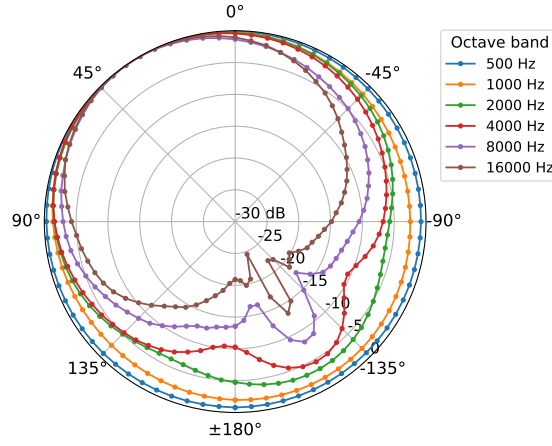


Figure 6.1: Top-view polar plot of the directivity response of microphone 2 of the Eigenmike em32 [77].

Additionally, an ideal SH receiver is modeled following the original HARP implementation. For each Ambisonics channel, a virtual microphone is placed at the center of the array and assigned the directivity pattern corresponding to the SH of that channel (see Sec. 3.1). This approach records the signal directly in the SH domain, avoiding the practical limitations of physical spherical microphone arrays discussed in Sec. 3.2 and eliminating the need for inverse radial filtering. Nevertheless, regularization remains an important consideration when DL models are trained and evaluated on different datasets, as they have been shown to be sensitive to differences in regularization [99]. When signals, recorded with a physical spherical microphone array, are encoded into the SH domain, regularization introduces an order-dependent attenuation at low frequencies, with the effect becoming increasingly pronounced at higher SH orders. To reproduce this behavior in the simulated data, the ARIRs are filtered using order-dependent high-pass filters derived from the ratio of the magnitude responses of the regularized and ideal inverse radial filters, as illustrated in Fig. 6.2. The filters are computed for an Eigenmike em32 using the Tikhonov regularization with the maximum gain set to 15 dB.

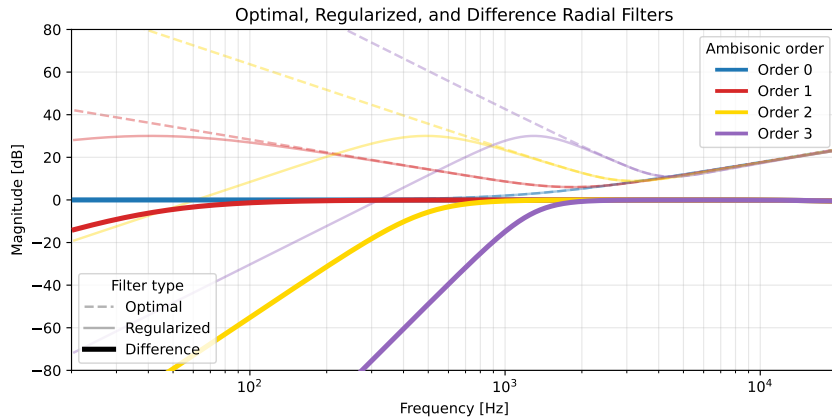


Figure 6.2: Difference between the inverse radial filters and their Tikhonov-regularized counterparts.

Reverberation Simulation

Early reflections are simulated with the ISM (Sec. 4.1) in shoebox-shaped rooms. The reflective properties of the walls are defined by frequency-dependent absorption coefficients. To approximate the reverberation characteristics of real acoustic environments, the absorption coefficients are obtained from the RT profiles of 271 real-world spaces measured by Traer et al. [113]. This dataset is intended to resemble the range and distribution of acoustic environments that people typically encounter in everyday life. The distribution of these profiles is shown in Fig. 6.3. For each randomly generated room, with dimensions sampled from a specified range, one measured RT profile is randomly selected, and the corresponding frequency-dependent mean absorption coefficients are calculated by inverting the Eyring equation (2.16). If any resulting coefficient is very close to 0 or 1, the selected profile is considered incompatible with the room size, and a new profile is drawn. The resulting mean absorption coefficients are assigned to all surfaces.

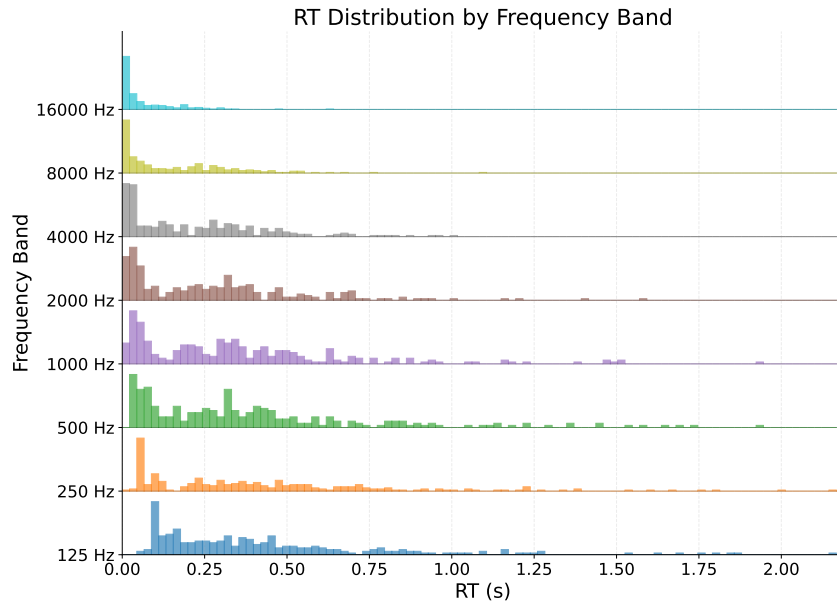


Figure 6.3: Distribution of reverberation times in the spaces measured by Traer et al. [113].

Late reverberation is synthetically added to selected simulations. The sound field is assumed to be diffuse, and late reverberation is modeled as frequency-dependent exponentially decaying noise (see Sec. 2.2.1), with decay rates derived from the corresponding RT profile. Uncorrelated noise signals are generated for each Ambisonics channel using the same gain and envelope. Under N3D normalization, this approach yields a diffuse sound field with equal energy in all directions [19]. The late reverberation is crossfaded with the simulated early reflections at the mixing time, which is estimated using

$$t_{\text{mp95}} = (0.0117V + 50.1) [\text{ms}], \quad (6.1)$$

as proposed in [55], where V denotes the room volume. This expression provides a conservative estimate of the perceptual mixing time corresponding to the 95th percentile of the listener distribution. The gain of the late reverberation is determined using an additional reference image-source simulation with a high reflection order of 12. This simulation, using the same room configuration, serves only as a reference for the late-reverberation energy and is not included in the final RIR. The gain of the late reverberation is calculated from the energy of the omnidirectional Ambisonics channel of the reference simulation after the estimated mixing time. The decaying noise used for the late reverberation is scaled to match this energy, with the same gain applied to all Ambisonics channels. Fig. 6.4 illustrates the late-reverberation generation procedure and the resulting RIRs for the individual Ambisonics channels.

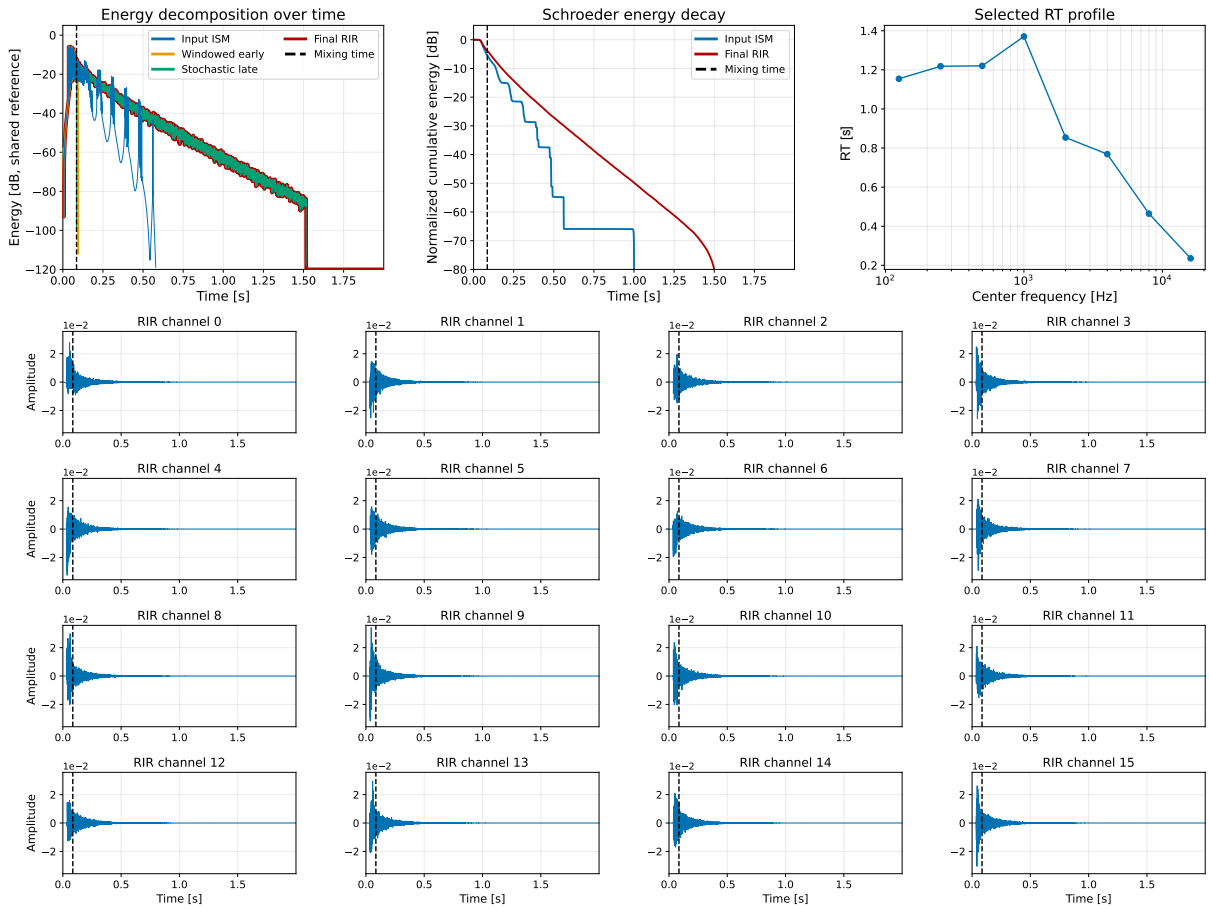


Figure 6.4: Addition of late reverberation. The top row shows, from left to right: the crossfade between the original room impulse response (RIR) and the synthesized late reverberation, a comparison of the Schroeder energy decay curves of the high-order image-source reference RIR and the synthesized RIR, and the reverberation time profile used for the synthesis. The bottom rows show the synthesized RIRs for the individual Ambisonics channels.

Room Geometry

The ISM simulations are based on shoebox geometry. To introduce geometric irregularity, the room corners are randomly shifted in the xy -plane for some datasets, resulting in non-shoebox room geometries. Corner pairs sharing the same x -coordinate are displaced jointly by a random amount within a predefined range.

Another approach to introduce geometric irregularity is randomizing the image-source positions. The shoebox geometry is kept unchanged, but each image source is shifted by a random amount in the x , y , and z directions. Image sources that are shifted closer to the receiver than the sound source are removed, since this would cause the corresponding reflections to arrive before the direct sound, which is physically impossible. This randomization approach is chosen because it is simple to implement in `pyroomacoustics` and provides an approximation of the randomized reverberation method proposed by Badeau [7]. Fig. 6.5 compares the original image-source positions with the randomized positions obtained using our proposed randomization approach.

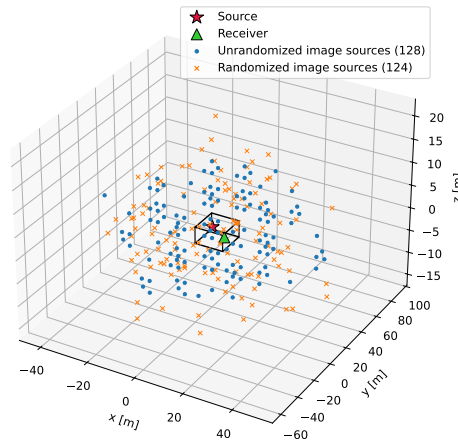


Figure 6.5: Calculated image sources vs. their counterparts with up to 10 m displacement in the x , y , and z directions for a shoebox room up to reflection order four.

Room, Receiver and Source Sampling

Datasets are generated by simulating 2,000 different rooms, with eight receiver positions per room and three source positions per receiver. The room generation parameters are summarized in Tab. 6.2. The speed of sound is fixed at $c = 343 \text{ m s}^{-1}$, and the relative air humidity is approximated as 45 % (see Sec. 2.1.3). Room dimensions are constrained to a range representative of most real-world rooms. The source positions are constrained to have a minimum angular separation of 15° , ensuring that two distinct sound events do not originate from nearly identical directions. In addition, the distances of sources and receivers from the room boundaries are restricted to avoid unrealistic or uncommon placements. A minimum source–receiver

distance of 1 m is enforced to reduce near-field effects that may not be adequately modeled by the plane-wave assumption in the simulation. Sound sources are modeled as ideal point sources, neglecting source directivity. Note that this simplification may limit the realism of the simulation, since real sound sources are often directional. The receiver’s azimuthal orientation is randomized independently at each receiver position. The resulting dataset is partitioned at the room level into training, validation, and test sets using a 70/20/10 split.

Parameter	Value
Room width and length	[2.5, 20.0] m, in steps of 0.5 m
Room height	[2.5, 5.0] m, in steps of 0.5 m
Minimum source-to-receiver distance	1.0 m
Minimum source-to-surface distance	0.5 m
Minimum receiver-to-surface distance	0.5 m
Number of rooms	2,000
Receiver positions per room	8
Source positions per receiver	3, separation $\geq 15^\circ$

Table 6.2: Sampling parameters for room generation.

Fig. 6.6 shows the distribution of source DOAs, RTs, DRRs, and room dimensions of the simulated dataset “SH-0₁₂-L₁-C₀-R₀”. The elevation distribution of the source DOAs is not uniform because the range of room heights is smaller than the range of room dimensions in the xy -plane, while the receivers’ elevation orientation is fixed at 0°. The RT distribution follows the measurements by Traer et al. [113], and the room dimensions are sampled uniformly within the ranges defined in Tab. 6.2.

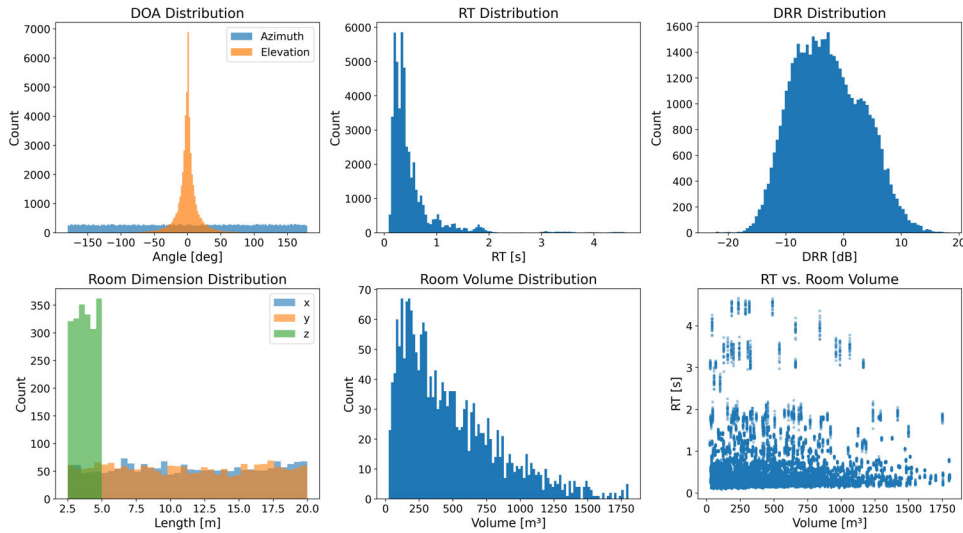


Figure 6.6: Distribution of directions of arrival, reverberation times, direct-to-reverberant ratios, and room dimensions for the simulated dataset “SH-0₁₂-L₁-C₀-R₀”.

Training Dataset Configurations

For training, 19 different datasets are simulated using varying acoustic and geometric simulation parameters. These datasets are designed to systematically investigate the effects of microphone simulation, the maximum ISM reflection order, diffuse late reverberation, room geometry, and image-source position randomization on model performance. An overview of the training datasets is provided in Fig. 6.7. To enable compact and consistent references throughout this work, the following naming convention is used:

$$\langle \text{Mic} \rangle - \text{O}_{\langle \text{Order} \rangle} - \text{L}_{\langle 0/1 \rangle} - \text{C}_{\langle \text{Corner-disp.} \rangle} - \text{R}_{\langle \text{Random-ISM} \rangle}.$$

Component	Values	Meaning
$\langle \text{Mic} \rangle$	SH, EM	Microphone simulation: SH or em32
$\text{O}_{\langle \text{Order} \rangle}$	$\text{O}_0, \text{O}_2, \text{O}_{12}$	ISM maximum reflection order (0: direct sound only)
$\text{L}_{\langle 0/1 \rangle}$	L_0, L_1	Diffuse late reverberation: disabled (0) / enabled (1)
$\text{C}_{\langle \text{Corner-disp.} \rangle}$	e.g., $\text{C}_0, \text{C}_{0.1}$	Shoebox room-corner displacement range [m]
$\text{R}_{\langle \text{Random-ISM} \rangle}$	e.g., $\text{R}_0, \text{R}_{0.1}$	Random image-source displacement range [m]

Table 6.3: Naming convention and parameter meanings for the simulated training datasets.

The individual components of the dataset names are summarized in Tab. 6.3. For example, “SH- O_{12} - L_0 - C_0 - R_0 ” denotes a dataset simulated using the SH microphone simulation, an ISM reflection order of 12, no diffuse late reverberation added, shoebox geometry, and no random image-source displacement.

ISM order 0	ISM order 2	ISM order 12	Corner displacement	Random ISM positions
SH- O_0 - L_0 - C_0 - R_0	SH- O_2 - L_0 - C_0 - R_0	SH- O_{12} - L_0 - C_0 - R_0	SH- O_{12} - L_0 - $\text{C}_{0.5}$ - R_0	SH- O_{12} - L_0 - C_0 - $\text{R}_{0.25}$
EM- O_0 - L_0 - C_0 - R_0	EM- O_2 - L_0 - C_0 - R_0	EM- O_{12} - L_0 - C_0 - R_0	SH- O_{12} - L_0 - C_1 - R_0	SH- O_{12} - L_0 - C_0 - R_1
SH- O_0 - L_1 - C_0 - R_0	SH- O_2 - L_1 - C_0 - R_0	SH- O_{12} - L_1 - C_0 - R_0	SH- O_{12} - L_0 - C_3 - R_0	SH- O_{12} - L_0 - C_0 - R_5
EM- O_0 - L_1 - C_0 - R_0	EM- O_2 - L_1 - C_0 - R_0	EM- O_{12} - L_1 - C_0 - R_0		SH- O_{12} - L_0 - C_0 - R_{10}

Figure 6.7: Overview of all simulated training datasets.

6.1.3 Measured Evaluation RIRs

Real-world measured data is used to evaluate the model. Ideally, the dataset should be sufficiently diverse to assess performance under a wide range of acoustic conditions. However, as shown in Tab. 6.1, the number of available measured HOA datasets is limited. The selection becomes even more restricted when requiring multiple source positions for each receiver position, together with accurately measured source–receiver positions.

Arni Dataset

One dataset that satisfies these criteria is the Arni em32 dataset [26]. It was recorded in the variable-acoustics room Arni at Aalto University and contains measurements for three sources at 65 receiver positions under three room configurations: dry, anisotropic, and reverberant. Genelec 8010A loudspeakers were used as sound sources, while the responses were captured with an Eigenmike em32 microphone array, which supports Ambisonics encoding up to fourth order [8]. The source–receiver positions and orientations were measured precisely using an OptiTrack system.

Agrid Dataset

The Arni dataset is limited in scope, as it includes only three acoustic configurations of the same variable-acoustics room, which does not represent a broad range of real-world environments. For this reason, an additional dataset is used for evaluation. This dataset is referred to as “Agrid” in this work and is not yet publicly available. It was measured with an Eigenmike em32 in 16 different rooms at Aalto University and contains measurements at two receiver positions in each room, with two source positions for each receiver.

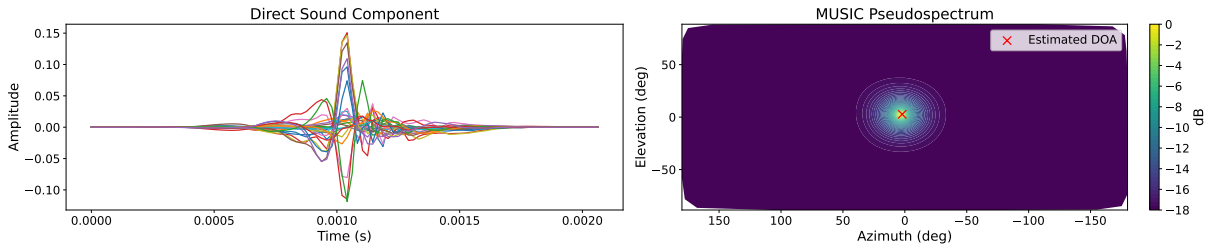


Figure 6.8: Direction-of-arrival estimation of the direct sound using the MUSIC algorithm.

The corresponding source–receiver positions were not precisely measured. The DOAs were therefore estimated from the direct-sound component of each RIR using the SHD-MUSIC algorithm (see Sec. 5.1). For DOA estimation, the signals were limited to the frequency range from 2 to 8 kHz, where the Eigenmike em32 provides the best directivity (see Fig. 3.6). The filtered broadband signal was then divided into ten logarithmically spaced frequency bands. The SCM was calculated for each band, and the resulting MUSIC pseudospectra were averaged to obtain the final DOA estimate (see Fig. 6.8). This proposed labeling procedure was validated using the Arni em32 dataset, for which accurate reference source–receiver positions are available. The MUSIC-based estimates exhibit a small deviation from the corresponding reference directions, as shown in Fig. 6.9. The systematic offsets observed in azimuth and elevation support the assumption that MUSIC provides consistent direction estimates. The offsets may be caused by calibration inaccuracies in the OptiTrack system. The mean angu-

lar distance between the estimated and reference directions was calculated to be 4.9° with a standard deviation of 2.0° , indicating a non-negligible labeling error. This labeling uncertainty should be taken into account when interpreting the localization error, as it may impose a lower bound on the achievable localization performance.

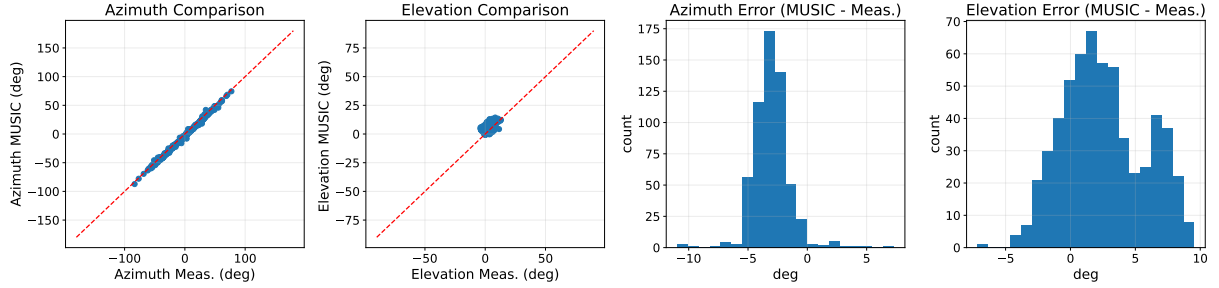


Figure 6.9: Angular differences between the measured source directions and the corresponding MUSIC estimates for the Arni dataset.

Motus Dataset

The Motus dataset [29] comprises Eigenmike em32 recordings from 830 different room configurations. For each configuration, RIRs for four distinct sources were recorded at a single receiver position, using Genelec 1030A loudspeakers. All measurements were conducted in the same room, with different acoustic conditions created by rearranging furniture and placing absorptive materials at various locations. The dataset also provides extensive metadata, including three-dimensional room geometry and 360° images captured from the receiver position (Fig. 6.10). In this project, only source positions 1–3 are considered, as the loudspeaker at source position 4 was directed towards the wall. Measurements in which a source is occluded are also excluded.

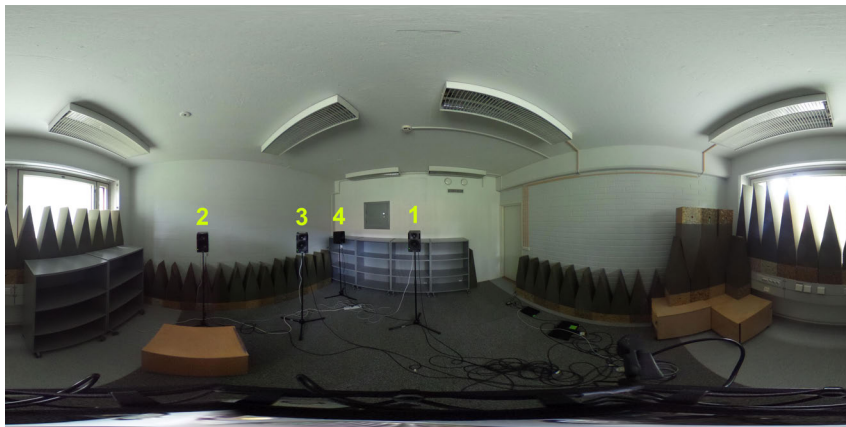


Figure 6.10: 360° photo from the receiver position in room configuration 20 of the Motus dataset [29].

Although the microphone and source positions remained fixed throughout all recordings, the tape measurements are not expected to be very accurate, and small changes may have occurred between room configurations. For this reason, the same MUSIC-based approach used for the Agrid data was applied to obtain ground-truth DOA labels.

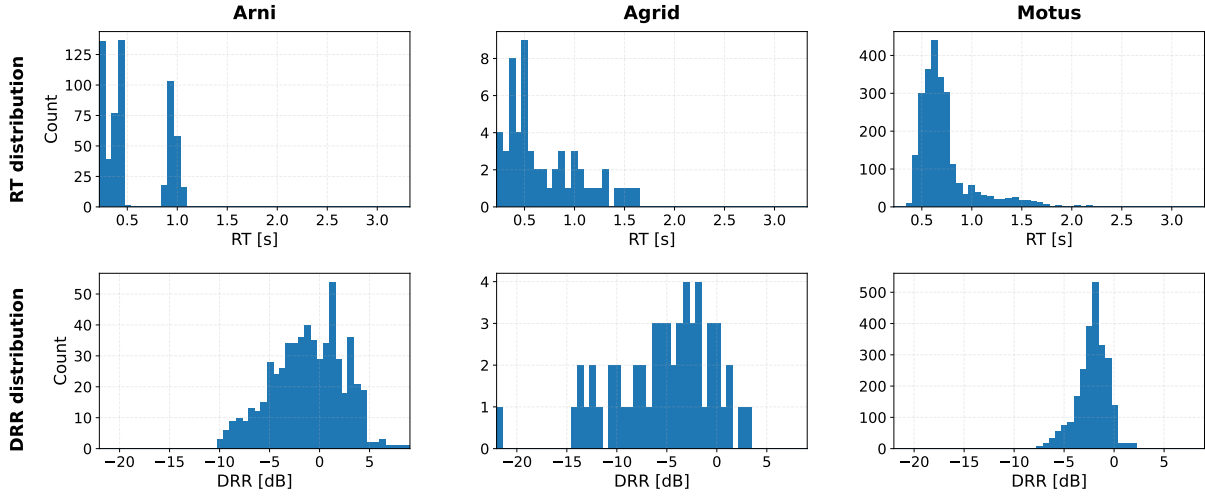


Figure 6.11: Reverberation time and direct-to-reverberant ratio distribution per measured dataset.

The RIRs of all datasets were truncated using the method proposed by Lundeby et al. [56], implemented in the `intersection_time_lundeby` function in `pyrato` [112]. This method also provided an estimate of the RT. Fig. 6.11 shows the RT and DRR distribution in the measured datasets. Similar ranges are covered by the simulated datasets (see Fig. 6.6 for comparison).

6.1.4 Audio Sample Generation

The training and evaluation samples are generated using an approach adapted from [83]. For each receiver position, three audio tracks are created, with one source RIR assigned to each track. A speech sample from the TIMIT dataset [27] is randomly selected for each track. To avoid data leakage, the TIMIT recordings are split into separate sets for generating training and test data. Since the TIMIT recordings are sampled at 16 kHz, this sampling rate is used throughout the data-generation pipeline. Each speech recording is repeated twice within its track, with pauses of random duration between 0 and 1.5s inserted before, between, and after the repetitions. The resulting signal is then convolved with the RIR assigned to that track. The voice-activity regions provided with the TIMIT recordings are shifted to account for the RIR delay and extended by one-sixth of the corresponding RT. Finally, all convolved tracks are zero-padded to the duration of the longest track and rounded up to the nearest whole second. An example of the resulting signals for each speaker track, together with their adjusted voice-

activity regions, is shown in Fig. 6.13. This procedure results in a nearly uniform distribution of frames containing one to three overlapping sources, as shown in Fig. 6.12. Frames without an active source occur less frequently. This imbalance is considered acceptable because the absence of active sources is expected to be easier for the model to detect.

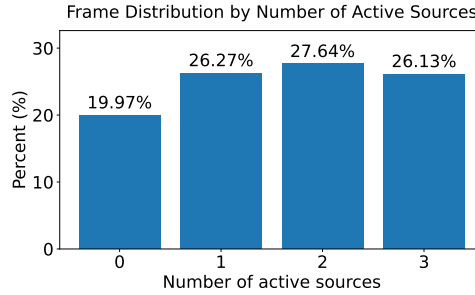


Figure 6.12: Distribution of the number of simultaneously active sources across all time frames in the generated datasets.

The three HOA speech tracks are combined using randomly selected signal-to-interference ratios (SIRs) between 0 and 10 dB, with the first track used as the reference. Ambient babble noise is then added to the speech mixture. The babble noise is generated by overlapping 50 speech recordings randomly selected from the TIMIT dataset, with the recordings repeated as needed to match the desired duration. This anechoic babble signal is convolved with a diffuse SRIR generated from uncorrelated exponentially decaying noise with equal energy in all Ambisonics channels, resulting in a diffuse sound field under N3D normalization. For each octave band, the decay time is estimated as the average of the corresponding RT values of the three source RIRs. The convolved babble noise is added with an SNR of 20 dB to the speech mixture. Both SIR and SNR are calculated only over active speech regions of the omnidirectional Ambisonics channel. The final Ambisonics mixture is peak-normalized and divided into one-second segments, which serve as input to the SSLD model. Features are extracted from these segments as described in the following section.

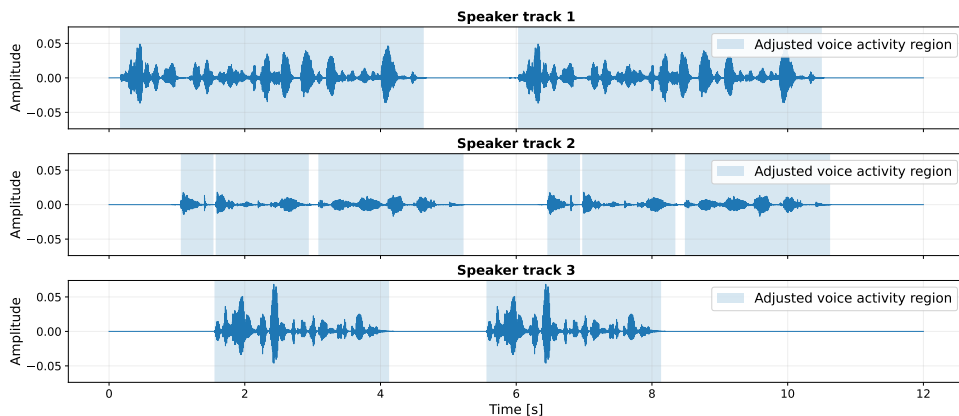


Figure 6.13: Example of the three generated speech tracks for a single receiver position, together with the voice-activity regions adjusted after convolution with the corresponding room impulse responses.

6.2 Deep-Learning SSLD Model

6.2.1 Input Feature Extraction

The HO-PIV is selected as the input feature representation, since it provides a convenient extension of the commonly used FO-PIV features [81, 118, 121] to HOA signals (see Sec. 3.1.2). Moreover, the suitability of HO-PIV features for DL-based SSLD was demonstrated by Poschadel et al. [83]. Third-order Ambisonics is used throughout this work, corresponding to $C = 16$ Ambisonics channels according to Eq. (3.2). This order provides a high spatial resolution while keeping the feature dimensionality manageable. Furthermore, Poschadel et al. reported better performance with third-order than with fourth-order Ambisonics in their experiments.

The one-second HOA samples (generated as described in the previous section) are processed using the STFT. A Hann window of 640 samples is applied with a hop size of 320 samples, and each frame is zero-padded to an FFT size of 1024 samples. This results in $F = 513$ frequency bins and $L = 50$ time frames per sample. The active and reactive HO-PIV components are then calculated for each time–frequency bin as described in Sec. 3.1.2. The HO-PIV components are normalized separately within each time–frequency bin to minimize variations caused by the overall signal level and to keep the resulting features within a consistent range [74]. Following the normalization proposed by Poschadel et al. [83], the normalized active and reactive HO-PIVs are given by

$$\frac{-1}{E_N(\nu, \ell)} \begin{bmatrix} \hat{\mathbf{I}}_{N,a}(\nu, \ell) \\ \hat{\mathbf{I}}_{N,r}(\nu, \ell) \end{bmatrix} \in \mathbb{R}^{L \times F \times 2(C-1)}. \quad (6.2)$$

The energy-based normalization term is defined as

$$E_N(\nu, \ell) = \sum_{n=0}^N \frac{1}{2n+1} \sum_{m=-n}^n |a_{nm}(\nu, \ell)|^2, \quad (6.3)$$

where $a_{nm}(\nu, \ell)$ is the complex-valued Ambisonics coefficient of order n and degree m at frequency bin ν and time frame ℓ , and N is the maximum Ambisonics order. The 513 normalized frequency bins of the HO-PIV features are reduced to 128 mel bands by averaging the corresponding bins. This differs from the approach of Poschadel et al. [83], who retained the original 513-bin frequency representation. The reduced representation lowers storage requirements and computational cost. A resolution of 128 mel bands is considered sufficient, as several established SSLD systems use only 64 frequency bands [81, 118, 121].

In addition to the HO-PIV features, the log-mel magnitude spectra of all Ambisonics channels are concatenated with the HO-PIVs. These spectra are also computed using 128 mel bands and are included to provide additional information about the spectral energy distribution, which may help the model estimate the number of simultaneously active sound sources.

The input features are standardized using z -score normalization. The HO-PIV and log-mel magnitude features are normalized separately based on the statistics of the corresponding training dataset. For each feature type, the mean μ and standard deviation σ are computed, and each feature value x is transformed according to

$$z = \frac{x - \mu}{\sigma}. \quad (6.4)$$

This transformation centers the feature distribution around zero and scales it to a standard deviation of one. The HO-PIV and log-mel features are normalized separately since their value ranges differ considerably. This avoids one feature type having a stronger influence on the model simply because of its larger numerical scale.

The resulting input feature tensor contains $L = 50$ time frames and 128 mel-frequency bands, with a feature-channel dimension consisting of $2(C - 1)$ HO-PIV components and C log-mel magnitude spectra. For third-order Ambisonics ($C = 16$), this yields $2(16 - 1) + 16 = 46$ feature channels and an input tensor of size $50 \times 128 \times 46$.

6.2.2 Network Architecture and Training

The model architecture is adapted from [83], which in turn is based on the SELDnet architecture [81]. The network consists of three convolutional blocks, followed by a recurrent unit and an FC layer. Exponential linear units (ELUs) are used as activation functions to introduce non-linearity into the network. The batch normalization layers help stabilize and accelerate training by normalizing the feature activations within each batch, reducing shifts in their distribution as the network parameters change. The dropout layers randomly deactivate a fraction of neurons during each training step. This reduces overfitting and improves generalization by preventing the model from becoming overly reliant on specific neurons. The convolutional layers extract local time–frequency patterns from the spectrograms and HO-PIV features. The frequency dimension is progressively reduced through max-pooling operations, while the temporal resolution is preserved. Starting from an input tensor of shape $(50, 128, 46)$, the convolutional blocks produce a feature representation of shape $(50, 2, 128)$. The frequency and feature dimensions are then flattened, resulting in a 256-dimensional feature vector for each of the 50 time frames. The resulting sequence is passed through two *bidirectional long short-term memory* (BiLSTM) layers, which model temporal dependencies using information from both preceding and subsequent time frames. Their output is mapped to the final prediction space by an FC layer. The network uses the ACCDOA representation, which allows SAD and SSL to be handled jointly with a single output (see Sec. 5.2.3). The final layer has nine outputs per time frame, corresponding to three 3D (x, y, z) ACCDOA vectors. The model can therefore detect and localize up to three simultaneous sound sources for each frame. Since the order

of the sources is arbitrary, a permutation-invariant MSE is used. An overview of the network architecture is given in Tab. 6.4.

Layer	Details	Output shape
Input		(50, 128, 46)
Conv2D	3×3	(50, 128, 128)
BatchNorm		(50, 128, 128)
Activation	ELU	(50, 128, 128)
MaxPool	1×2	(50, 64, 128)
Dropout	0.1	(50, 64, 128)
Conv2D	3×3	(50, 64, 128)
BatchNorm		(50, 64, 128)
Activation	ELU	(50, 64, 128)
MaxPool	1×8	(50, 8, 128)
Dropout	0.1	(50, 8, 128)
Conv2D	3×3	(50, 8, 128)
BatchNorm		(50, 8, 128)
Activation	ELU	(50, 8, 128)
MaxPool	1×4	(50, 2, 128)
Dropout	0.1	(50, 2, 128)
Reshape		(50, 256)
BiLSTM		(50, 256)
BiLSTM		(50, 256)
TimeDist FC	ELU	(50, 256)
Dropout	0.1	(50, 256)
TimeDist FC	linear	(50, 9)

Table 6.4: Network architecture used in this work, adapted from [83].

The hyperparameters were tuned to achieve fast convergence while avoiding underfitting or overfitting. A dropout rate of 0.1 is used for regularization. Preliminary experiments indicated that larger batch sizes yielded faster and more stable training. Accordingly, the batch size is set to 800, which is the maximum supported by the available hardware. The model is trained using the Adam optimizer with an initial learning rate of 10^{-3} and a weight decay of $3 \cdot 10^{-4}$. The learning rate is reduced by a factor of 0.5 whenever the validation loss fails to improve for five consecutive epochs. In addition, early stopping is applied if no improvement in the validation loss is observed for 20 consecutive epochs. With these settings, the training process typically converges after approximately 100 epochs.

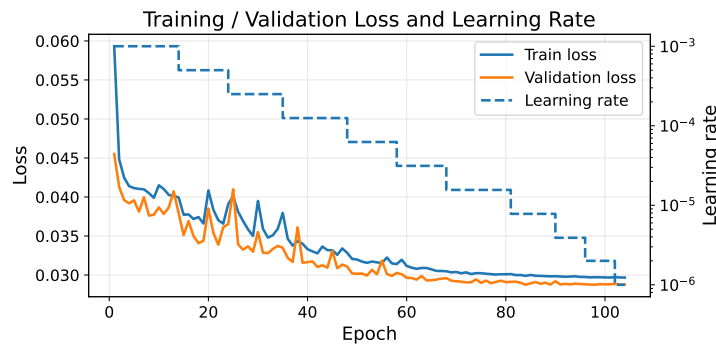


Figure 6.14: Training and validation loss together with the learning-rate schedule for the model trained on the simulated dataset EM- O_{12} - L_0 - C_0 - R_0 .

Fig. 6.14 shows the training and validation losses across epochs for a representative example. The losses are aligned and gradually decrease toward convergence, indicating stable and ef-

fective training. The learning rate is continuously reduced throughout training, which leads to smaller fluctuations in the losses and eventually to convergence. The validation loss remains well below the training loss for most epochs. A possible explanation is the use of dropout during training, which randomly deactivates neurons and thereby reduces the effective model capacity.

6.2.3 Evaluation Methodology

Evaluation Metrics

The evaluation metrics used are based on [83], following [60, 61]. These metrics evaluate SSL and SAD separately. For SSL, the localization error (LE) and localization recall (LR) are computed. For SAD, the location-sensitive error rate (ER_{15°) and F_1 -score (F_{15°) are computed.

First, active predictions are determined using an ACCDOA activity threshold of $\tau = 0.5$ (see Sec. 5.2.3). These predictions are matched to the active ground-truth sources using the Hungarian algorithm [47], which finds the assignment that minimizes the total angular distance between prediction–reference pairs. Let $\hat{\mathbf{x}}_i$ and $\mathbf{x}_j$ denote the Cartesian direction vectors of the i -th predicted and j -th reference source, respectively. Their angular distance is

$$d(\hat{\mathbf{x}}_i, \mathbf{x}_j) = \arccos\left(\frac{\hat{\mathbf{x}}_i \cdot \mathbf{x}_j}{\|\hat{\mathbf{x}}_i\| \|\mathbf{x}_j\|}\right). \quad (6.5)$$

Each matched active prediction–reference pair is counted as a true positive (TP). Active predictions that remain unmatched are counted as false positives (FP), whereas active reference sources that remain unmatched are counted as false negatives (FN). The number of active reference sources can therefore be written as $N_{\text{ref}} = \text{TP} + \text{FN}$.

Localization performance is measured by the LE, defined as the mean angular distance over all TP pairs. Angular distances are not computed for non-TP samples because at least one of the corresponding DOA vectors is a zero vector, making the angular distance undefined or meaningless. The LE over all evaluation samples K can be written as

$$\text{LE} = \frac{\sum_k^K \sum_n^{\text{TP}(k)} d(\hat{\mathbf{x}}_{kn}, \mathbf{x}_{kn})}{\sum_k^K \text{TP}(k)}, \quad (6.6)$$

where n indexes the matched TP prediction–reference pairs within evaluation sample k , and $d(\cdot, \cdot)$ denotes their angular distance. Since LE is calculated only for correctly detected events,

it does not account for missed detections or false alarms. Therefore, LE is typically reported together with the localization recall as defined in Eq. (6.7), which indicates how many of the actual reference sound sources are considered in the LE.

$$\text{LR} = \frac{\sum_k^K \text{TP}(k)}{\sum_k^K \text{N}_{\text{ref}}(k)} \quad (6.7)$$

For the SAD metrics, a matched predicted–reference source pair is considered correctly detected only if both sources are active and their angular distance does not exceed the threshold of 15° . Such a pair is counted as a location-sensitive true positive (TP_{15°). If both sources are active but their angular distance exceeds the threshold, the match is counted as a location-sensitive false positive (FP_{15°). Based on these counts, the location-sensitive error rate (ER_{15°) and F_1 -score (F_{15°) are defined as

$$\text{ER}_{15^\circ} = \frac{\sum_k^K (\text{FP}(k) + \text{FP}_{15^\circ}(k) + \text{FN}(k))}{\sum_k^K \text{N}_{\text{ref}}(k)}, \quad (6.8)$$

$$F_{15^\circ} = \frac{\sum_k^K 2 \text{TP}_{15^\circ}(k)}{\sum_k^K (2 \text{TP}_{15^\circ}(k) + \text{FP}_{15^\circ}(k) + \text{FP}(k) + \text{FN}(k))}. \quad (6.9)$$

Finally, the metrics are combined into the aggregated sound detection and localization score $\mathcal{E}_{\text{SDL}}$ as defined in Eq. (6.10). Lower values of $\mathcal{E}_{\text{SDL}}$ indicate better overall performance.

$$\mathcal{E}_{\text{SDL}} = \frac{1}{4} \left[\text{ER}_{15^\circ} + (1 - F_{15^\circ}) + \frac{\text{LE}}{180^\circ} + (1 - \text{LR}) \right] \quad (6.10)$$

Statistical Analysis

The evaluation metrics introduced above result in a single value for the complete evaluation set. Estimating the statistical uncertainty of these values is not straightforward, since the metrics are nonlinear functions of aggregated quantities. Consequently, the variability of a metric cannot in general be obtained from the variance of independently calculated per-sample metric values. Furthermore, repeatedly training the same model with different random initializations

would be computationally expensive and would primarily quantify training variability rather than uncertainty caused by the finite evaluation set.

Therefore, the statistical uncertainty of the evaluation metrics is estimated using the jackknife procedure described in [62]. The jackknife is a leave-one-out resampling method that does not require knowledge of the underlying distribution of the evaluated metric. In the present work, each unique room–recording position is treated as one approximately independent observation X_i . All time samples belonging to the same recording position in the same room are therefore removed jointly in one jackknife iteration.

Let $\hat{\psi}$ denote the value of a metric computed from all N observations

$$\hat{\psi} = f(X_1, X_2, \dots, X_N), \quad (6.11)$$

where X_i contains all evaluation samples belonging to recording position i , and $f(\cdot)$ denotes the complete metric computation. For each jackknife iteration, one observation unit is removed and the metric is recomputed from the remaining data:

$$\hat{\psi}_{(i)} = f(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_N), \quad i = 1, \dots, N. \quad (6.12)$$

The corresponding jackknife pseudovalues are

$$\tilde{\psi}_i = N\hat{\psi} - (N-1)\hat{\psi}_{(i)}. \quad (6.13)$$

The mean of the pseudovalues gives the bias-corrected jackknife estimate:

$$\hat{\psi}_{\text{jack}} = \frac{1}{N} \sum_{i=1}^N \tilde{\psi}_i. \quad (6.14)$$

Following the standard leave-one-out jackknife formulation of [34], the estimated variance of the jackknife estimator is

$$\widehat{\text{Var}}_{\text{jack}} = \frac{1}{N(N-1)} \sum_{i=1}^N (\tilde{\psi}_i - \hat{\psi}_{\text{jack}})^2. \quad (6.15)$$

The corresponding jackknife standard error is

$$\hat{\sigma}_{\text{jack}} = \sqrt{\widehat{\text{Var}}_{\text{jack}}}. \quad (6.16)$$

6 Methodology

Using an approximation based on Student's t -distribution [106] for the standardized jackknife estimator, an approximate $(1 - \alpha)$ confidence interval is

$$\hat{\psi}_{\text{jack}} \pm t_{1-\alpha/2, N-1} \hat{\sigma}_{\text{jack}}, \quad (6.17)$$

where $t_{1-\alpha/2, N-1}$ denotes the corresponding quantile of Student's t -distribution with $N - 1$ degrees of freedom. We use $\alpha = 0.05$ for all reported 95 % confidence intervals. These confidence intervals capture uncertainty caused by the finite composition of the evaluation set. They do not capture variability due to model initialization, training randomness, or other sources of stochasticity in the training procedure.

Following Roff [88], the models are compared using a paired t -test of their jackknife pseudovalues. Since the same evaluation unit X is removed from both models in each jackknife iteration, the pseudovalues $\tilde{\psi}_{A,i}$ and $\tilde{\psi}_{B,i}$ form paired observations. For models A and B , the paired pseudo-value differences are defined as

$$\tilde{\Delta}_{A,B,i} = \tilde{\psi}_{A,i} - \tilde{\psi}_{B,i}, \quad i = 1, \dots, N. \quad (6.18)$$

Analogously to Eq. (6.14), their mean gives the jackknife estimate of the difference between the two models:

$$\hat{\Delta}_{A,B,\text{jack}} = \frac{1}{N} \sum_{i=1}^N \tilde{\Delta}_{A,B,i}. \quad (6.19)$$

The corresponding jackknife standard error is

$$\hat{\sigma}_{A,B,\text{jack}} = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^N \left(\tilde{\Delta}_{A,B,i} - \hat{\Delta}_{A,B,\text{jack}} \right)^2}. \quad (6.20)$$

To test the null hypothesis that the underlying metric difference $\Delta_{A,B}$ between models A and B is zero:

$$H_0 : \Delta_{A,B} = 0, \quad (6.21)$$

we use the paired t statistic

$$t_{A,B} = \frac{\hat{\Delta}_{A,B,\text{jack}}}{\hat{\sigma}_{A,B,\text{jack}}}. \quad (6.22)$$

Under the null hypothesis, $t_{A,B}$ is approximated by Student's t -distribution with $N - 1$ degrees of freedom. The corresponding two-sided p -value is

$$p_{A,B} = 2 \left[1 - F_{t_{N-1}}(|t_{A,B}|) \right], \quad (6.23)$$

where $F_{t_{N-1}}$ denotes the cumulative distribution function of Student's t -distribution with $N - 1$ degrees of freedom.

The p -values from all pairwise model comparisons are corrected for multiple comparisons using Holm’s procedure [36]. Specifically, the raw p -values are sorted in ascending order, $p_{(1)} \leq \dots \leq p_{(m)}$, where m is the number of comparisons. Starting from the smallest p -value, the i -th ordered hypothesis is rejected if

$$p_{(i)} \leq \frac{\alpha}{m - i + 1},$$

with $\alpha = 0.05$. Testing proceeds sequentially until the first hypothesis fails this criterion. That hypothesis and all subsequent hypotheses are not rejected.

The statistical comparison procedure is applied separately to every evaluation dataset (Sec. 6.1.3) and metric (Sec. 6.2.3). The sign of $\hat{\Delta}_{A,B,\text{jack}}$ indicates which model performs better. For LE, ER_{15° , and $\mathcal{E}_{\text{SDL}}$, lower values are better, so $\hat{\Delta}_{A,B,\text{jack}} < 0$ favors model A . For LR and F_{15° , higher values are better, so $\hat{\Delta}_{A,B,\text{jack}} > 0$ favors model A .

6.3 MUSIC Baseline

To contextualize the performance relative to classical signal-processing-based SSLD algorithms, we also evaluated the datasets using the MUSIC algorithm (see Sec. 5.1). Since estimating the number n of active sources with MUSIC is unreliable in reverberant environments and often requires additional heuristics, the number of active sources is taken from the ground truth and the algorithm is only used for localization. Therefore, SAD performance is not evaluated. MUSIC is applied independently to each time frame in each dataset. For every frame, SCMs are computed from the third-order Ambisonics signals across 128 mel-frequency bands. The corresponding MUSIC pseudospectra are then averaged over frequency, yielding a single pseudospectrum for each frame. The source directions are estimated from the MUSIC pseudospectrum by selecting the n highest peaks. To avoid multiple detections of the same source, a minimum angular separation of 15° between peaks is specified.

7 Results

This chapter presents and discusses the results of the experiments conducted to evaluate the proposed methods for room acoustics simulation. A complete overview of the resulting SSLD metric scores is provided in Tab. A.2 in the appendix, while the results of the statistical significance tests are shown in Figs. A.1–A.5.

Fig. 7.1 summarizes the performance of the trained models for different microphone simulation methods, image-source reflection orders, and late-reverberation characteristics. All datasets were simulated using shoebox-shaped rooms. Each colored column corresponds to a different simulated dataset used for training. The dataset labels follow the naming convention defined in Sec. 6.1.2. The plots report the SSLD metrics (introduced in Sec. 6.2.3) for all evaluation datasets, including the test split of the corresponding simulation dataset as well as the Arni, Motus, and Agrid datasets (see Sec. 6.1). The 95 % confidence intervals, computed as described in Sec. 6.2.3, are indicated by the whiskers. In addition to the SSLD metrics, Fig. 7.1 presents confusion matrices for each evaluation dataset. These matrices compare the number of ground-truth sound sources with the number of sources predicted by the SSLD model, with the entries expressed as percentages.

Overall, Fig. 7.1 reveals substantial variability in model performance depending on both the evaluated dataset and metric. The aggregated $\mathcal{E}_{\text{SDL}}$ -score therefore provides a useful summary measure for comparing model performance. The confidence intervals are generally considerably wider for the Agrid dataset, indicating greater uncertainty in the corresponding metric estimates. This is likely due to the substantially smaller size and high variability within the Agrid dataset. The confusion matrices reveal that source-number estimation becomes increasingly challenging as the number of simultaneously active sources increases. In particular, the prediction of three active sources is substantially less accurate on the measured evaluation datasets than on the corresponding simulated test sets. Models with relatively balanced predictions correctly estimate three active sources in only approximately 50 % of cases. Models that achieve higher accuracy for three active sources generally exhibit a stronger tendency to overestimate the number of sources. Tab. A.2 further demonstrates that the performance degradation with increasing source count also applies to DOA estimation, with localization errors generally increasing as more sources overlap.

7 Results

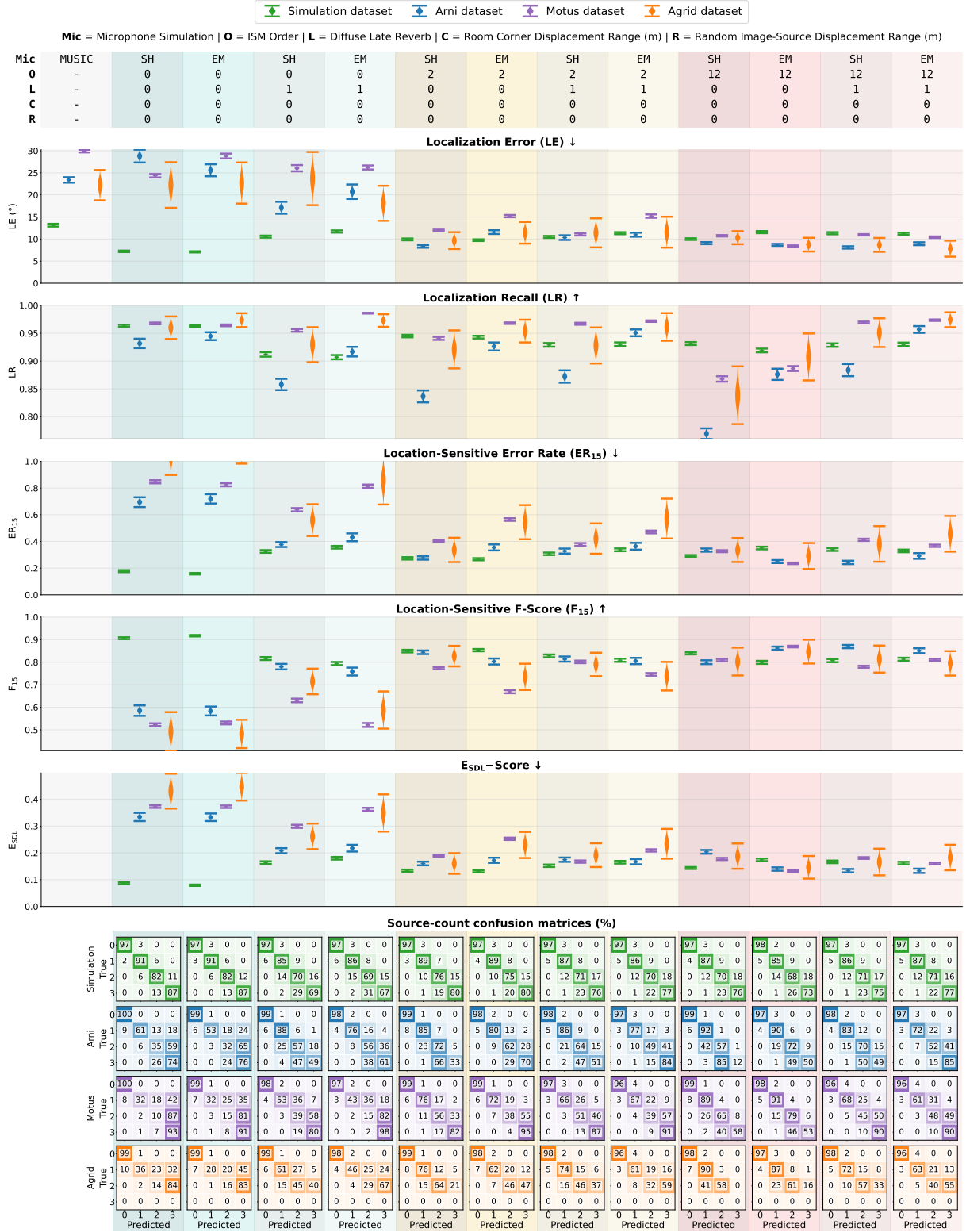


Figure 7.1: Overview of the results for microphone simulation, image-source order, and diffuse late reverberation. Colored columns correspond to the different simulated training datasets. The plots show the SSLD metrics for all evaluation datasets, with whiskers indicating 95 % confidence intervals, as well as the corresponding confusion matrices.

7.1 Effect of Microphone Simulation

First, the influence of the microphone simulation approach is investigated. As described in Sec. 6.1.2, an ideal SH receiver is compared with a measurement-based simulation of the Eigenmike em32. The results shown in Fig. 7.1 do not reveal a consistent overall advantage of either approach. At ISM reflection order 2, the model trained with the ideal SH receiver generalizes better to the measured evaluation datasets. At higher ISM order 12, the em32-based simulation yields better performance. According to the statistical significance analysis presented in Figs. A.1–A.5, the model trained on the “EM- O_{12} - L_0 - C_0 - R_0 ” dataset performs significantly better on most SSLD metrics than almost all of the other configurations across the Arni and Motus evaluation datasets. However, the magnitude of the observed differences should also be considered when interpreting these results. For example, a relatively small improvement, such as a 1° reduction in localization error (LE), may be statistically significant while providing only limited practical benefit. Thus, statistical significance alone does not necessarily indicate a practically meaningful improvement in SSLD performance. Moreover, small differences in LE should be interpreted with caution, as the reference DOAs in the measured datasets are themselves subject to uncertainty. As discussed in Sec. 6.1.3, deviations of approximately 5° may be expected.

Despite these differences in performance, the em32 simulation has two important practical disadvantages. First, dataset generation is substantially more computationally expensive. This is primarily due to the additional convolution operations required to apply the measured microphone directivity filters to each individual image-source reflection. As the number of reflections increases rapidly with increasing ISM order, the computational cost associated with these operations also grows considerably. In the experiments conducted for this work, generating the “EM- O_{12} - L_0 - C_0 - R_0 ” dataset required approximately 40 times the computation time of the SH receiver simulation. Second, the em32 simulation is inherently tied to the characteristics of the Eigenmike em32, which may limit its generalizability to recordings obtained with other microphone arrays. In contrast, the ideal SH receiver represents the sound field independently of a specific microphone implementation. Models trained using this simulation can therefore, in principle, be applied to recordings from different microphone arrays, provided that the recorded signals are transformed into a compatible SH representation.

7.2 Effect of Image-Source Order and Diffuse Late Reverberation

Next, the influence of the ISM reflection order and the addition of diffuse late reverberation are investigated. Simulations were generated using maximum ISM reflection orders of 0, 2, and 12, where order 0 contains only the direct sound. For each configuration, the RIRs were either used as simulated or synthesized with artificial diffuse late reverberation as described in Sec. 6.1.2.

Training on direct sound only results in substantially worse SSL and SAD performance on the measured evaluation datasets compared with models trained on data that includes room reflections. It tends to overestimate the number of sources, as shown by the confusion matrices. However, the direct-sound-only model performs very well on its corresponding simulated test set. This is expected, since the simulated test data contains neither reflections nor reverberation and therefore represents a considerably simpler acoustic condition. At the same time, the large performance difference between simulated and real data highlights the mismatch between the two domains. Despite degrading performance on the simulated test set, adding diffuse late reverberation to the direct-sound simulation improves generalization to the measured datasets, indicating that reverberation during training is important for matching real acoustic conditions. The improvement is particularly noticeable for source-number estimation.

Including second-order image-source reflections in the training data already yields a substantial improvement over the direct-sound datasets for both SSL and SAD on the measured evaluation datasets. Localization performance improves considerably, with localization errors reaching approximately 10° . Increasing the maximum ISM order from 2 to 12 does not improve performance on all evaluation datasets when using the ideal SH receiver, whereas the em32-based simulation benefits significantly from including higher-order reflections.

Once early image-source reflections are included, the additional diffuse late reverberation has comparatively little influence on the results. This suggests that the presence of early reflections is particularly important for generalization to real reverberant recordings, whereas the late reverberant tail appears to be less critical for the investigated SSLD model. Moreover, most of the improvement is already achieved with relatively low ISM orders. This could be explained by the fact that early reflections generally contain more energy than later, higher-order reflections, making them more influential for the spatial and spectral cues used by the model.

On the evaluation data, the MUSIC baseline achieves localization accuracy comparable to that of the model trained on non-reverberant data. This result is not surprising, as reverberation violates the free-field assumptions underlying the algorithm. To assess the achievable performance under free-field conditions, MUSIC was additionally evaluated on the simulated

non-reverberant dataset “SH- O_0 - L_0 - C_0 - R_0 ”. In this case, localization improves substantially, with a localization error below 15° . Nevertheless, the model trained with image-source simulations achieves even lower localization errors on the reverberant datasets, highlighting its strong robustness to reverberation.

Fig. 7.2 illustrates a reverberant sample from the Arni dataset, plotting the ground-truth and predicted source directions from MUSIC, a model trained on direct sound only, and a model trained on 12th-order image-source reflections. Predictions were assigned to the ground-truth sources using the Hungarian algorithm based on angular distance (see Sec. 6.2.3). As MUSIC estimates the source directions independently for each frame, without any temporal context from preceding or subsequent frames, its predictions are highly variable. While some estimates are close to the ground truth, others deviate substantially, likely in frames where the direct sound is weak relative to the reverberant energy. Temporal averaging over a longer time window could potentially reduce this variability and stabilize the estimates. The model trained on direct sound only produces less noisy predictions than MUSIC, but its DOA estimates are consistently offset from the ground truth. In addition, the model tends to overestimate the number of active sources, resulting in a large number of unmatched predictions. The model trained with 12th-order image-source reflections clearly performs best in this example. Both the estimated number of sources and their directions are generally closely aligned with the ground truth. Discontinuities in the DOA estimates are visible at the boundaries between one-second segments. These jumps are expected, since each segment was evaluated independently and the model had no temporal context across segment boundaries.

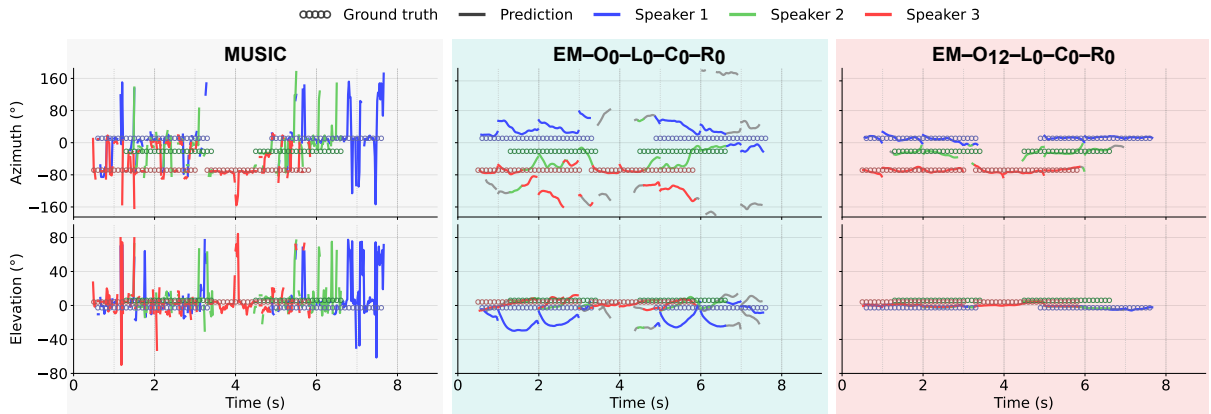


Figure 7.2: Predictions and ground truth over time for three selected models, evaluated on a sample from the Arni dataset (reverberant condition, receiver position 65).

7.3 Effect of Reverberation Time

To further investigate the dependence of model performance on reverberation time, Fig. 7.3 presents the evaluation metrics as a function of RT. For this analysis, the dataset samples were grouped into RT bins. The SSLD metrics and their corresponding 95 % confidence intervals were then calculated separately for each bin. The wider confidence intervals at longer RTs are mainly due to the smaller number of evaluation samples in this range. The figure also highlights the different RT ranges represented by the individual datasets. In particular, the simulated datasets reach different maximum RT values depending on the maximum ISM reflection order and whether late reverberation is added. As expected, localization performance decreases noticeably with increasing RT for both the MUSIC algorithm and the models trained on direct sound only. This dependence becomes progressively weaker with increasing ISM order. In contrast, adding synthesized diffuse late reverberation does not appear to provide an additional benefit. Instead, it tends to degrade performance at larger RT values, despite these training datasets covering the longest RTs. Overall, the most consistent performance across all datasets and RTs is observed for the model trained on the “EM-0₁₂-L₀-C₀-R₀” dataset.

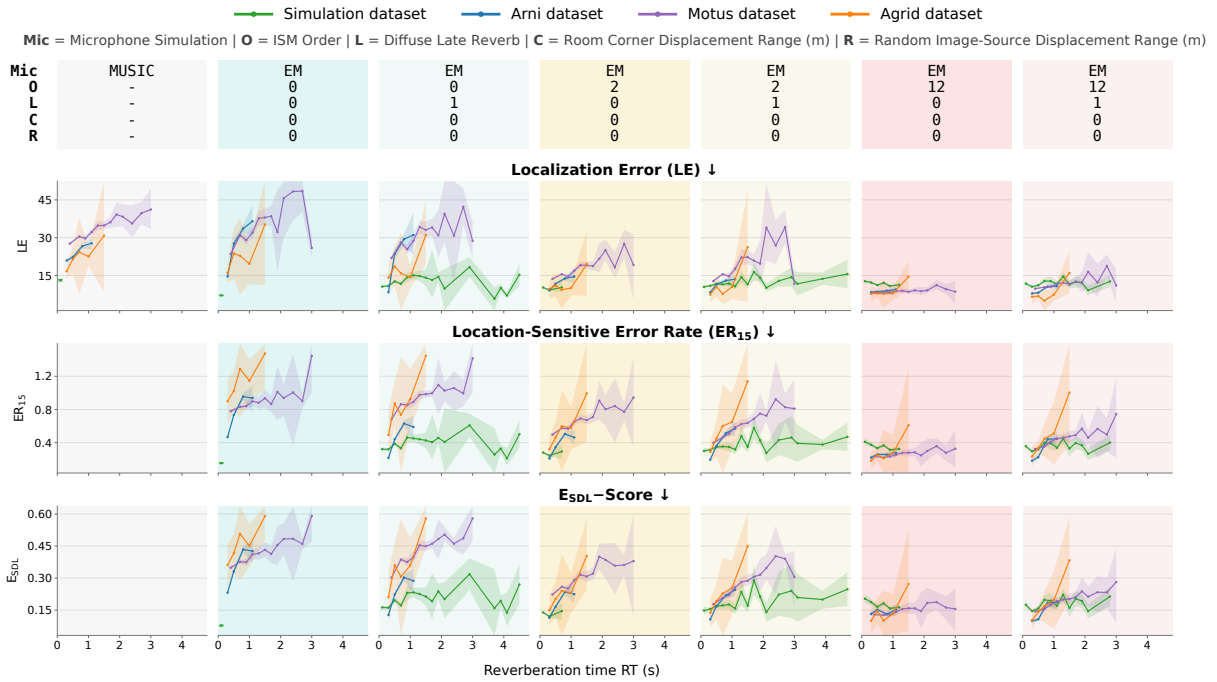


Figure 7.3: Metrics with 95 % confidence intervals as a function of reverberation time for the Eigenmike em32 simulations.

7.4 Effect of Room Geometry and Random Image-Source Positions

This section investigates the influence of the room geometry on the SSLD performance. A set of models was trained on datasets with different parameters related to room geometry and reflection directions. Here, all datasets were simulated using an ideal SH receiver, with image-source reflections up to order 12. No additional reverberation based on decaying noise was synthesized. Fig. 7.4 shows the corresponding results. Two types of modifications were considered. In one case, the shoebox corners were shifted in the xy -plane to create non-shoebox room geometries (blue columns). In the other case, the shoebox room geometry was kept unchanged, while the image-source positions were randomized (green columns), as explained in Sec. 6.1.2.

As shown in Fig. 7.4, the results of all models are very similar. Changing the room geometry from shoebox to non-shoebox configurations does not substantially affect performance on either the simulated or the measured test sets. At most, the results suggest a slight improvement with increasingly randomized geometry.

Likewise, randomizing the image-source positions also has only a small effect on the results. One noticeable trend is that increasing the degree of randomization leads to a decrease in performance on the simulated test sets. One possible explanation is that models trained on shoebox rooms exploit correlations associated with the more regular image-source structure, and that these correlations become less pronounced as the image-source positions are randomized. In contrast, performance on the measured test sets does not change or even improves slightly with increasing randomization. Interestingly, this trend is also observed for the Arni dataset, despite the fact that it was recorded in a shoebox-shaped room. Models trained on randomized image-source positions are generally competitive with, and in many cases outperform, models trained using precise shoebox geometry. However, the differences are generally small and may partly reflect variability across individual training runs.

Overall, the results suggest that the precise room geometry and the exact directions of early reflections are not critical for model performance. Moreover, the comparable results obtained on simulated and measured data indicate that modeling early reflections with randomized positions provides a sufficiently realistic approximation for the evaluated SSLD model.

7 Results

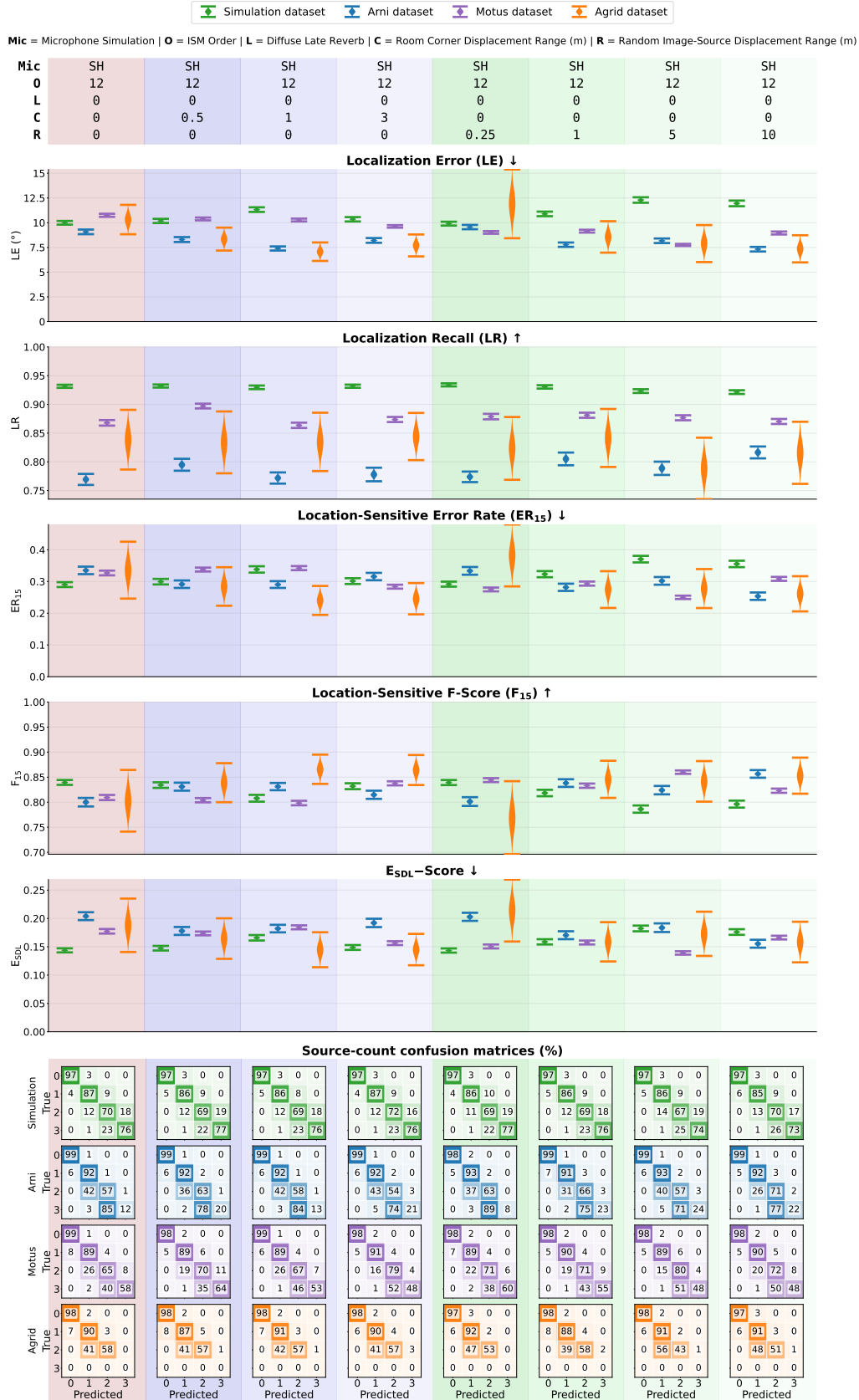


Figure 7.4: Overview of the results for non-shoebox room geometries and randomly displaced image sources. Colored columns correspond to the different simulated training datasets. The plots show the SSLD metrics for all evaluation datasets, with whiskers indicating 95 % confidence intervals, as well as the corresponding confusion matrices.

8 Final Discussion

8.1 Conclusions

This work investigated how the choice of room simulation parameters used to generate training data affects the performance of an SSLD model when evaluated on real-world measurements. Training data containing only the direct sound was found to be insufficient for generalization to real measurements and resulted in poor localization and detection performance. Adding synthesized diffuse late reverberation led to some improvements when combined with the direct-sound data, although these improvements were no longer observed once early reflections were present. Instead, performance tended to decrease slightly at longer reverberation times when diffuse late reverberation was synthesized together with image-source reflections.

Early reflections were shown to be an essential component of the simulated training data. Even the inclusion of low-order early reflections led to a substantial improvement in both localization and detection performance. Models trained on image-source simulations achieved comparable performance on simulated and measured test sets, despite the simulations representing only simplified approximations of real room acoustics. This indicates that the resulting simulation-to-real domain gap was sufficiently small for the model to generalize effectively to real measurements. Higher-order reflections yielded further improvements, with the largest gains observed for the Eigenmike em32 microphone simulation.

Moreover, the computationally more expensive microphone simulation based on a measured ATF achieved the best overall performance when combined with high image-source reflection orders, although the gains over the more generic SH receiver simulation were relatively small given the substantially higher computational cost. At low image-source orders, however, it performed worse than the SH receiver. These results therefore indicate a trade-off between computational efficiency and maximum achievable SSLD performance. Using an ideal SH receiver provides a less expensive simulation approach and may also improve generalization across different microphone arrays, as it is less tied to the characteristics of a particular array. In contrast, when the goal is to achieve the best possible performance, incorporating a detailed microphone-array simulation with high-order image-source reflections may still be beneficial.

Another finding of this work was that the geometric shape of the simulated rooms and the exact directions of the corresponding image sources do not appear to be critical for the SSLD model. In fact, randomizing image-source positions may even slightly improve generalization to measured data. While early reflections in the training data improved performance, their precise directions and times of arrival seemed to play a secondary role, at least within the investigated range. The reflections therefore do not need to originate from a physically consistent room model to be beneficial. This is particularly relevant for future dataset generation, as simpler simulation approaches may provide similar SSLD performance while substantially reducing computational cost.

Even when the model was evaluated on the simulated test sets, the $\mathcal{E}_{\text{SDL}}$ -score remained well above the optimal value of zero. This indicates that the task remains challenging even under domain-matched conditions. This may reflect limitations of the model architecture, the input features, the amount of training data, or a combination of these factors. In domain-matched scenarios, the SSLD model achieved the best performance under anechoic conditions (direct-sound simulation only), indicating that reflections and reverberation make both source detection and localization more challenging. Even with a perfectly matched simulation, the achievable performance is still bounded by the limitations of the SSLD model itself.

Overall, the results suggest that the presence of early reflections in the training data is substantially more important than their exact geometrical configuration or the characteristics of the late reverberation. However, these findings should be interpreted with some caution. The experiments were conducted on a simplified version of the SELD task involving only a single sound class (“speech”) and did not include additional subtasks such as classification or distance estimation. Furthermore, the evaluation was limited to a specific model architecture and set of input features. Therefore, the observed effects cannot necessarily be assumed to generalize to other SELD tasks, model architectures, or feature representations.

8.2 Future Work

Future work should further investigate the use of randomized image-source positions by incorporating algorithms that do not rely on shoebox-based simulations. Instead, image-source positions could be generated using a stochastic model, such as the approach proposed in [7]. Such methods would be computationally lightweight and could potentially enable real-time room simulation.

The findings should also be validated on more complex SELD tasks, including scenarios with moving sound sources. It would further be useful to investigate whether the observed sensitivity to room simulation parameters extends to other deep learning tasks that depend on simulated

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acoustic data, such as sound source separation or speech enhancement. This would clarify whether the findings are specific to SSLD or reflect a broader phenomenon.

Further work should examine additional simulation parameters, such as anisotropy or source directivity, as well as more advanced simulation methods that account for acoustic effects such as scattering and diffraction. However, for the model investigated in this work, increasing the physical realism of the simulation is not expected to improve performance, as the model performed similarly on domain-matched simulated test data and out-of-domain real-world measurements. Future studies should therefore also consider models with higher spatial resolution, increased capacity, and larger training datasets, and evaluate whether such models benefit from more detailed and physically accurate room simulations.

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List of Abbreviations

ACDDOA activity-coupled Cartesian direction of arrival 34, 48, 50

ACN Ambisonic Channel Number 16, 17

AIC Akaike information criterion 29

ARIR Ambisonic room impulse response 36, 37

ATF array transfer function 36, 63

BiLSTM bidirectional long short-term memory 48, 49

CNN convolutional neural network 31

DCASE Detection and Classification of Acoustic Scenes and Events 30, 32, 34

DL deep learning 1, 2, 23, 30, 35, 37, 47

DOA direction of arrival 17, 28, 29, 31, 33–35, 41, 43, 45, 50, 55, 57, 59

DRR direct-to-reverberant ratio 12, 13, 41, 45

EDC energy decay curve 11

ELU exponential linear unit 48, 49

FC fully connected 32, 48, 49

FFT fast Fourier transform 47

FO-PIV first-order pseudointensity vector 17, 18, 47

FOA first-order Ambisonics 17

GRU gated recurrent unit 32

HO-PIV higher-order pseudointensity vector 18, 31, 47, 48

HOA higher-order Ambisonics 18, 31, 35, 42, 46, 47

List of Abbreviations

- ISM** image-source method 23, 24, 26, 27, 38, 40, 42, 57, 58, 60
- MDL** minimum description length 29
- MSE** mean squared error 34, 49
- MUSIC** multiple signal classification 28–30, 43, 45, 54, 58–60
- PIT** permutation-invariant training 34
- RIR** room impulse response 9–12, 23, 26, 35, 39, 43–46, 58
- RNN** recurrent neural network 31
- RT** reverberation time 11, 12, 38, 41, 45, 46, 60
- SAD** sound activity detection 28, 34, 48, 50, 51, 54, 58
- SALSA** spatial cue-augmented log-spectrogram 31, 32
- SCM** spatial covariance matrix 28, 29, 31, 43, 54
- SELD** sound event localization and detection 30–32, 34, 35, 48, 64
- SH** spherical harmonic 14–20, 22, 29, 31, 36, 37, 42, 57, 58, 61, 63
- SHD-MUSIC** spherical harmonic domain multiple signal classification 29, 43
- SIR** signal-to-interference ratio 46
- SNR** signal-to-noise ratio 10, 46
- SRIR** spatial room impulse response 10, 11, 46
- SSL** sound source localization 28, 34, 48, 50, 58
- SSLD** sound source localization and detection i, 1, 28, 31–33, 35, 36, 46, 47, 54, 55, 57, 58, 60, 61, 63–65
- STFT** short-time Fourier transform 4, 47

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Table A.1: Spherical harmonics $Y_n^m(\theta, \phi)$ up to order $n = 4$, with the corresponding Ambisonic Channel Numbers [86].

n	m	ACN	$Y_n^m(\theta, \phi)$
0	0	0	$\sqrt{\frac{1}{4\pi}}$
1	-1	1	$\sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\phi}$
1	0	2	$\sqrt{\frac{3}{4\pi}} \cos \theta$
1	1	3	$-\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}$
2	-2	4	$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{-2i\phi}$
2	-1	5	$\sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{-i\phi}$
2	0	6	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$
2	1	7	$-\sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{i\phi}$
2	2	8	$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{2i\phi}$
3	-3	9	$\sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{-3i\phi}$
3	-2	10	$\sqrt{\frac{105}{32\pi}} \cos \theta \sin^2 \theta e^{-2i\phi}$
3	-1	11	$\sqrt{\frac{21}{64\pi}} (5 \cos^2 \theta - 1) \sin \theta e^{-i\phi}$
3	0	12	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$
3	1	13	$-\sqrt{\frac{21}{64\pi}} (5 \cos^2 \theta - 1) \sin \theta e^{i\phi}$
3	2	14	$\sqrt{\frac{105}{32\pi}} \cos \theta \sin^2 \theta e^{2i\phi}$
3	3	15	$-\sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{3i\phi}$
4	-4	16	$\sqrt{\frac{315}{512\pi}} \sin^4 \theta e^{-4i\phi}$
4	-3	17	$\sqrt{\frac{315}{64\pi}} \cos \theta \sin^3 \theta e^{-3i\phi}$
4	-2	18	$\sqrt{\frac{45}{128\pi}} (7 \cos^2 \theta - 1) \sin^2 \theta e^{-2i\phi}$
4	-1	19	$\sqrt{\frac{45}{64\pi}} (7 \cos^3 \theta - 3 \cos \theta) \sin \theta e^{-i\phi}$
4	0	20	$\sqrt{\frac{9}{256\pi}} (35 \cos^4 \theta - 30 \cos^2 \theta + 3)$
4	1	21	$-\sqrt{\frac{45}{64\pi}} (7 \cos^3 \theta - 3 \cos \theta) \sin \theta e^{i\phi}$
4	2	22	$\sqrt{\frac{45}{128\pi}} (7 \cos^2 \theta - 1) \sin^2 \theta e^{2i\phi}$
4	3	23	$-\sqrt{\frac{315}{64\pi}} \cos \theta \sin^3 \theta e^{3i\phi}$
4	4	24	$\sqrt{\frac{315}{512\pi}} \sin^4 \theta e^{4i\phi}$

A Additional Tables and Figures

Table A.2: Evaluation results for the SSLD metrics across all trained models and test datasets. Each metric was computed over the entire test set (**Total**) and independently recomputed on the subsets of frames containing one, two, or three ground-truth sources. Bold values indicate the best-performing model for each dataset–metric combination.

Train-/ Test-Dataset	LE [°] ↓		LR ↑		ER _{15°} ↓		F _{15°} ↑		E _{SDL} ↓	
	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3
MUSIC	Simulated	5.11	–	–	–	–	–	–	–	–
		13.14	–	–	–	–	–	–	–	–
		13.38	–	–	–	–	–	–	–	–
	Arni	23.34	–	–	–	–	–	–	–	–
		23.39	–	–	–	–	–	–	–	–
		23.20	–	–	–	–	–	–	–	–
	Motus	21.64	–	–	–	–	–	–	–	–
		29.93	–	–	–	–	–	–	–	–
		29.10	–	–	–	–	–	–	–	–
	Agrid	24.10	–	–	–	–	–	–	–	–
		22.23	–	–	–	–	–	–	–	–
		21.34	–	–	–	–	–	–	–	–
SH-0 ₀ -L ₀ -C ₀ -R ₀	Simulated	4.05	0.96	0.98	0.18	0.11	0.91	0.94	0.09	0.05
		7.22	6.90	0.97	0.18	0.18	0.91	0.91	0.09	0.09
		8.70	–	0.96	0.19	0.19	0.90	0.90	0.10	0.10
	Arni	11.15	0.93	0.91	0.69	0.76	0.59	0.66	0.33	0.31
		28.80	16.16	0.97	0.73	0.73	0.61	0.61	0.33	0.31
		44.67	–	0.91	0.64	0.64	0.52	0.52	0.36	0.36
	Motus	8.34	0.97	0.92	0.85	1.25	0.52	0.55	0.37	0.46
		24.35	23.73	0.99	1.05	1.05	0.42	0.42	0.37	0.45
		29.55	–	0.97	0.58	0.58	0.59	0.59	0.29	0.29
	Agrid	15.03	0.96	0.90	1.05	1.26	0.49	0.48	0.43	0.49
		22.28	25.42	0.99	0.95	0.95	0.50	0.50	0.43	0.40
		–	–	–	–	–	–	–	–	–
EM-0 ₀ -L ₀ -C ₀ -R ₀	Simulated	4.72	0.96	0.97	0.16	0.11	0.92	0.95	0.08	0.05
		6.55	–	0.97	0.15	0.15	0.92	0.92	0.08	0.07
		8.45	–	0.95	0.17	0.17	0.91	0.91	0.09	0.09
	Arni	10.05	0.94	0.94	0.72	0.82	0.58	0.67	0.33	0.32
		25.58	17.76	0.98	0.74	0.74	0.61	0.61	0.33	0.31
		37.09	–	0.92	0.67	0.67	0.50	0.50	0.36	0.36
	Motus	8.85	0.96	0.93	0.82	1.17	0.53	0.57	0.37	0.43
		28.81	24.03	0.98	0.99	0.99	0.46	0.46	0.37	0.42
		38.14	–	0.96	0.60	0.60	0.57	0.57	0.32	0.32
	Agrid	12.42	0.97	0.93	1.12	1.40	0.48	0.50	0.45	0.51
		22.71	27.29	0.99	0.98	0.98	0.47	0.47	0.45	0.42
		–	–	–	–	–	–	–	–	–
SH-0 ₀ -L ₁ -C ₀ -R ₀	Simulated	7.78	0.91	0.94	0.32	0.27	0.82	0.86	0.16	0.13
		10.55	10.03	0.93	0.32	0.32	0.82	0.83	0.16	0.16
		11.92	–	0.89	0.34	0.34	0.80	0.80	0.18	0.18
	Arni	5.71	0.86	0.94	0.38	0.14	0.78	0.93	0.21	0.08
		17.09	12.39	0.87	0.39	0.39	0.78	0.78	0.21	0.20
		25.25	–	0.82	0.45	0.45	0.71	0.71	0.26	0.26
	Motus	5.16	0.96	0.96	0.64	0.58	0.63	0.76	0.30	0.22
		26.05	16.30	0.98	0.62	0.62	0.63	0.68	0.30	0.26
		39.73	–	0.93	0.66	0.66	0.50	0.50	0.36	0.36
	Agrid	6.20	0.93	0.94	0.56	0.47	0.71	0.79	0.26	0.19
		23.73	32.20	0.93	0.60	0.60	0.71	0.67	0.26	0.30
		–	–	–	–	–	–	–	–	–

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A Additional Tables and Figures

Table A.2 – continued

Train- / Test-Dataset	LE [°] ↓			LR ↑		ER _{15°} ↓		F _{15°} ↑		E _{SDL} ↓	
	Total	Sources 1 / 2 / 3		Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3
EM-O ₀ -L ₁ -C ₀ -R ₀	Simulated	11.71	8.90 11.12 13.16	0.91	0.94 0.92 0.88	0.36	0.28 0.35 0.38	0.79	0.85 0.80 0.77	0.18	0.13 0.17 0.20
	Arni	20.72	6.71 12.91 32.32	0.92	0.96 0.96 0.87	0.43	0.31 0.42 0.48	0.76	0.86 0.79 0.69	0.22	0.13 0.19 0.27
	Motus	26.22	6.18 19.07 37.16	0.99	0.97 0.99 0.99	0.81	0.83 0.89 0.75	0.52	0.68 0.54 0.40	0.36	0.30 0.37 0.39
	Agrid	18.13	10.28 21.77 –	0.97	0.96 0.98 –	0.86	0.98 0.80 –	0.59	0.61 0.58 –	0.35	0.37 0.34 –
SH-O ₂ -L ₀ -C ₀ -R ₀	Simulated	9.92	6.47 9.41 11.60	0.95	0.97 0.95 0.93	0.27	0.18 0.27 0.30	0.85	0.91 0.85 0.82	0.13	0.08 0.13 0.15
	Arni	8.33	5.57 8.38 9.42	0.84	0.92 0.88 0.77	0.28	0.16 0.24 0.34	0.84	0.92 0.87 0.80	0.16	0.09 0.13 0.21
	Motus	11.96	5.32 11.63 14.29	0.94	0.94 0.94 0.94	0.40	0.28 0.44 0.41	0.77	0.87 0.76 0.74	0.19	0.13 0.20 0.20
	Agrid	9.66	6.28 11.26 –	0.92	0.92 0.92 –	0.34	0.33 0.34 –	0.83	0.84 0.82 –	0.16	0.15 0.16 –
EM-O ₂ -L ₀ -C ₀ -R ₀	Simulated	9.75	6.70 9.31 11.22	0.94	0.96 0.95 0.93	0.27	0.17 0.27 0.29	0.85	0.91 0.86 0.83	0.13	0.08 0.13 0.15
	Arni	11.57	7.69 11.00 13.43	0.93	0.95 0.95 0.90	0.35	0.26 0.38 0.36	0.80	0.87 0.80 0.78	0.17	0.12 0.17 0.19
	Motus	15.20	5.42 13.17 19.49	0.97	0.94 0.96 0.98	0.56	0.34 0.62 0.60	0.67	0.84 0.68 0.58	0.25	0.15 0.26 0.29
	Agrid	11.45	7.43 13.30 –	0.95	0.93 0.96 –	0.54	0.55 0.54 –	0.73	0.76 0.72 –	0.23	0.23 0.23 –
SH-O ₂ -L ₁ -C ₀ -R ₀	Simulated	10.48	7.80 9.79 11.90	0.93	0.95 0.94 0.91	0.31	0.25 0.31 0.32	0.83	0.87 0.83 0.81	0.15	0.12 0.15 0.16
	Arni	10.33	5.84 10.27 12.13	0.87	0.95 0.89 0.83	0.33	0.17 0.35 0.36	0.81	0.92 0.81 0.78	0.17	0.08 0.18 0.21
	Motus	11.06	5.14 9.11 14.30	0.97	0.97 0.98 0.96	0.38	0.42 0.39 0.35	0.80	0.82 0.81 0.79	0.17	0.16 0.16 0.17
	Agrid	11.41	5.77 14.19 –	0.93	0.95 0.92 –	0.42	0.33 0.46 –	0.79	0.85 0.76 –	0.19	0.14 0.21 –

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A Additional Tables and Figures

Table A.2 – continued

Train- / Test-Dataset	LE [°] ↓			LR ↑		ER _{15°} ↓		F _{15°} ↑		E _{SDL} ↓	
	Total	Sources 1 / 2 / 3		Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3
EM-O ₂ -L ₁ -C ₀ -R ₀	Simulated	8.95		0.95		0.29		0.85		0.13	
		11.33	10.63	0.93	0.94	0.34	0.34	0.81	0.82	0.17	0.16
			12.65		0.92		0.35		0.79		0.18
	Arni	7.09		0.97		0.30		0.86		0.13	
		10.99	11.13	0.95	0.95	0.36	0.47	0.81	0.76	0.17	0.21
			12.26		0.95		0.30		0.83		0.15
	Motus	5.17		0.97		0.44		0.81		0.17	
		15.20	10.92	0.97	0.98	0.47	0.48	0.75	0.77	0.21	0.20
			21.28		0.97		0.46		0.70		0.23
	Agrid	6.58		0.97		0.58		0.77		0.22	
		11.58	13.99	0.96	0.96	0.57	0.56	0.74	0.73	0.23	0.24
			–		–		–		–		–
SH-O ₁₂ -L ₀ -C ₀ -R ₀	Simulated	6.54		0.96		0.20		0.90		0.09	
		9.99	9.37	0.93	0.94	0.29	0.30	0.84	0.84	0.14	0.14
			11.62		0.92		0.31		0.82		0.16
	Arni	6.73		0.94		0.11		0.94		0.07	
		9.07	9.82	0.77	0.79	0.34	0.34	0.80	0.80	0.20	0.20
			9.54		0.70		0.41		0.75		0.25
	Motus	5.45		0.92		0.13		0.93		0.08	
		10.75	10.27	0.87	0.87	0.33	0.31	0.81	0.82	0.18	0.17
			12.89		0.85		0.40		0.75		0.22
	Agrid	8.64		0.93		0.20		0.89		0.11	
		10.32	11.26	0.84	0.79	0.34	0.39	0.80	0.76	0.19	0.23
			–		–		–		–		–
EM-O ₁₂ -L ₀ -C ₀ -R ₀	Simulated	9.17		0.95		0.30		0.84		0.14	
		11.57	10.99	0.92	0.93	0.35	0.36	0.80	0.80	0.17	0.17
			12.80		0.90		0.36		0.78		0.18
	Arni	6.80		0.96		0.12		0.94		0.06	
		8.68	9.02	0.88	0.90	0.25	0.26	0.86	0.86	0.14	0.14
			9.16		0.83		0.28		0.84		0.17
	Motus	4.59		0.95		0.11		0.95		0.06	
		8.43	8.33	0.89	0.92	0.24	0.20	0.87	0.89	0.13	0.11
			9.89		0.84		0.30		0.82		0.17
	Agrid	6.07		0.96		0.16		0.92		0.08	
		8.74	10.12	0.91	0.88	0.29	0.34	0.85	0.81	0.15	0.17
			–		–		–		–		–
SH-O ₁₂ -L ₁ -C ₀ -R ₀	Simulated	8.33		0.95		0.26		0.86		0.12	
		11.32	10.66	0.93	0.94	0.34	0.34	0.81	0.81	0.17	0.16
			12.84		0.91		0.35		0.78		0.18
	Arni	6.08		0.96		0.19		0.91		0.09	
		8.10	8.40	0.88	0.93	0.24	0.25	0.87	0.87	0.13	0.12
			8.65		0.83		0.25		0.85		0.15
	Motus	4.57		0.97		0.38		0.83		0.15	
		10.96	10.26	0.97	0.97	0.41	0.47	0.78	0.77	0.18	0.19
			13.48		0.97		0.38		0.77		0.18
	Agrid	6.22		0.95		0.40		0.82		0.17	
		8.68	9.85	0.95	0.95	0.38	0.37	0.81	0.81	0.17	0.16
			–		–		–		–		–

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A Additional Tables and Figures

Table A.2 – continued

Train- / Test-Dataset	LE [°] ↓			LR ↑		ER _{15°} ↓		F _{15°} ↑		E _{SDL} ↓	
	Total	Sources 1 / 2 / 3		Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3
EM-0 ₁₂ -L ₁ -C ₀ -R ₀	Simulated	11.21	8.73 10.46 12.59	0.93	0.95 0.94 0.92	0.33	0.27 0.33 0.34	0.81	0.86 0.82 0.79	0.16	0.13 0.16 0.17
	Arni	8.94	6.48 8.47 10.16	0.96	0.97 0.96 0.95	0.29	0.33 0.34 0.23	0.85	0.85 0.83 0.87	0.13	0.14 0.15 0.12
	Motus	10.42	4.68 8.38 13.63	0.97	0.97 0.99 0.97	0.37	0.44 0.36 0.34	0.81	0.81 0.83 0.80	0.16	0.17 0.15 0.16
	Agrid	7.83	6.10 8.65 –	0.97	0.97 0.97 –	0.46	0.52 0.41 –	0.80	0.78 0.81 –	0.18	0.20 0.17 –
SH-0 ₁₂ -L ₀ -C _{0.5} -R ₀	Simulated	10.18	6.65 9.68 11.74	0.93	0.95 0.94 0.92	0.30	0.22 0.31 0.31	0.83	0.89 0.83 0.82	0.15	0.10 0.15 0.16
	Arni	8.30	5.62 8.59 9.24	0.79	0.94 0.82 0.73	0.29	0.09 0.27 0.37	0.83	0.95 0.84 0.77	0.18	0.06 0.16 0.23
	Motus	10.38	4.50 10.19 12.53	0.90	0.95 0.90 0.88	0.34	0.13 0.32 0.41	0.80	0.94 0.82 0.74	0.17	0.07 0.16 0.22
	Agrid	8.34	5.88 9.70 –	0.83	0.92 0.79 –	0.28	0.16 0.34 –	0.84	0.92 0.80 –	0.16	0.09 0.20 –
SH-0 ₁₂ -L ₀ -C ₁ -R ₀	Simulated	11.32	8.89 10.57 12.69	0.93	0.95 0.94 0.92	0.34	0.28 0.34 0.35	0.81	0.85 0.81 0.79	0.17	0.13 0.16 0.18
	Arni	7.39	5.48 7.54 8.15	0.77	0.94 0.79 0.70	0.29	0.09 0.28 0.37	0.83	0.95 0.84 0.78	0.18	0.06 0.17 0.23
	Motus	10.27	4.57 10.25 12.31	0.86	0.94 0.87 0.84	0.34	0.12 0.33 0.42	0.80	0.94 0.81 0.74	0.18	0.07 0.18 0.23
	Agrid	7.07	5.62 7.88 –	0.83	0.93 0.79 –	0.24	0.11 0.30 –	0.87	0.94 0.83 –	0.14	0.07 0.18 –
SH-0 ₁₂ -L ₀ -C ₃ -R ₀	Simulated	10.34	7.08 9.93 11.81	0.93	0.96 0.94 0.92	0.30	0.22 0.31 0.32	0.83	0.89 0.84 0.81	0.15	0.10 0.15 0.16
	Arni	8.21	5.21 8.31 9.48	0.78	0.94 0.79 0.72	0.32	0.09 0.31 0.39	0.81	0.95 0.82 0.76	0.19	0.06 0.19 0.24
	Motus	9.62	4.96 8.73 12.00	0.87	0.95 0.92 0.82	0.28	0.10 0.21 0.39	0.84	0.95 0.89 0.76	0.16	0.06 0.11 0.22
	Agrid	7.71	5.82 8.78 –	0.84	0.94 0.80 –	0.25	0.11 0.30 –	0.86	0.94 0.82 –	0.15	0.06 0.18 –

Continued on next page

A Additional Tables and Figures

Table A.2 – continued

Train- / Test-Dataset		LE [°] ↓		LR ↑		ER _{15°} ↓		F _{15°} ↑		E _{SDL} ↓	
		Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3	Total	Sources 1 / 2 / 3
SH-0 ₁₂ -L ₀ -C ₀ -R _{0.25}	Simulated	9.90	6.23	0.93	0.96	0.29	0.20	0.84	0.90	0.14	0.10
			9.37		0.94		0.30		0.84		0.14
			11.55		0.92		0.31		0.82		0.16
	Arni	9.56	7.45	0.77	0.95	0.33	0.12	0.80	0.94	0.20	0.07
			10.41		0.81		0.33		0.80		0.19
			9.82		0.69		0.41		0.74		0.26
	Motus	9.01	5.08	0.88	0.93	0.27	0.14	0.84	0.93	0.15	0.08
			8.32		0.88		0.25		0.86		0.14
			10.84		0.86		0.33		0.80		0.18
	Agrid	11.93	9.74	0.82	0.94	0.38	0.24	0.77	0.87	0.21	0.12
			13.21		0.77		0.44		0.72		0.26
			–		–		–		–		–
SH-0 ₁₂ -L ₀ -C ₀ -R ₁	Simulated	10.89	8.29	0.93	0.95	0.32	0.27	0.82	0.86	0.16	0.12
			10.43		0.94		0.34		0.82		0.16
			12.11		0.92		0.32		0.81		0.17
	Arni	7.78	5.26	0.81	0.93	0.28	0.11	0.84	0.94	0.17	0.07
			8.01		0.84		0.26		0.86		0.15
			8.68		0.73		0.36		0.78		0.22
	Motus	9.14	4.00	0.88	0.95	0.29	0.12	0.83	0.94	0.16	0.06
			8.91		0.90		0.28		0.85		0.15
			11.13		0.85		0.36		0.78		0.20
	Agrid	8.57	5.62	0.84	0.92	0.27	0.14	0.85	0.93	0.16	0.08
			10.19		0.80		0.33		0.81		0.19
			–		–		–		–		–
SH-0 ₁₂ -L ₀ -C ₀ -R ₅	Simulated	12.29	9.55	0.92	0.95	0.37	0.32	0.79	0.83	0.18	0.15
			11.84		0.93		0.39		0.79		0.18
			13.58		0.91		0.37		0.77		0.19
	Arni	8.17	5.57	0.79	0.94	0.30	0.09	0.82	0.95	0.18	0.06
			8.43		0.80		0.30		0.82		0.18
			9.12		0.73		0.37		0.77		0.23
	Motus	7.75	4.92	0.88	0.95	0.25	0.13	0.86	0.94	0.14	0.07
			7.53		0.92		0.20		0.89		0.11
			8.95		0.82		0.32		0.81		0.18
	Agrid	7.91	5.62	0.79	0.94	0.28	0.11	0.84	0.94	0.17	0.06
			9.33		0.72		0.35		0.79		0.23
			–		–		–		–		–
SH-0 ₁₂ -L ₀ -C ₀ -R ₁₀	Simulated	11.95	8.95	0.92	0.94	0.36	0.29	0.80	0.84	0.18	0.14
			11.37		0.93		0.35		0.80		0.17
			13.43		0.91		0.37		0.77		0.19
	Arni	7.32	5.34	0.82	0.95	0.25	0.09	0.86	0.95	0.16	0.06
			7.50		0.87		0.22		0.88		0.13
			8.04		0.73		0.33		0.80		0.21
	Motus	8.96	3.82	0.87	0.95	0.31	0.12	0.82	0.94	0.17	0.06
			9.20		0.90		0.28		0.84		0.15
			10.67		0.82		0.38		0.77		0.21
	Agrid	7.37	5.29	0.82	0.94	0.26	0.11	0.85	0.94	0.16	0.06
			8.58		0.76		0.33		0.81		0.20
			–		–		–		–		–

A Additional Tables and Figures

Localization Error (LE) ↓

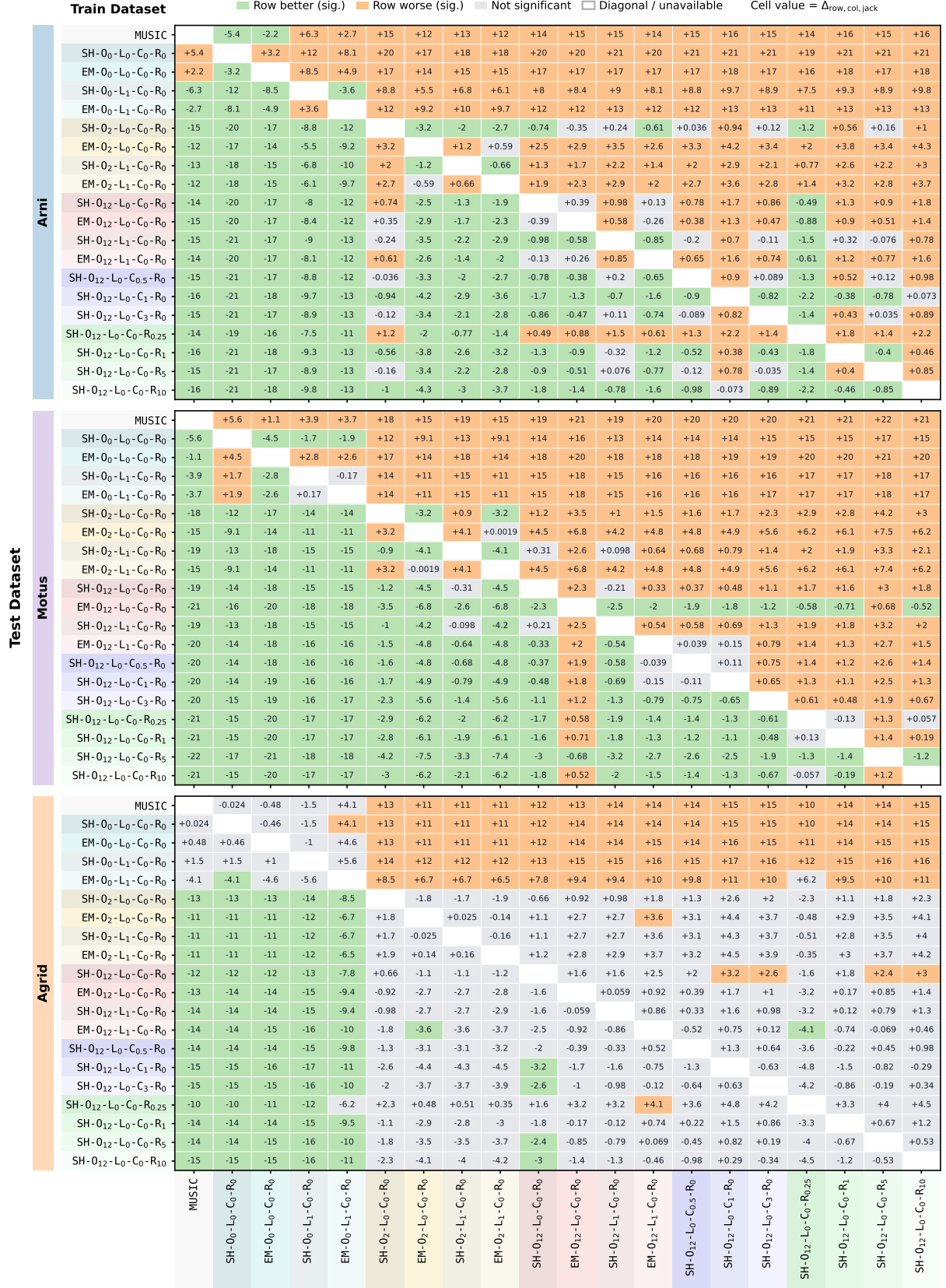


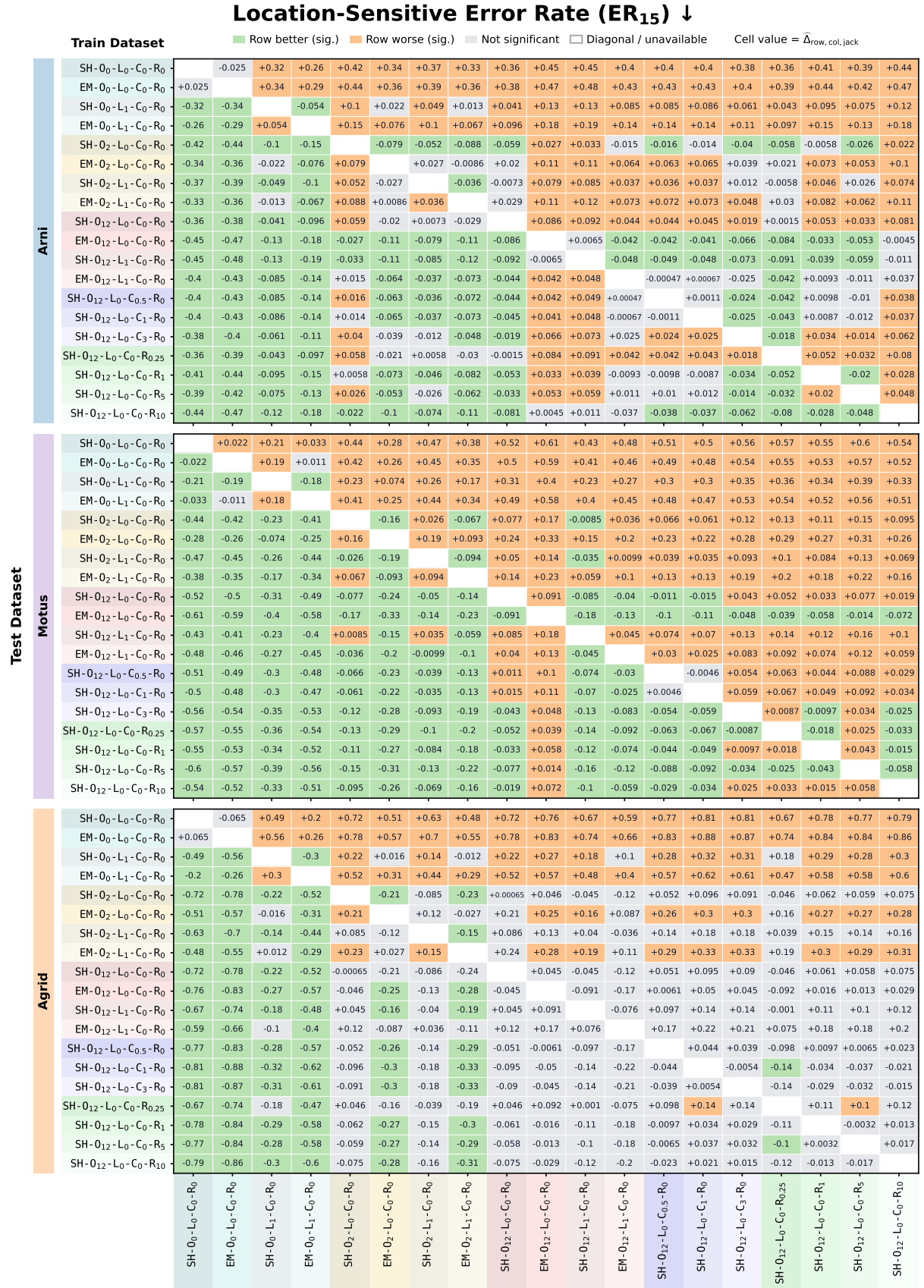
Figure A.1: Significance matrices for localization error (LE) across all evaluation datasets. Green cells indicate that the row model performs significantly better than the column model, whereas orange cells indicate that the row model performs significantly worse, based on pairwise comparisons with Holm correction at $\alpha = 0.05$. Cell values show the jackknife estimate of the difference between the row and column models, $\hat{\Delta}_{\text{row, col, jack}}$.

A Additional Tables and Figures

Localization Recall (LR) ↑

Train Dataset		Row better (sig.) Row worse (sig.) Not significant Diagonal / unavailable																		Cell value = $\Delta_{row, col, jack}$	
Armi	SH-0 ₀ -L ₀ -C ₀ -R ₀		-0.013	+0.074	+0.015	+0.095	+0.0055	+0.06	-0.019	+0.16	+0.056	+0.048	-0.025	+0.14	+0.16	+0.15	+0.16	+0.13	+0.14	+0.12	
	EM-0 ₀ -L ₀ -C ₀ -R ₀	+0.013		+0.087	+0.028	+0.11	+0.018	+0.073	-0.006	+0.18	+0.069	+0.061	-0.012	+0.15	+0.17	+0.17	+0.17	+0.14	+0.16	+0.13	
	SH-0 ₀ -L ₁ -C ₀ -R ₀	-0.074	-0.087		-0.059	+0.021	-0.068	-0.014	-0.093	+0.089	-0.018	-0.026	-0.099	+0.063	+0.086	+0.08	+0.084	+0.053	+0.069	+0.042	
	EM-0 ₀ -L ₁ -C ₀ -R ₀	-0.015	-0.028	+0.059		+0.08	-0.0094	+0.045	-0.034	+0.15	+0.041	+0.033	-0.04	+0.12	+0.15	+0.14	+0.14	+0.11	+0.13	+0.1	
	SH-0 ₂ -L ₀ -C ₀ -R ₀	-0.095	-0.11	-0.021	-0.08		-0.09	-0.036	-0.11	+0.067	-0.04	-0.047	-0.12	+0.042	+0.065	+0.058	+0.063	+0.031	+0.048	+0.02	
	EM-0 ₂ -L ₀ -C ₀ -R ₀	-0.0055	-0.018	+0.068	+0.0094	+0.09		+0.054	-0.024	+0.16	+0.05	+0.043	-0.03	+0.13	+0.15	+0.15	+0.15	+0.12	+0.14	+0.11	
	SH-0 ₂ -L ₁ -C ₀ -R ₀	-0.06	-0.073	+0.014	-0.045	+0.036	-0.054		-0.079	+0.1	-0.004	-0.012	-0.085	+0.077	+0.1	+0.094	+0.098	+0.067	+0.083	+0.056	
	EM-0 ₂ -L ₁ -C ₀ -R ₀	+0.019	+0.006	+0.093	+0.034	+0.11	+0.024	+0.079		+0.18	+0.075	+0.067	-0.006	+0.16	+0.18	+0.17	+0.18	+0.15	+0.16	+0.13	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₀	-0.16	-0.18	-0.089	-0.15	-0.067	-0.16	-0.1	-0.18		-0.11	-0.11	-0.19	-0.026	-0.0025	-0.0086	-0.0045	-0.036	-0.019	-0.047	
	EM-0 ₁₂ -L ₀ -C ₀ -R ₀	-0.056	-0.069	+0.018	-0.041	+0.04	-0.05	+0.004	-0.075	+0.11		-0.0075	-0.081	+0.081	+0.1	+0.098	+0.1	+0.071	+0.087	+0.06	
	SH-0 ₁₂ -L ₁ -C ₀ -R ₀	-0.048	-0.061	+0.026	-0.033	+0.047	-0.043	+0.012	-0.067	+0.11	+0.0075		-0.073	+0.089	+0.11	+0.11	+0.11	+0.079	+0.095	+0.067	
	EM-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.025	+0.012	+0.099	+0.04	+0.12	+0.03	+0.085	+0.006	+0.19	+0.081	+0.073		+0.16	+0.18	+0.18	+0.18	+0.15	+0.17	+0.14	
	SH-0 ₁₂ -L ₀ -C _{0.5} -R ₀	-0.14	-0.15	-0.063	-0.12	-0.042	-0.13	-0.077	-0.16	+0.026	-0.081	-0.089	-0.16		+0.023	+0.017	+0.021	-0.01	+0.0062	-0.021	
	SH-0 ₁₂ -L ₀ -C ₁ -R ₀	-0.16	-0.17	-0.086	-0.15	-0.065	-0.15	-0.1	-0.18	+0.0025	-0.1	-0.11	-0.18	-0.023		-0.0061	-0.002	-0.033	-0.017	-0.045	
	SH-0 ₁₂ -L ₀ -C ₃ -R ₀	-0.15	-0.17	-0.08	-0.14	-0.058	-0.15	-0.094	-0.17	+0.0086	-0.098	-0.11	-0.18	-0.017	+0.0061		+0.0041	-0.027	-0.011	-0.038	
	SH-0 ₁₂ -L ₀ -C ₀ -R _{0.25}	-0.16	-0.17	-0.084	-0.14	-0.063	-0.15	-0.098	-0.18	+0.0045	-0.1	-0.11	-0.18	-0.021	+0.002	-0.0041		-0.031	-0.015	-0.043	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₁	-0.13	-0.14	-0.053	-0.11	-0.031	-0.12	-0.067	-0.15	+0.036	-0.071	-0.079	-0.15	+0.01	+0.033	+0.027	+0.031		+0.016	-0.011	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₅	-0.14	-0.16	-0.069	-0.13	-0.048	-0.14	-0.083	-0.16	+0.019	-0.087	-0.095	-0.17	-0.0062	+0.017	+0.011	+0.015	-0.016		-0.028	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₁₀	-0.12	-0.13	-0.042	-0.1	-0.02	-0.11	-0.056	-0.13	+0.047	-0.06	-0.067	-0.14	+0.021	+0.045	+0.038	+0.043	+0.011	+0.028		
Motus	SH-0 ₀ -L ₀ -C ₀ -R ₀		+0.0035	+0.013	-0.018	+0.027	-0.0004	+0.0006	-0.004	+0.1	+0.081	-0.0017	-0.0058	+0.071	+0.1	+0.094	+0.089	+0.087	+0.091	+0.097	
	EM-0 ₀ -L ₀ -C ₀ -R ₀	-0.0035		+0.009	-0.022	+0.023	-0.0039	-0.0029	-0.0075	+0.096	+0.078	-0.0051	-0.0093	+0.067	+0.1	+0.091	+0.085	+0.083	+0.087	+0.094	
	SH-0 ₀ -L ₁ -C ₀ -R ₀	-0.013	-0.009		-0.031	+0.014	-0.013	-0.012	-0.017	+0.087	+0.069	-0.014	-0.018	+0.058	+0.092	+0.081	+0.076	+0.074	+0.078	+0.085	
	EM-0 ₀ -L ₁ -C ₀ -R ₀	+0.018	+0.022	+0.031		+0.045	+0.018	+0.019	+0.014	+0.12	+0.1	+0.017	+0.013	+0.089	+0.12	+0.11	+0.11	+0.1	+0.11	+0.12	
	SH-0 ₂ -L ₀ -C ₀ -R ₀	-0.027	-0.023	-0.014	-0.045		-0.027	-0.026	-0.031	+0.073	+0.054	-0.029	-0.033	+0.044	+0.077	+0.067	+0.062	+0.06	+0.064	+0.071	
	EM-0 ₂ -L ₀ -C ₀ -R ₀	+0.0004	+0.0039	+0.013	-0.018	+0.027		+0.00099	-0.0037	+0.1	+0.082	-0.0013	-0.0054	+0.071	+0.1	+0.094	+0.089	+0.087	+0.091	+0.098	
	SH-0 ₂ -L ₁ -C ₀ -R ₀	-0.0006	+0.0029	+0.012	-0.019	+0.026	-0.00099		-0.0046	+0.099	+0.081	-0.0022	-0.0064	+0.07	+0.1	+0.093	+0.088	+0.086	+0.09	+0.097	
	EM-0 ₂ -L ₁ -C ₀ -R ₀	+0.0004	+0.0075	+0.017	-0.014	+0.031	+0.0037	+0.0046		+0.1	+0.085	+0.0024	-0.0017	+0.075	+0.11	+0.098	+0.093	+0.091	+0.095	+0.1	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₀	-0.1	-0.096	-0.087	-0.12	-0.073	-0.1	-0.099	-0.1		-0.019	-0.1	-0.11	-0.029	+0.0043	-0.0058	-0.011	-0.013	-0.0088	-0.0024	
	EM-0 ₁₂ -L ₀ -C ₀ -R ₀	-0.081	-0.078	-0.069	-0.1	-0.054	-0.082	-0.081	-0.085	+0.019		-0.083	-0.087	-0.011	+0.023	+0.013	+0.0078	+0.0054	+0.0098	+0.016	
	SH-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.0017	+0.0051	+0.014	-0.017	+0.029	+0.0013	+0.0022	-0.0024	+0.1	+0.083		-0.0041	+0.072	+0.11	+0.096	+0.091	+0.088	+0.093	+0.099	
	EM-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.0058	+0.0093	+0.018	-0.013	+0.033	+0.0054	+0.0064	+0.0017	+0.11	+0.087	+0.0041		+0.076	+0.11	+0.1	+0.095	+0.092	+0.097	+0.1	
	SH-0 ₁₂ -L ₀ -C _{0.5} -R ₀	-0.071	-0.067	-0.058	-0.089	-0.044	-0.071	-0.07	-0.075	+0.029	+0.011	-0.072	-0.076		+0.034	+0.023	+0.018	+0.016	+0.02	+0.027	
	SH-0 ₁₂ -L ₀ -C ₁ -R ₀	-0.1	-0.1	-0.092	-0.12	-0.077	-0.1	-0.1	-0.11	-0.0043	-0.023	-0.11	-0.11	-0.034		-0.01	-0.015	-0.018	-0.013	-0.0067	
	SH-0 ₁₂ -L ₀ -C ₃ -R ₀	-0.094	-0.091	-0.081	-0.11	-0.067	-0.094	-0.093	-0.098	+0.0058	-0.013	-0.096	-0.1	-0.023	+0.01		-0.0051	-0.0074	-0.003	+0.0034	
	SH-0 ₁₂ -L ₀ -C ₀ -R _{0.25}	-0.089	-0.085	-0.076	-0.11	-0.062	-0.089	-0.088	-0.093	+0.011	-0.0078	-0.091	-0.095	-0.018	+0.015	+0.0051		-0.0023	+0.0021	+0.0085	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₁	-0.087	-0.083	-0.074	-0.1	-0.06	-0.087	-0.086	-0.091	+0.013	-0.0054	-0.088	-0.092	-0.016	+0.018	+0.0074	+0.0023		+0.0044	+0.011	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₅	-0.091	-0.087	-0.078	-0.11	-0.064	-0.091	-0.09	-0.095	+0.0088	-0.0098	-0.093	-0.097	-0.02	+0.013	+0.003	-0.0021	-0.0044		+0.0064	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₁₀	-0.097	-0.094	-0.085	-0.12	-0.071	-0.098	-0.097	-0.1	+0.0024	-0.016	-0.099	-0.1	-0.027	+0.0067	-0.0034	-0.0085	-0.011	-0.0064		
Agrid	SH-0 ₀ -L ₀ -C ₀ -R ₀		-0.014	+0.03	-0.013	+0.039	+0.006	+0.032	-0.0014	+0.12	+0.052	+0.0089	-0.014	+0.13	+0.13	+0.12	+0.14	+0.12	+0.17	+0.14	
	EM-0 ₀ -L ₀ -C ₀ -R ₀	+0.014		+0.044	+0.0062	+0.053	+0.02	+0.045	+0.012	+0.13	+0.066	+0.022	-0.00077	+0.14	+0.14	+0.13	+0.15	+0.13	+0.18	+0.16	
	SH-0 ₀ -L ₁ -C ₀ -R ₀	-0.03	-0.044		-0.043	+0.0085	-0.024	+0.0015	-0.032	+0.091	+0.022	-0.022	-0.045	+0.096	+0.095	+0.085	+0.11	+0.088	+0.14	+0.11	
	EM-0 ₀ -L ₁ -C ₀ -R ₀	+0.013	-0.00062	+0.043		+0.052	+0.019	+0.045	+0.012	+0.13	+0.065	+0.022	-0.0014	+0.14	+0.14	+0.13	+0.15	+0.13	+0.18	+0.16	
	SH-0 ₂ -L ₀ -C ₀ -R ₀	-0.039	-0.053	-0.0085	-0.052		-0.033	-0.007	-0.04	+0.082	+0.014	-0.03	-0.053	+0.087	+0.086	+0.077	+0.098	+0.079	+0.13	+0.11	
	EM-0 ₂ -L ₀ -C ₀ -R ₀	-0.006	-0.02	+0.024	-0.019	+0.033		+0.026	-0.0073	+0.12	+0.047	+0.0029	-0.02	+0.12	+0.12	+0.11	+0.13	+0.11	+0.17	+0.14	
	SH-0 ₂ -L ₁ -C ₀ -R ₀	-0.032	-0.045	-0.0015	-0.045	+0.007	-0.026		-0.033	+0.089	+0.021	-0.023	-0.046	+0.094	+0.093	+0.084	+0.1	+0.086	+0.14	+0.11	
	EM-0 ₂ -L ₁ -C ₀ -R ₀	+0.0014	-0.012	+0.032	-0.012	+0.04	+0.0073	+0.033		+0.12	+0.054	+0.01	-0.013	+0.13	+0.13	+0.12	+0.14	+0.12	+0.17	+0.15	
	SH-0 ₁₂ -L ₀ -C ₀ -R ₀	-0.12	-0.13	-0.091	-0.13	-0.082	-0.12	-0.089	-0.12		-0.069	-0.11	-0.14	+0.0047	+0.0038	-0.0055	+0.015	-0.003	+0.05	+0.023	
	EM-0 ₁₂ -L ₀ -C ₀ -R ₀	-0.052	-0.066	-0.022	-0.065	-0.014	-0.047	-0.021	-0.054	+0.069		-0.044	-0.067	+0.074	+0.073	+0.063	+0.084	+0.066	+0.12	+0.092	
	SH-0 ₁₂ -L ₁ -C _{0</}																				

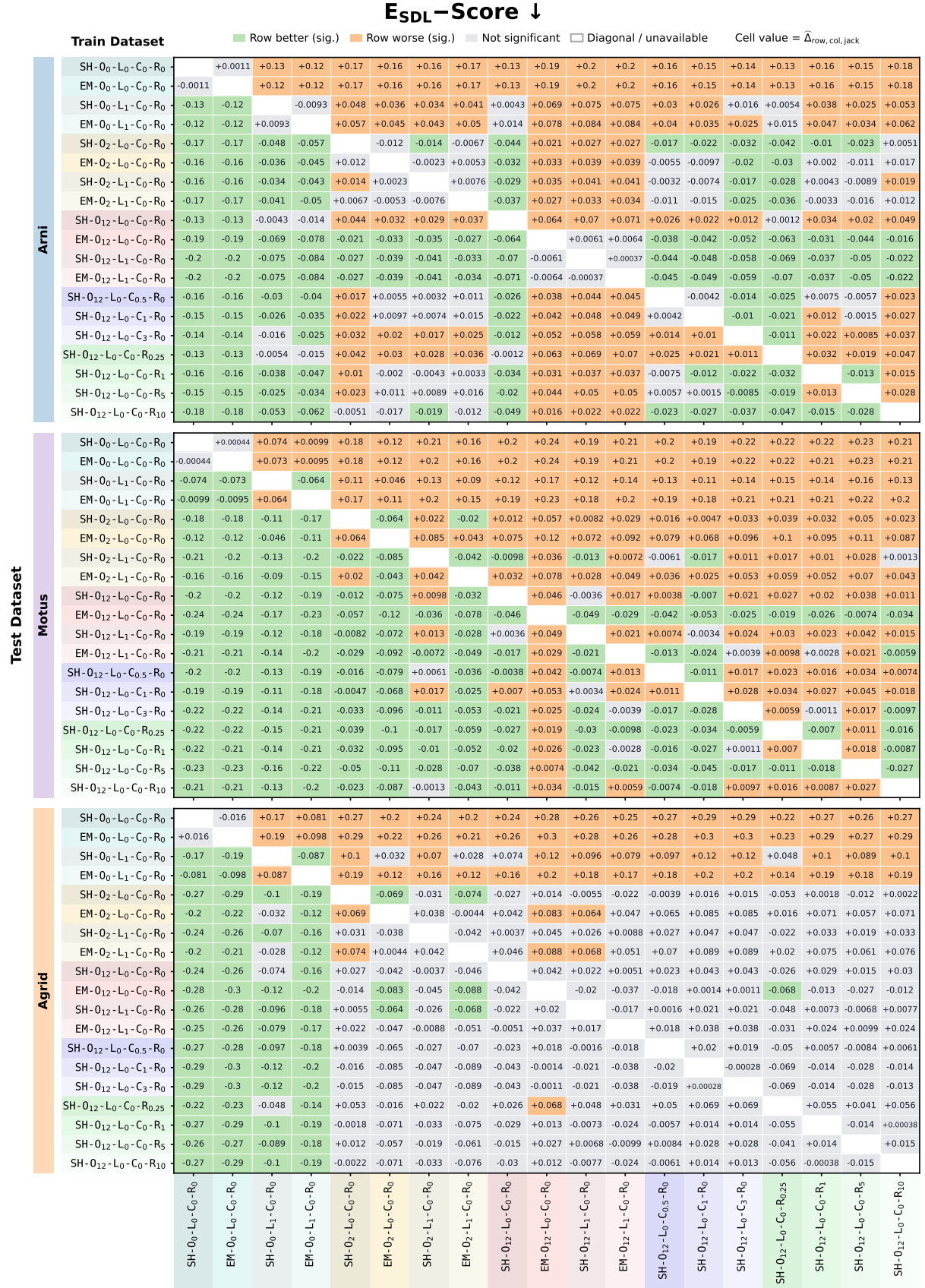
A Additional Tables and Figures



A Additional Tables and Figures

		Location-Sensitive F-Score (F ₁₅) ↑																			
Train Dataset		Row better (sig.) Row worse (sig.) Not significant Diagonal / unavailable																Cell value = $\hat{A}_{row, col, jack}$			
Test Dataset	Arni	SH-0 ₀ -L ₀ -C ₀ -R ₀		+0.0018	-0.19	-0.17	-0.26	-0.22	-0.23	-0.22	-0.21	-0.28	-0.28	-0.27	-0.25	-0.25	-0.23	-0.22	-0.25	-0.24	-0.27
		EM-0 ₀ -L ₀ -C ₀ -R ₀	-0.0018		-0.2	-0.18	-0.26	-0.22	-0.23	-0.22	-0.22	-0.28	-0.29	-0.27	-0.25	-0.25	-0.23	-0.22	-0.25	-0.24	-0.27
		SH-0 ₀ -L ₁ -C ₀ -R ₀	+0.19	+0.2		+0.022	-0.064	-0.023	-0.034	-0.025	-0.02	-0.082	-0.089	-0.071	-0.051	-0.051	-0.035	-0.021	-0.058	-0.044	-0.076
		EM-0 ₀ -L ₁ -C ₀ -R ₀	+0.17	+0.18	-0.022		-0.085	-0.045	-0.055	-0.047	-0.041	-0.1	-0.11	-0.092	-0.072	-0.073	-0.056	-0.043	-0.08	-0.066	-0.098
		SH-0 ₂ -L ₀ -C ₀ -R ₀	+0.26	+0.26	+0.064	+0.085		+0.041	+0.03	+0.039	+0.044	-0.018	-0.025	-0.0069	+0.013	+0.013	+0.029	+0.043	+0.0058	+0.02	-0.013
		EM-0 ₂ -L ₀ -C ₀ -R ₀	+0.22	+0.22	+0.023	+0.045	-0.041		-0.011	-0.0021	+0.0032	-0.059	-0.066	-0.048	-0.028	-0.028	-0.012	+0.002	-0.035	-0.021	-0.053
		SH-0 ₂ -L ₁ -C ₀ -R ₀	+0.23	+0.23	+0.034	+0.055	-0.03	+0.011		+0.0086	+0.014	-0.048	-0.055	-0.037	-0.017	-0.017	-0.00086	+0.013	-0.024	-0.01	-0.043
		EM-0 ₂ -L ₁ -C ₀ -R ₀	+0.22	+0.22	+0.025	+0.047	-0.039	+0.0021	-0.0086		+0.0053	-0.057	-0.064	-0.046	-0.026	-0.026	-0.0095	+0.004	-0.033	-0.019	-0.051
		SH-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.21	+0.22	+0.02	+0.041	-0.044	-0.0032	-0.014	-0.0053		-0.062	-0.069	-0.051	-0.031	-0.031	-0.015	-0.0013	-0.038	-0.024	-0.056
		EM-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.28	+0.28	+0.082	+0.1	+0.018	+0.059	+0.048	+0.057	+0.062		-0.007	+0.011	+0.031	+0.031	+0.047	+0.061	+0.024	+0.038	+0.0058
		SH-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.28	+0.29	+0.089	+0.11	+0.025	+0.066	+0.055	+0.064	+0.069	+0.007		+0.018	+0.038	+0.038	+0.054	+0.068	+0.031	+0.045	+0.013
		EM-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.27	+0.27	+0.071	+0.092	+0.0069	+0.048	+0.037	+0.046	+0.051	-0.011	-0.018		+0.02	+0.02	+0.036	+0.05	+0.013	+0.027	-0.0056
		SH-0 ₁₂ -L ₀ -C _{0.5} -R ₀	+0.25	+0.25	+0.051	+0.072	-0.013	+0.028	+0.017	+0.026	+0.031	-0.031	-0.038	-0.02		-0.0002	+0.016	+0.03	-0.0072	+0.0069	-0.025
		EM-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.25	+0.25	+0.051	+0.073	-0.013	+0.028	+0.017	+0.026	+0.031	-0.031	-0.038	-0.02	+0.0002		+0.016	+0.03	-0.007	+0.0071	-0.025
		SH-0 ₁₂ -L ₀ -C ₃ -R ₀	+0.23	+0.23	+0.035	+0.056	-0.029	+0.012	+0.00086	+0.0095	+0.015	-0.047	-0.054	-0.036	-0.016	-0.016		+0.014	-0.023	-0.0092	-0.042
		SH-0 ₁₂ -L ₀ -C ₀ -R _{0.25}	+0.22	+0.22	+0.021	+0.043	-0.043	-0.002	-0.013	-0.004	+0.0013	-0.061	-0.068	-0.05	-0.03	-0.03	-0.014		-0.037	-0.023	-0.055
		SH-0 ₁₂ -L ₀ -C ₀ -R ₁	+0.25	+0.25	+0.058	+0.08	-0.0058	+0.035	+0.024	+0.033	+0.038	-0.024	-0.031	-0.013	+0.0072	+0.007	+0.023	+0.037		+0.014	-0.018
		SH-0 ₁₂ -L ₀ -C ₀ -R ₅	+0.24	+0.24	+0.044	+0.066	-0.02	+0.021	+0.01	+0.019	+0.024	-0.038	-0.045	-0.027	-0.0069	-0.0071	+0.0092	+0.023	-0.014		-0.032
		SH-0 ₁₂ -L ₀ -C ₀ -R ₁₀	+0.27	+0.27	+0.076	+0.098	+0.013	+0.053	+0.043	+0.051	+0.056	-0.0058	-0.013	+0.0056	+0.025	+0.025	+0.042	+0.055	+0.018	+0.032	
		Test Dataset	Motus	SH-0 ₀ -L ₀ -C ₀ -R ₀		-0.0077	-0.11	+0.0013	-0.25	-0.15	-0.28	-0.22	-0.29	-0.35	-0.26	-0.29	-0.28	-0.28	-0.31	-0.32	-0.31
EM-0 ₀ -L ₀ -C ₀ -R ₀	+0.0077				-0.1	+0.009	-0.24	-0.14	-0.27	-0.22	-0.28	-0.34	-0.25	-0.28	-0.27	-0.27	-0.31	-0.31	-0.3	-0.33	-0.29
SH-0 ₀ -L ₁ -C ₀ -R ₀	+0.11			+0.1		+0.11	-0.14	-0.039	-0.17	-0.12	-0.18	-0.24	-0.15	-0.18	-0.17	-0.17	-0.21	-0.21	-0.2	-0.23	-0.19
EM-0 ₀ -L ₁ -C ₀ -R ₀	-0.0013			-0.009	-0.11		-0.25	-0.15	-0.28	-0.22	-0.29	-0.35	-0.26	-0.29	-0.28	-0.28	-0.32	-0.32	-0.31	-0.34	-0.3
SH-0 ₂ -L ₀ -C ₀ -R ₀	+0.25			+0.24	+0.14	+0.25		+0.1	-0.029	+0.027	-0.037	-0.096	-0.0071	-0.037	-0.031	-0.026	-0.065	-0.071	-0.06	-0.087	-0.05
EM-0 ₂ -L ₀ -C ₀ -R ₀	+0.15			+0.14	+0.039	+0.15	-0.1		-0.13	-0.077	-0.14	-0.2	-0.11	-0.14	-0.13	-0.13	-0.17	-0.17	-0.16	-0.19	-0.15
SH-0 ₂ -L ₁ -C ₀ -R ₀	+0.28			+0.27	+0.17	+0.28	+0.029	+0.13		+0.055	-0.008	-0.068	+0.022	-0.0088	-0.0026	+0.0031	-0.036	-0.043	-0.032	-0.059	-0.022
EM-0 ₂ -L ₁ -C ₀ -R ₀	+0.22			+0.22	+0.12	+0.22	-0.027	+0.077	-0.055		-0.063	-0.12	-0.034	-0.064	-0.058	-0.052	-0.092	-0.098	-0.087	-0.11	-0.077
SH-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.29			+0.28	+0.18	+0.29	+0.037	+0.14	+0.008	+0.063		-0.06	+0.029	-0.00083	+0.0054	+0.011	-0.028	-0.035	-0.024	-0.051	-0.014
EM-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.35			+0.34	+0.24	+0.35	+0.096	+0.2	+0.068	+0.12	+0.06		+0.089	+0.059	+0.065	+0.071	+0.032	+0.025	+0.036	+0.0092	+0.046
SH-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.26			+0.25	+0.15	+0.26	+0.0071	+0.11	-0.022	+0.034	-0.029	-0.089		-0.03	-0.024	-0.018	-0.058	-0.064	-0.053	-0.08	-0.043
EM-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.29			+0.28	+0.18	+0.29	+0.037	+0.14	+0.0088	+0.064	+0.00083	-0.059	+0.03		+0.0062	+0.012	-0.027	-0.034	-0.023	-0.05	-0.013
SH-0 ₁₂ -L ₀ -C _{0.5} -R ₀	+0.28			+0.27	+0.17	+0.28	+0.031	+0.13	+0.0026	+0.058	-0.0054	-0.065	+0.024	-0.0062		+0.0057	-0.034	-0.04	-0.029	-0.056	-0.019
SH-0 ₁₂ -L ₀ -C ₁ -R ₀	+0.28			+0.27	+0.17	+0.28	+0.026	+0.13	-0.0031	+0.052	-0.011	-0.071	+0.018	-0.012	-0.0057		-0.039	-0.046	-0.035	-0.062	-0.025
SH-0 ₁₂ -L ₀ -C ₃ -R ₀	+0.31			+0.31	+0.21	+0.32	+0.065	+0.17	+0.036	+0.092	+0.028	-0.032	+0.058	+0.027	+0.034	+0.039		-0.0064	+0.0047	-0.022	+0.015
SH-0 ₁₂ -L ₀ -C ₀ -R _{0.25}	+0.32			+0.31	+0.21	+0.32	+0.071	+0.17	+0.043	+0.098	+0.035	-0.025	+0.064	+0.034	+0.04	+0.046	+0.0064		+0.011	-0.016	+0.021
SH-0 ₁₂ -L ₀ -C ₀ -R ₁	+0.31			+0.3	+0.2	+0.31	+0.06	+0.16	+0.032	+0.087	+0.024	-0.036	+0.053	+0.023	+0.029	+0.035	-0.0047	-0.011		-0.027	+0.01
SH-0 ₁₂ -L ₀ -C ₀ -R ₅	+0.34			+0.33	+0.23	+0.34	+0.087	+0.19	+0.059	+0.11	+0.051	-0.0092	+0.08	+0.05	+0.056	+0.062	+0.022	+0.016	+0.027		+0.037
SH-0 ₁₂ -L ₀ -C ₀ -R ₁₀	+0.3			+0.29	+0.19	+0.3	+0.05	+0.15	+0.022	+0.077	+0.014	-0.046	+0.043	+0.013	+0.019	+0.025	-0.015	-0.021	-0.01	-0.037	
Test Dataset	Agrid			SH-0 ₀ -L ₀ -C ₀ -R ₀		+0.011	-0.22	-0.095	-0.33	-0.24	-0.3	-0.24	-0.31	-0.35	-0.32	-0.3	-0.35	-0.37	-0.37	-0.28	-0.35
		EM-0 ₀ -L ₀ -C ₀ -R ₀	-0.011		-0.23	-0.11	-0.35	-0.25	-0.31	-0.26	-0.32	-0.37	-0.33	-0.31	-0.36	-0.38	-0.38	-0.29	-0.36	-0.36	-0.37
		SH-0 ₀ -L ₁ -C ₀ -R ₀	+0.22	+0.23		+0.13	-0.11	-0.02	-0.076	-0.023	-0.088	-0.13	-0.099	-0.08	-0.12	-0.15	-0.15	-0.054	-0.13	-0.13	-0.14
		EM-0 ₀ -L ₁ -C ₀ -R ₀	+0.095	+0.11	-0.13		-0.24	-0.15	-0.2	-0.15	-0.21	-0.26	-0.23	-0.21	-0.25	-0.28	-0.28	-0.18	-0.26	-0.25	-0.27
		SH-0 ₂ -L ₀ -C ₀ -R ₀	+0.33	+0.35	+0.11	+0.24		+0.092	+0.037	+0.089	+0.024	-0.02	+0.013	+0.032	-0.012	-0.039	-0.037	+0.058	-0.019	-0.015	-0.026
		EM-0 ₂ -L ₀ -C ₀ -R ₀	+0.24	+0.25	+0.02	+0.15	-0.092		-0.055	-0.0029	-0.068	-0.11	-0.079	-0.06	-0.1	-0.13	-0.13	-0.034	-0.11	-0.11	-0.12
		SH-0 ₂ -L ₁ -C ₀ -R ₀	+0.3	+0.31	+0.076	+0.2	-0.037	+0.055		+0.052	-0.013	-0.057	-0.024	-0.0048	-0.049	-0.076	-0.074	+0.021	-0.055	-0.051	-0.063
		EM-0 ₂ -L ₁ -C ₀ -R ₀	+0.24	+0.26	+0.023	+0.15	-0.089	+0.0029	-0.052		-0.065	-0.11	-0.076	-0.057	-0.1	-0.13	-0.13	-0.031	-0.11	-0.1	-0.12
		SH-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.31	+0.32	+0.088	+0.21	-0.024	+0.068	+0.013	+0.065		-0.044	-0.011	+0.0078	-0.036	-0.063	-0.061	+0.038	-0.043	-0.039	-0.05
		EM-0 ₁₂ -L ₀ -C ₀ -R ₀	+0.35	+0.37	+0.13	+0.26	+0.02	+0.11	+0.057	+0.11	+0.044		+0.033	+0.052	+0.0079	-0.019	-0.017	+0.078	+0.0012	+0.0053	-0.0061
		SH-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.32	+0.33	+0.099	+0.23	+0.013	+0.079	+0.024	+0.076	+0.011	-0.033		+0.019	-0.025	-0.052	-0.05	+0.045	-0.032	-0.028	-0.039
EM-0 ₁₂ -L ₁ -C ₀ -R ₀	+0.3	+0.31	+0.08	+0.21	-0.032	+0.06	+0.0048	+0.057	-0.0078	-0.052	-0.019		-0.044	-0.071	-0.069	+0.026	-0.051	-0.047	-0.058		
SH-0 ₁₂ -L ₀ -C _{0.5} -R ₀	+0.35	+0.36	+0.12	+0.25	+0.012	+0.1	+0.0														

A Additional Tables and Figures



B Digital Appendix

The source code used for data generation, model training, and evaluation in this thesis is available in the following GitHub repository:

`https://github.com/Fabian-Staub/acoustic-simulation-fidelity-ssld`

The repository contains the main scripts used for the experiments presented in this thesis, together with additional supporting code and documentation.

Use of AI

Artificial intelligence (AI) tools were used during this work, primarily to support text formulation, programming tasks, and the search for potentially relevant literature. All referenced sources were reviewed manually, and the relevant information was independently verified. All AI-generated outputs were critically assessed and carefully validated for correctness before being incorporated into this work. The AI tool used predominantly was ChatGPT, with GPT-5.6 Sol as the underlying model.

Declaration

I hereby declare that I have completed and written this work independently. All sources, literature, and other aids used have been properly acknowledged and identified as such.

A handwritten signature in black ink, reading "F. Staub". The signature is written in a cursive style with a large, stylized 'F' and 'S'. It is positioned above a horizontal line.

Fabian Staub

Ilmenau, August 31, 2026